

# LIGHT FIELDS

Computational Imaging – Spring 2024

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# Slide credits

Some of these slides are based on the following content:

- [Computational Photography \(15-463\)](#). Carnegie Mellon University.
- Computational Plenoptic Imaging. Wetzstein et al. SIGGRAPH Course.
- Computational Imaging and Display. Stanford University.
- Diego Gutierrez.

# Overview

Basis:

- ✓ Digital photography (Lecture #1)
- ✓ Mathematical foundations (Lecture #2)

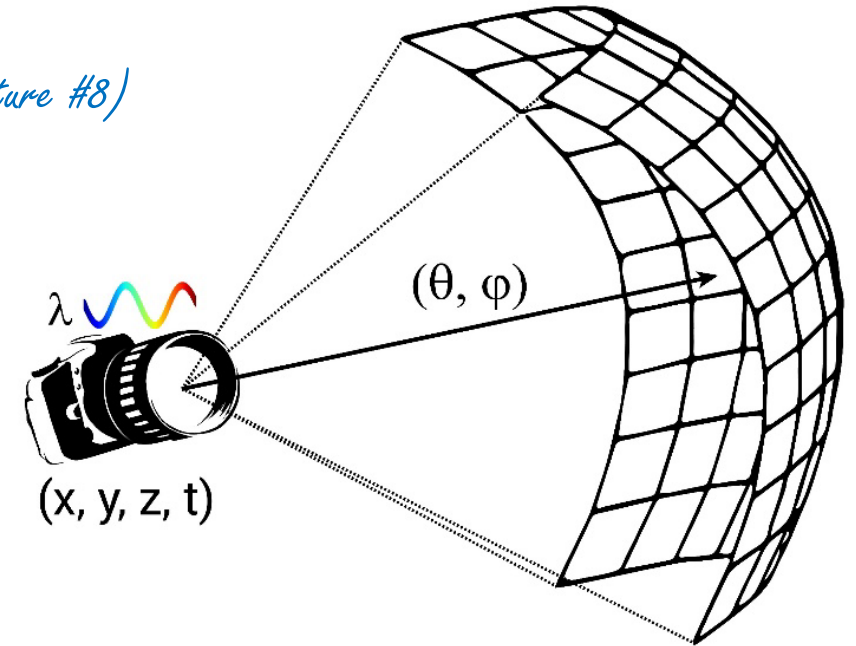
$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

*Hyperspectral imaging (Lecture #6)* →  $\lambda$

*Transient imaging (Lecture #8)* →  $t$

*Light field imaging (Lecture #4)* →  $(x, y, z)$

*High dynamic range imaging (Lecture #5)* →  $P$



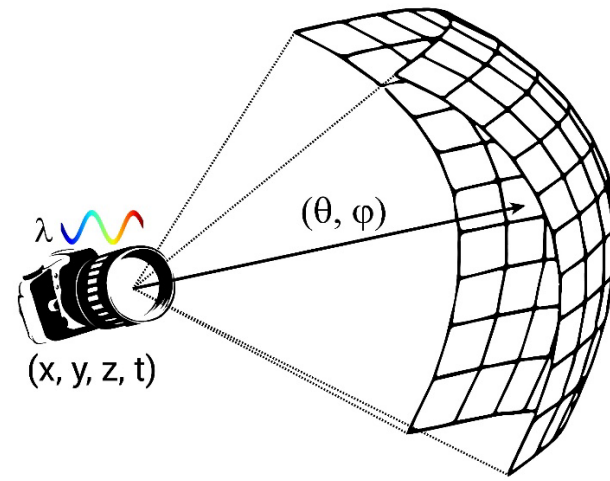
Plus common tools and techniques:

- ✓ Coded photography (Lecture #3)
- ✓ Compressive sensing (Lecture #3)
- Computational illumination (Lecture #7)

# Overview

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

$$P(\theta, \Phi, \lambda, t, x, y, z)$$



## Grayscale snapshot:

- Seen from a single view point
- At a single time
- Averaged over wavelengths

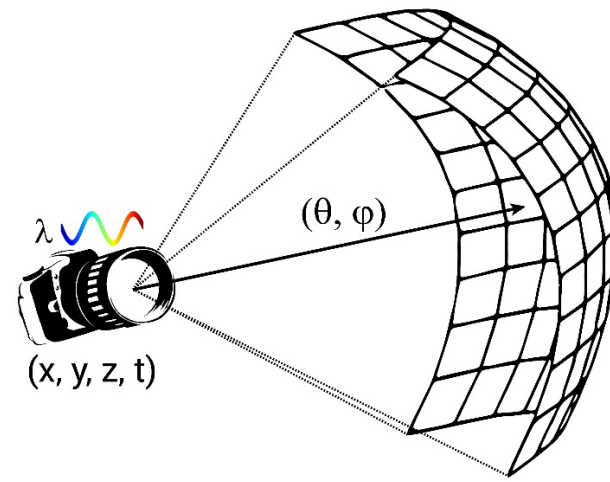




# Overview

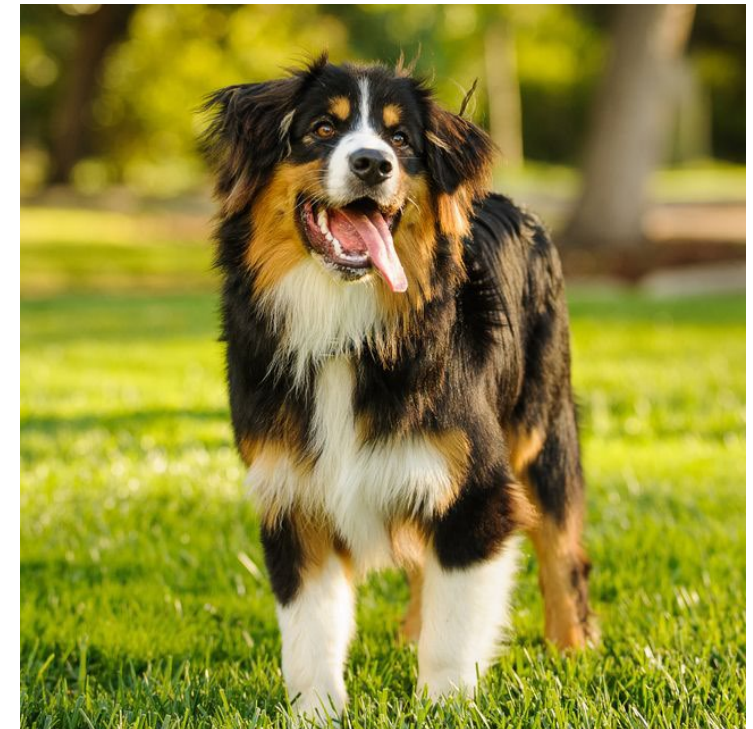
$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

$$P(\theta, \Phi, \lambda, t, x, y, z)$$



## Color snapshot:

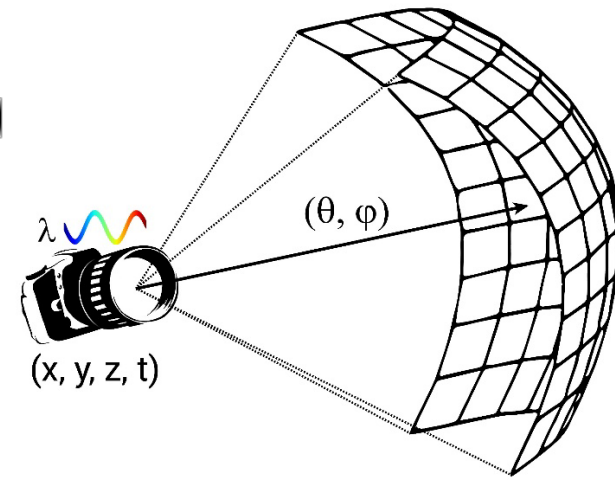
- Seen from a single view point
- At a single time
- As a function of wavelength



# Overview

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

$$P(\theta, \Phi, \lambda, t, x, y, z)$$



## Color video:

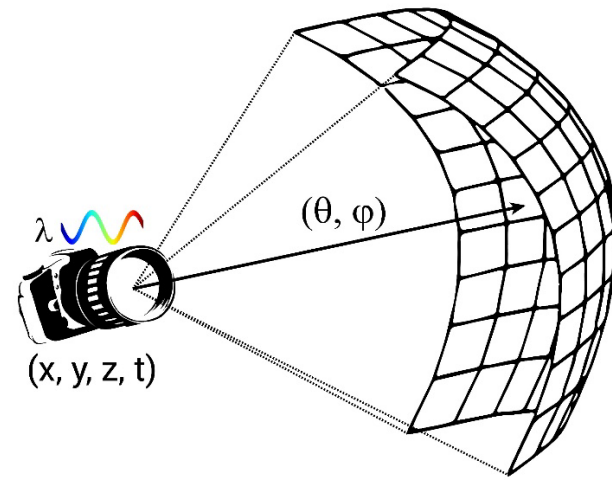
- Seen from a single view point
- Over time
- As a function of wavelength



# Overview

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

$$P(\theta, \Phi, \lambda, t, x, y, z)$$



## Holographic Color video:

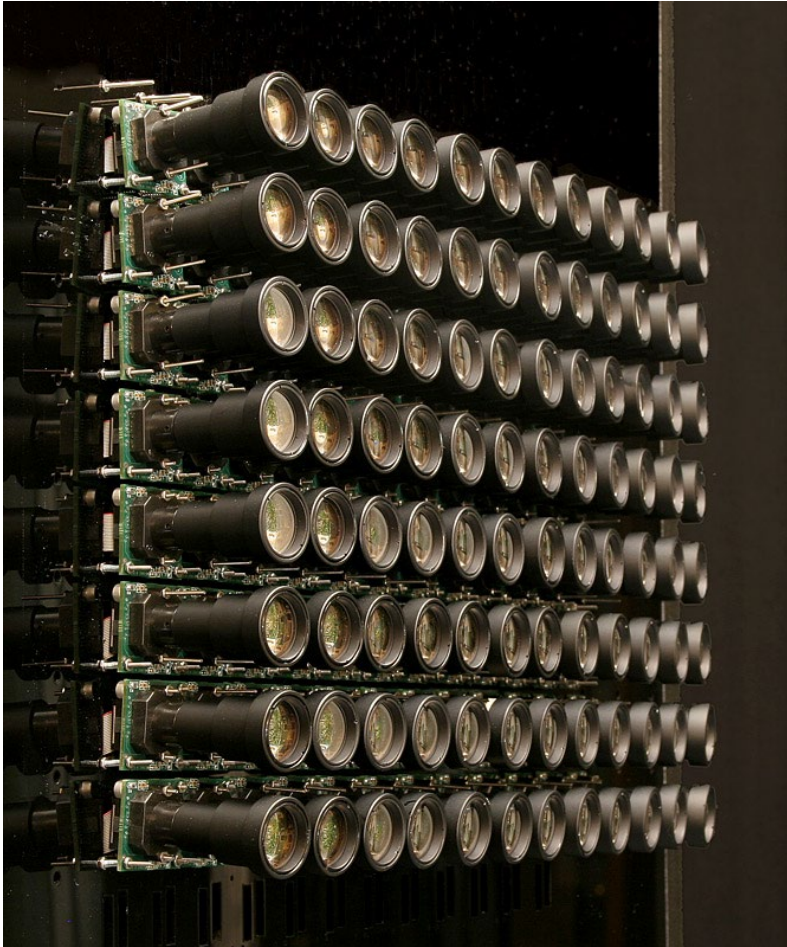
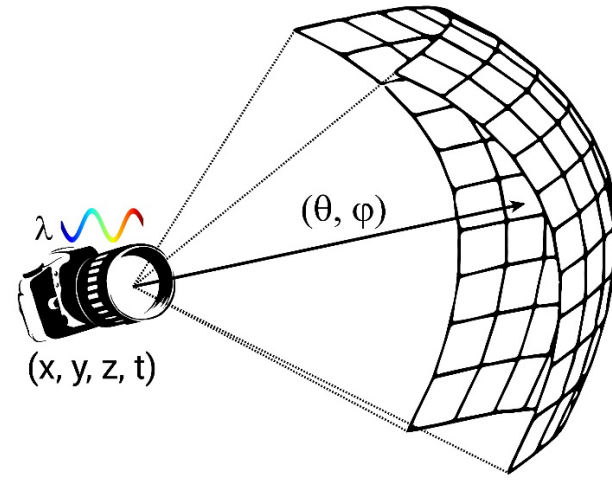
- Seen from any viewpoint
- Over time
- As a function of wavelength





# Overview

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

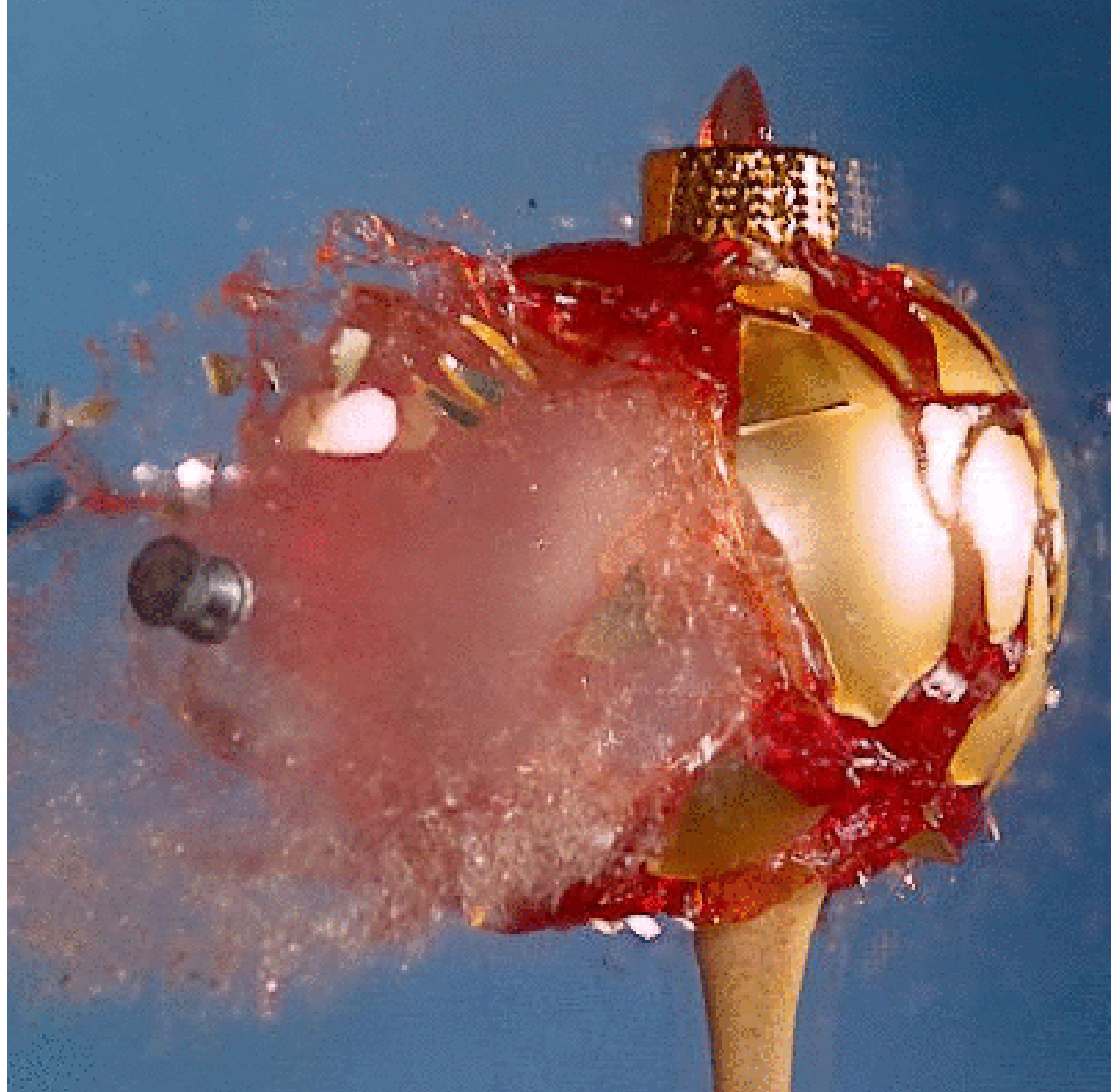


Stanford multi-camera array



Lytro Illum (micro-lens array)

A '*new kind*'  
of pictures



# *Not just view shifting...*

Source: LightField Forum



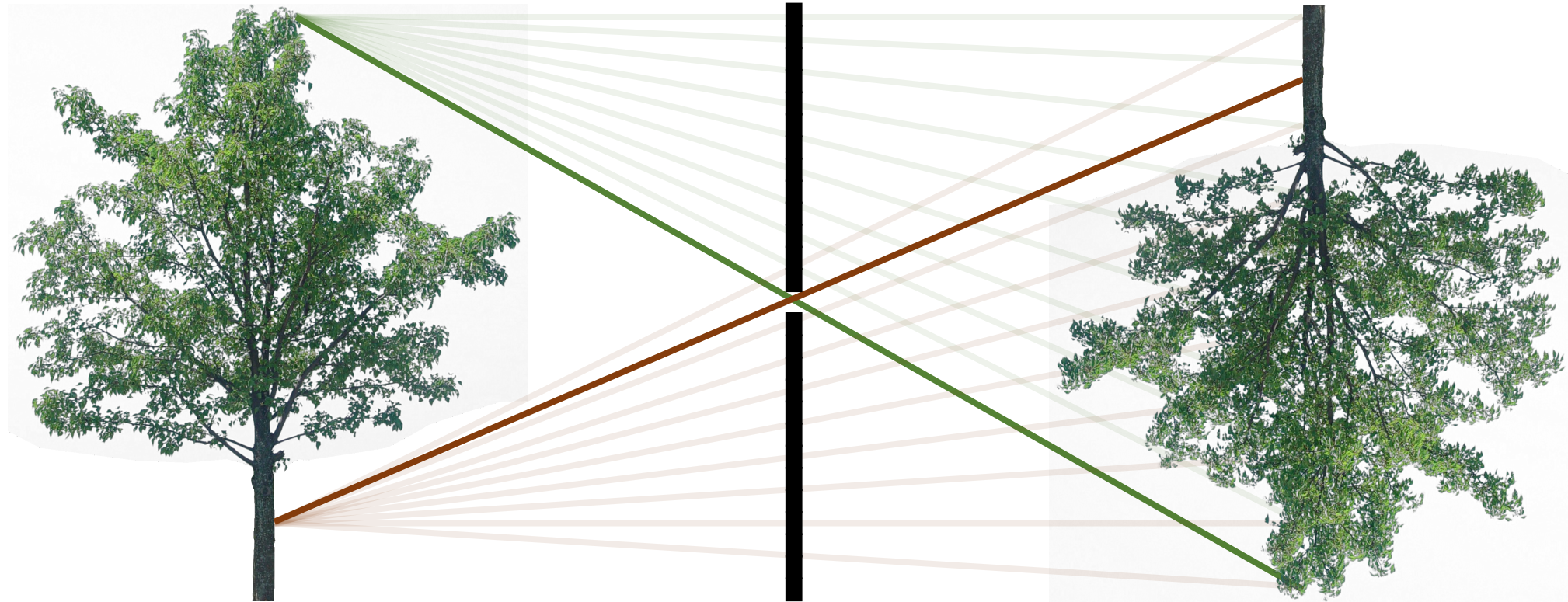
## Focus shifting

# Light Field Imaging

- Concept
- Parameterization
- Acquisition
- Rendering



# Pinhole camera

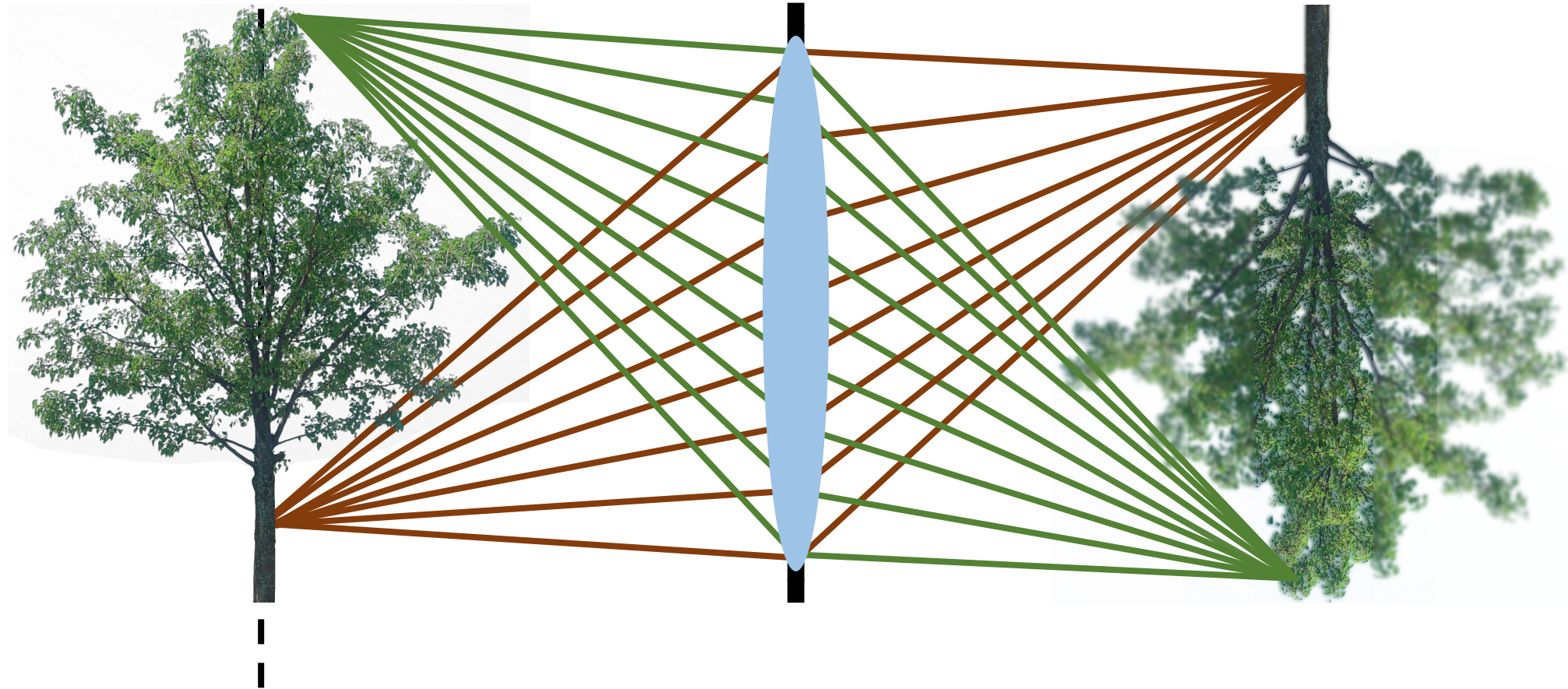


- Everything is in focus.
- Very light inefficient.



# Lens camera

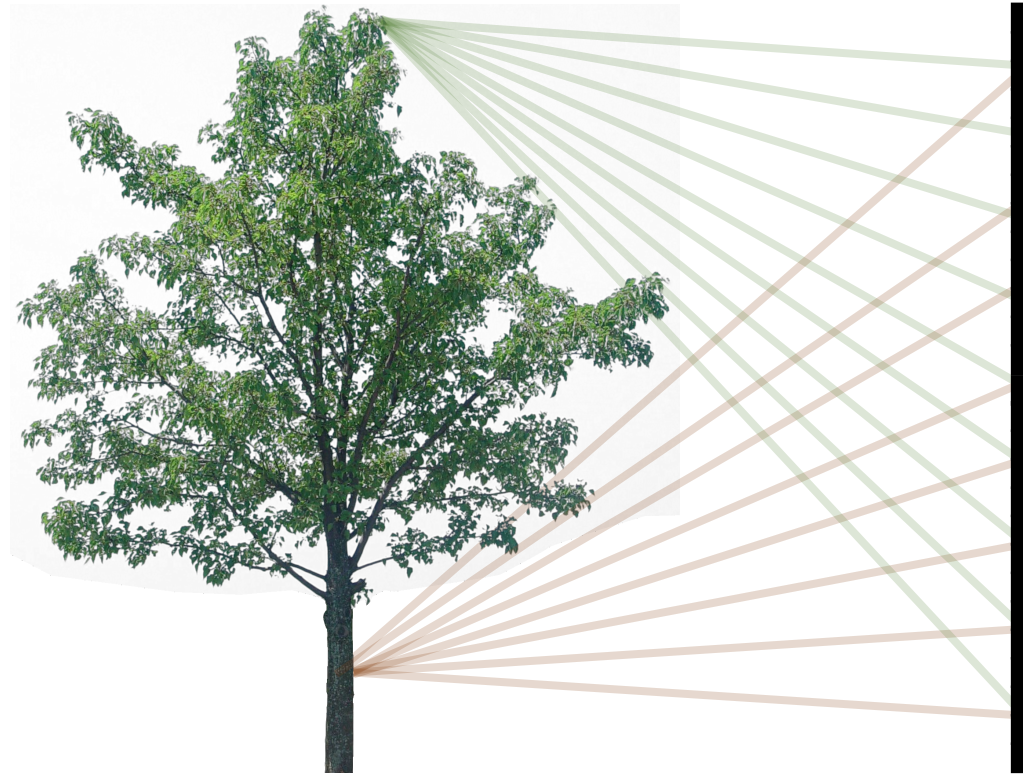
focus plane



- Only one plane is in focus.
- Very light efficient.

A lens measures all rays radiated from the object (up to aperture size).

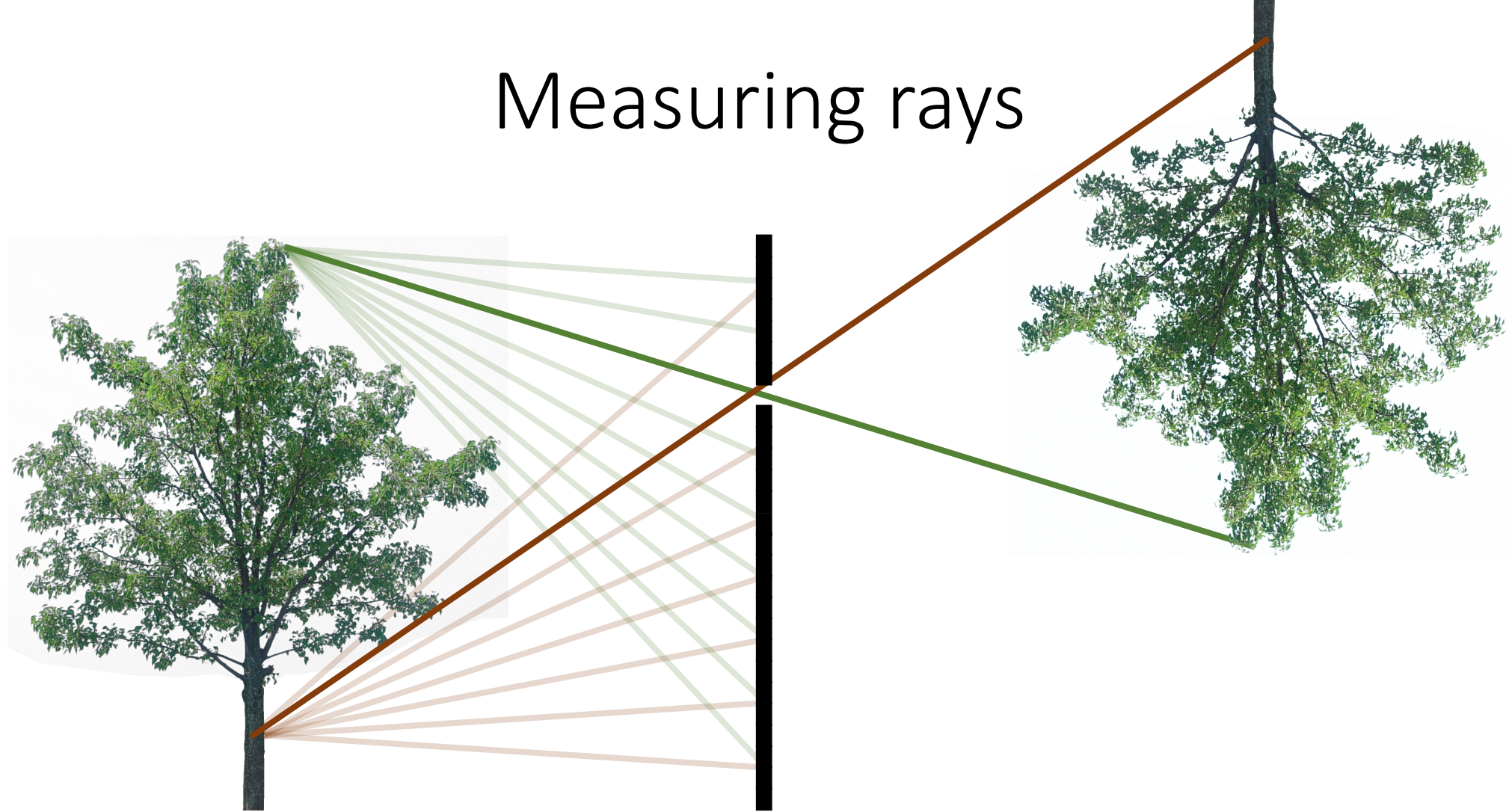
# Measuring rays



We can capture the same set of rays by using a pinhole camera from multiple viewpoints

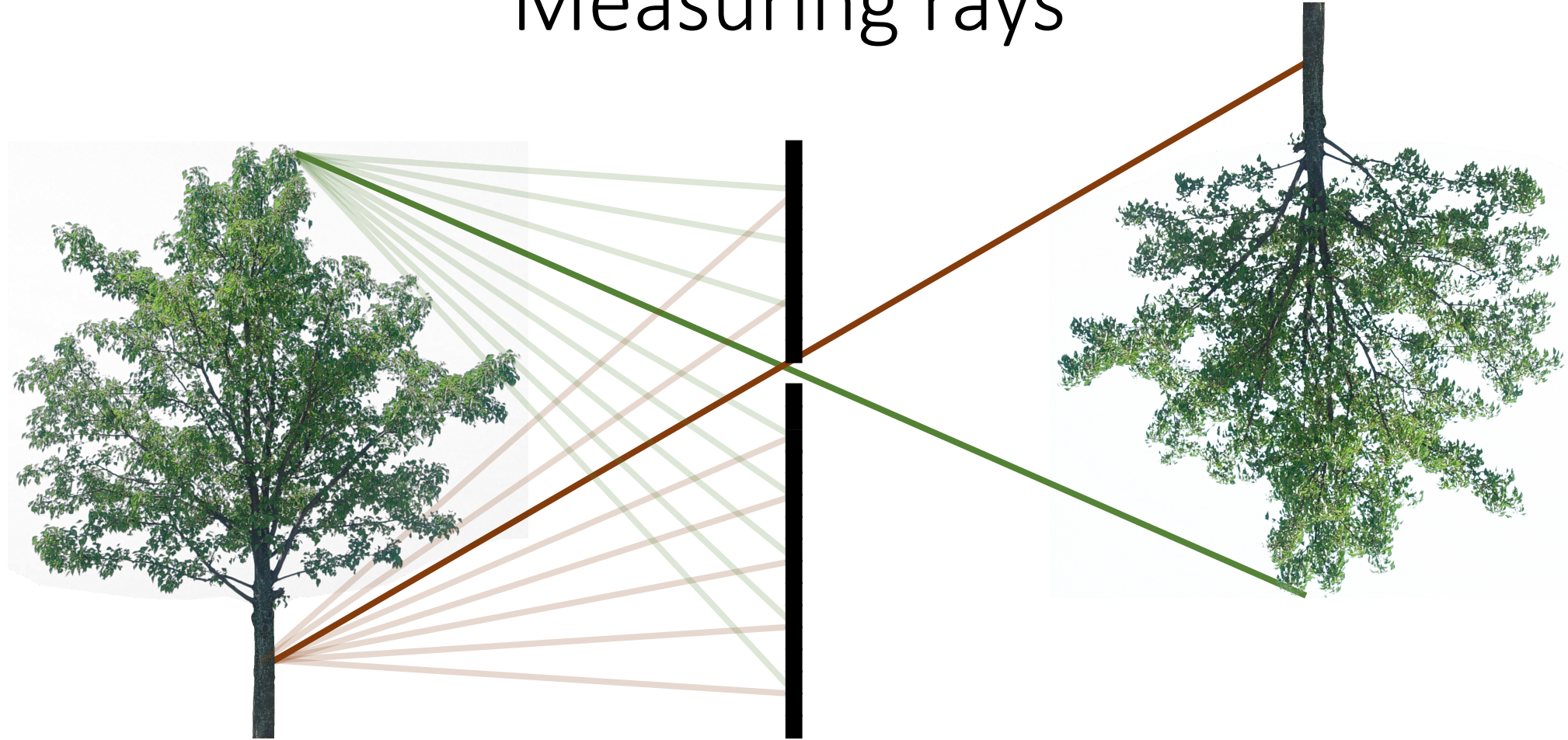


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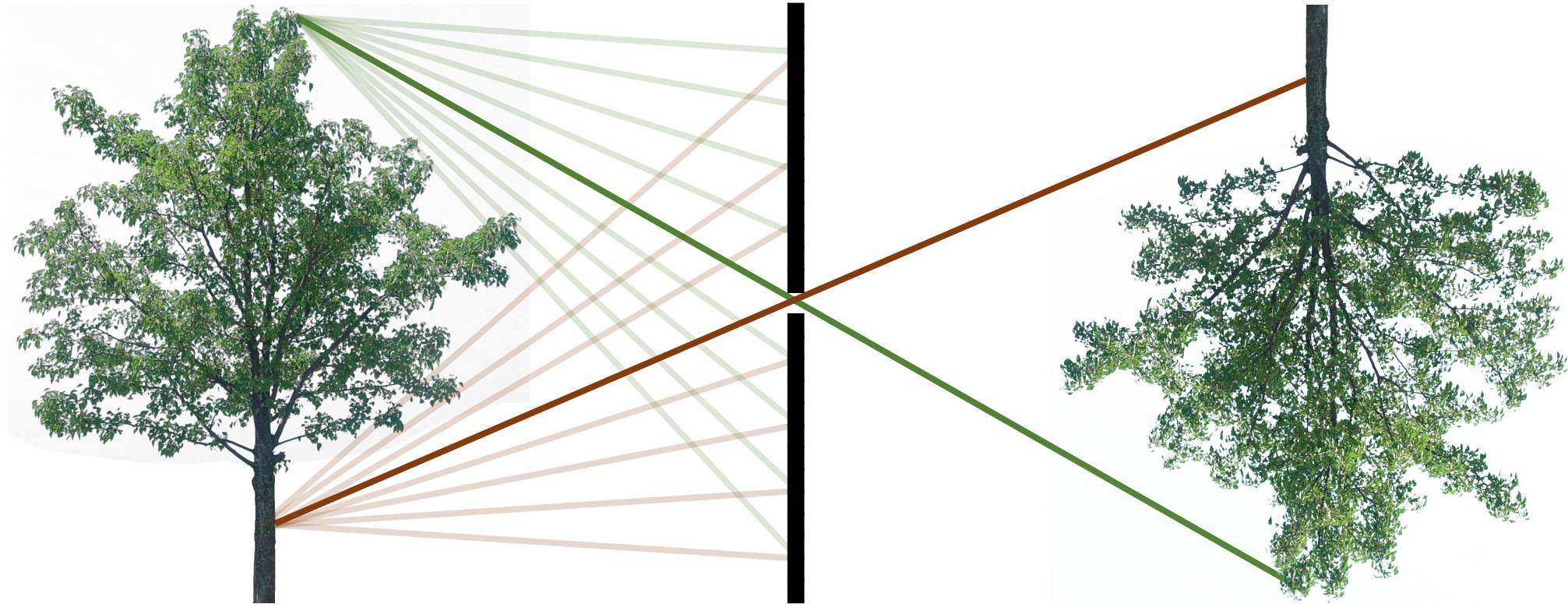
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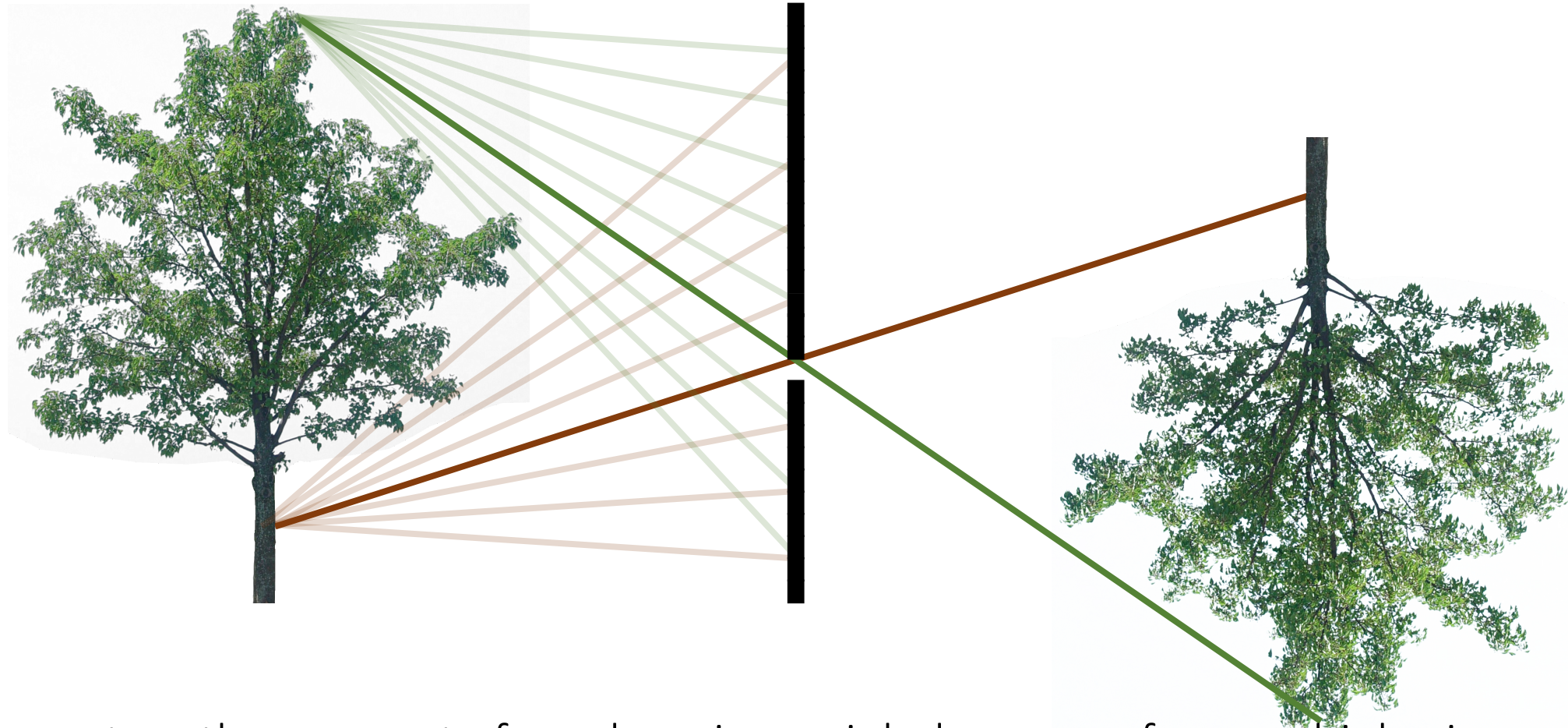
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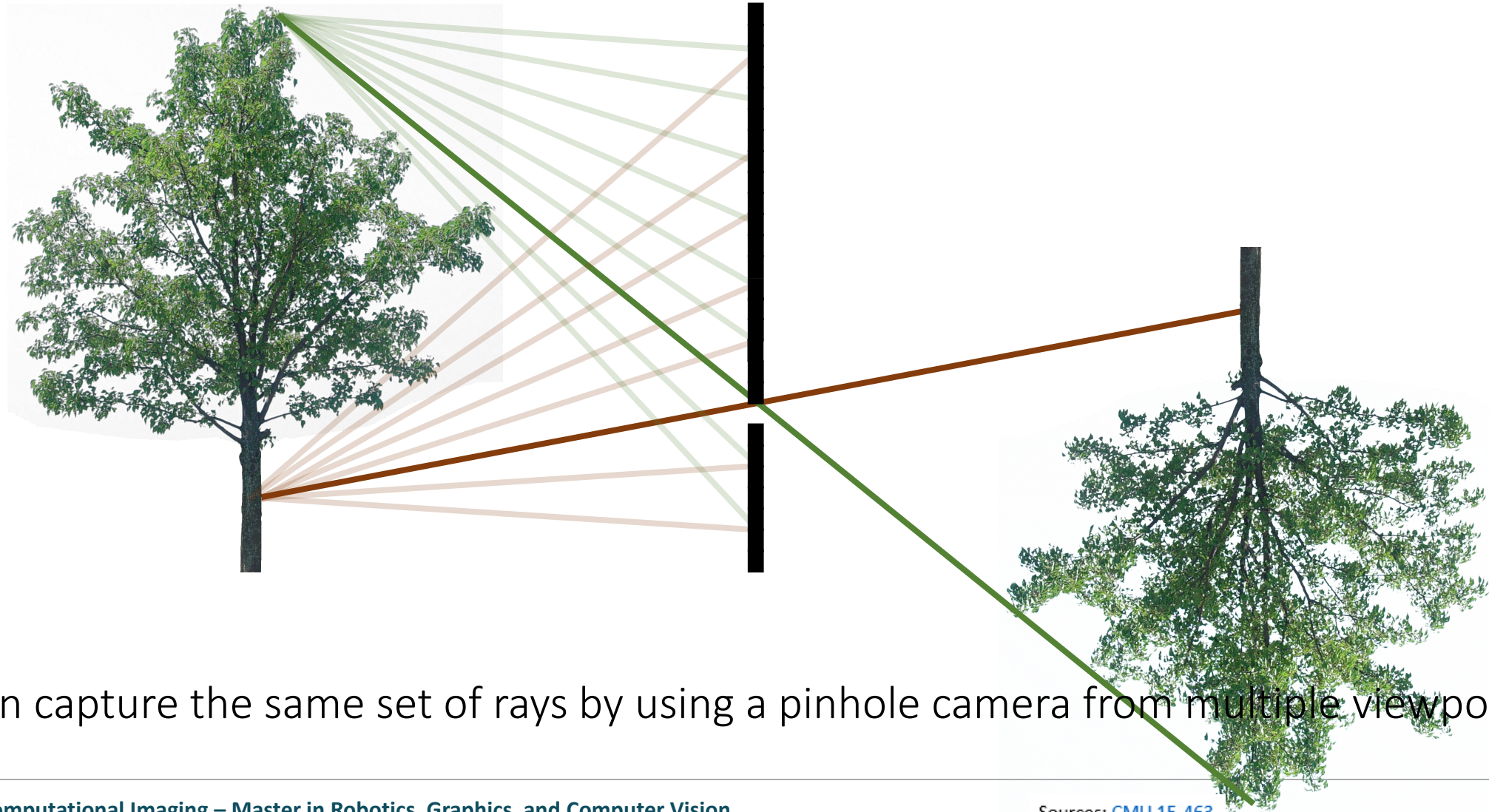


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We can capture the same set of rays by using a pinhole camera from multiple viewpoints

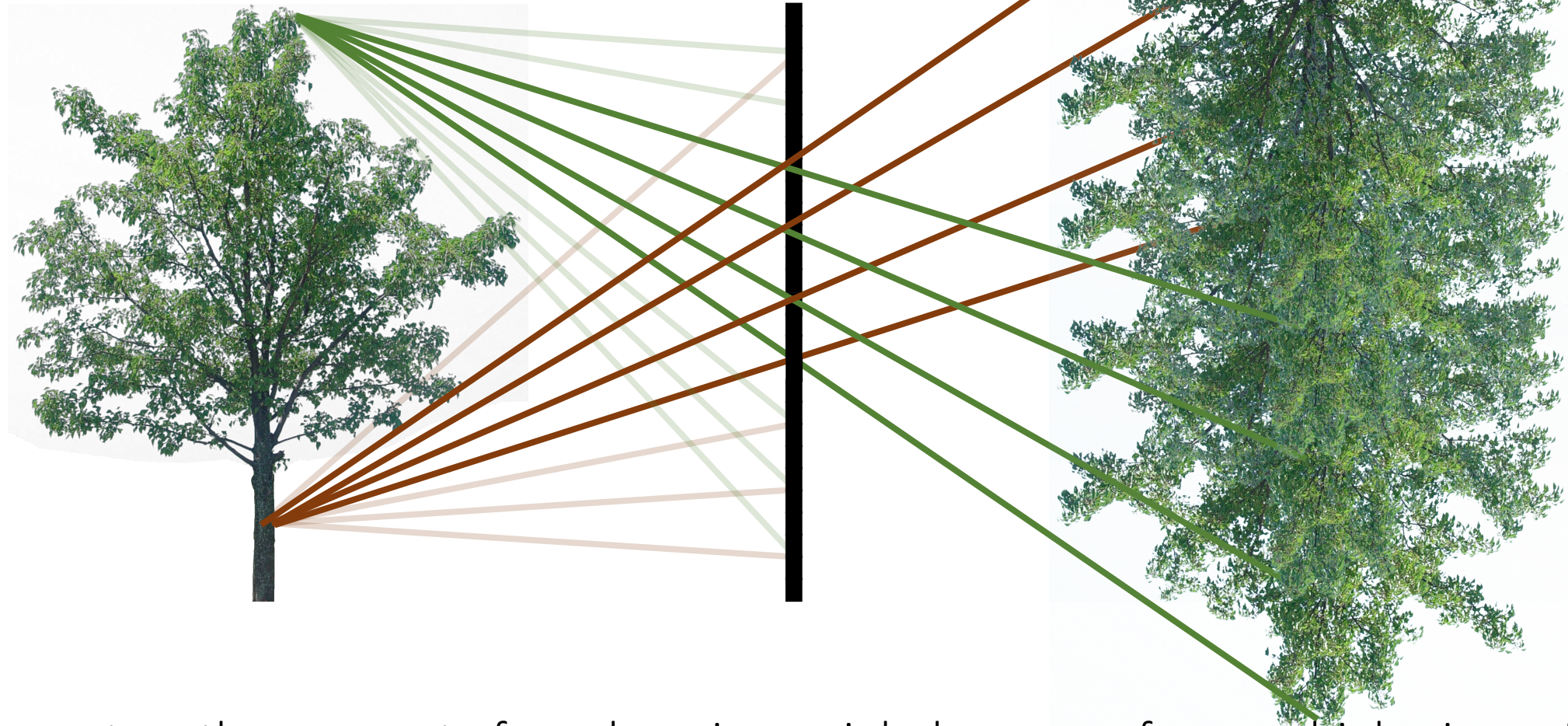
# Measuring rays



We can capture the same set of rays by using a pinhole camera from multiple viewpoints



# Measuring rays

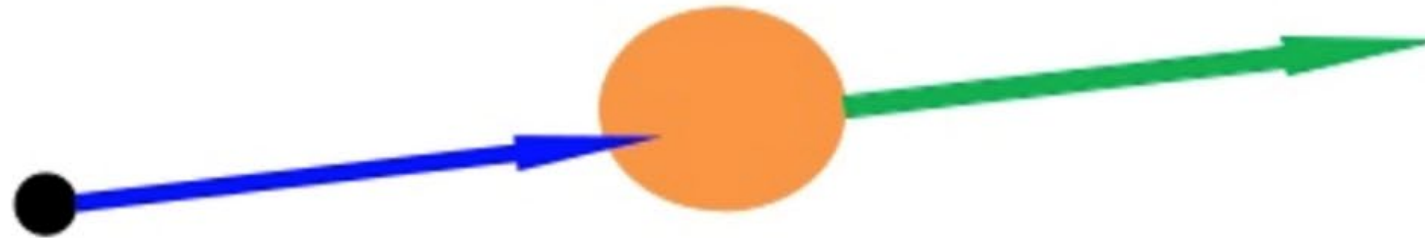


We can capture the same set of rays by using a pinhole camera from multiple viewpoints  
How can we merge these images into a lens-based, defocused image?

# Light Field Parameterization



Let's ignore color and time (i.e., these are attributes of rays)...



$$P(\theta, \phi, p_x, p_y, p_z)$$

The plenoptic function is 5D:

- 3D position
- 2D direction

Require 5D to represent attributes across occlusions

Consider a region free of occluding objects...



$$P(\theta, \phi, p_x, p_y, p_z)$$

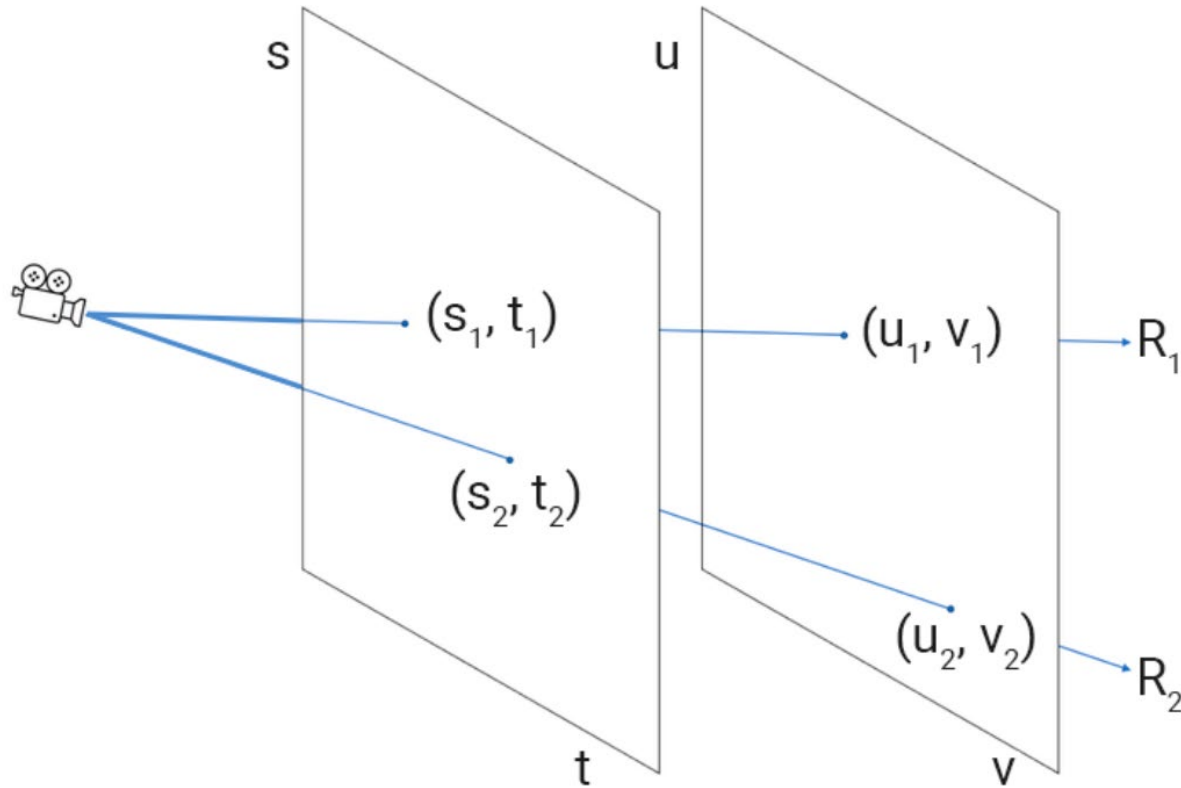
The function contains **redundant information**, because the radiance along a ray remains constant from point to point along its length.

This redundant information is one-dimensional, leaving us with a **four-dimensional function**

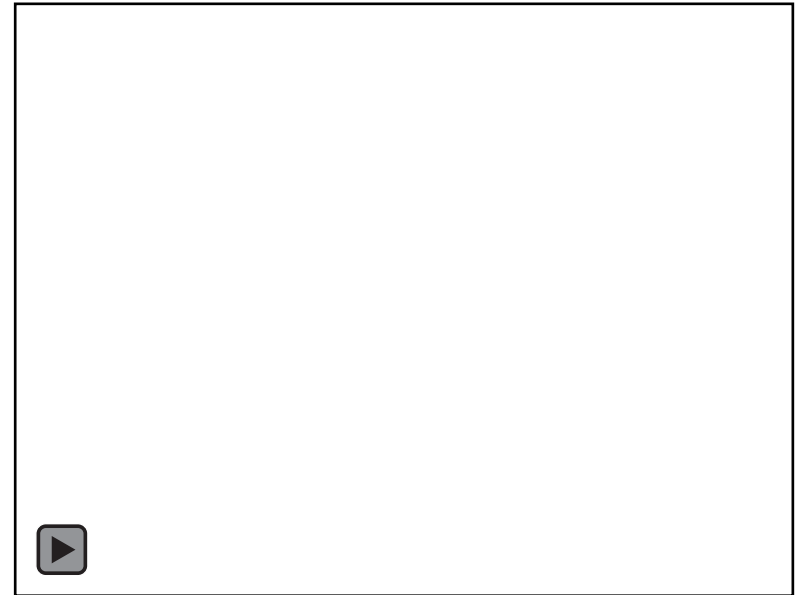


# 4D parameterization of light fields

- We can reparameterize the light field in 4D using just two planes:



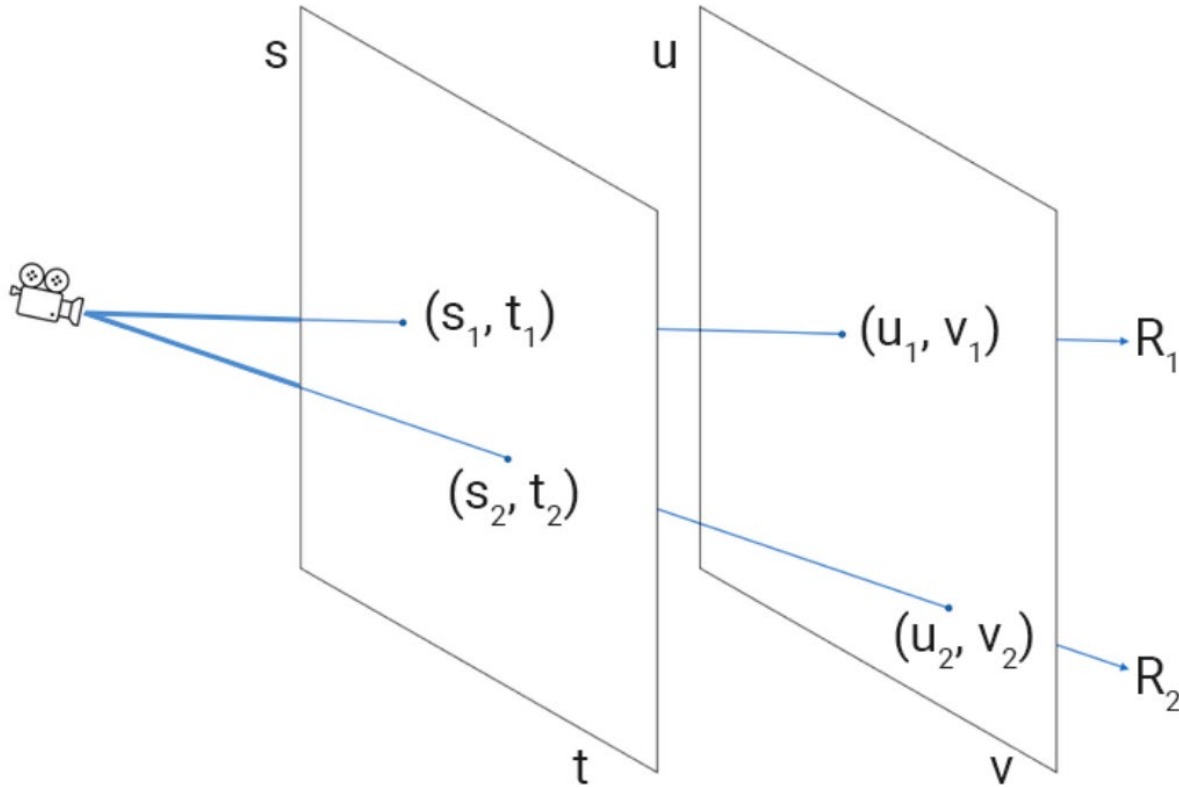
Light field: Representation of the amount of light, flowing in every direction, for every point in space.





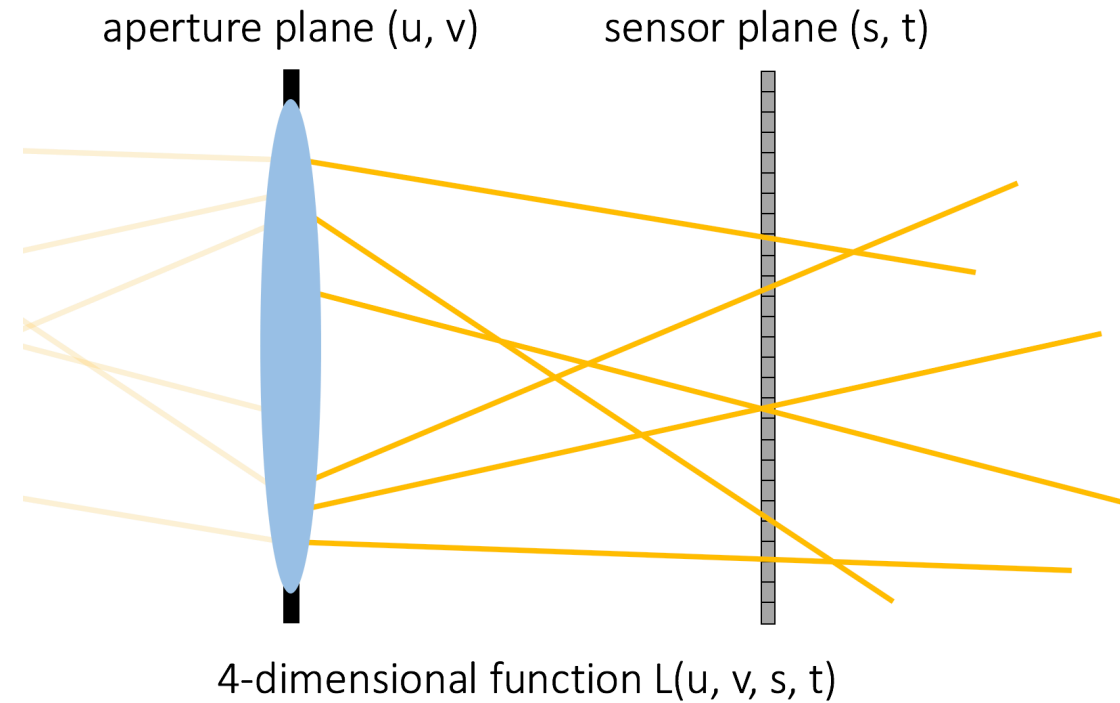
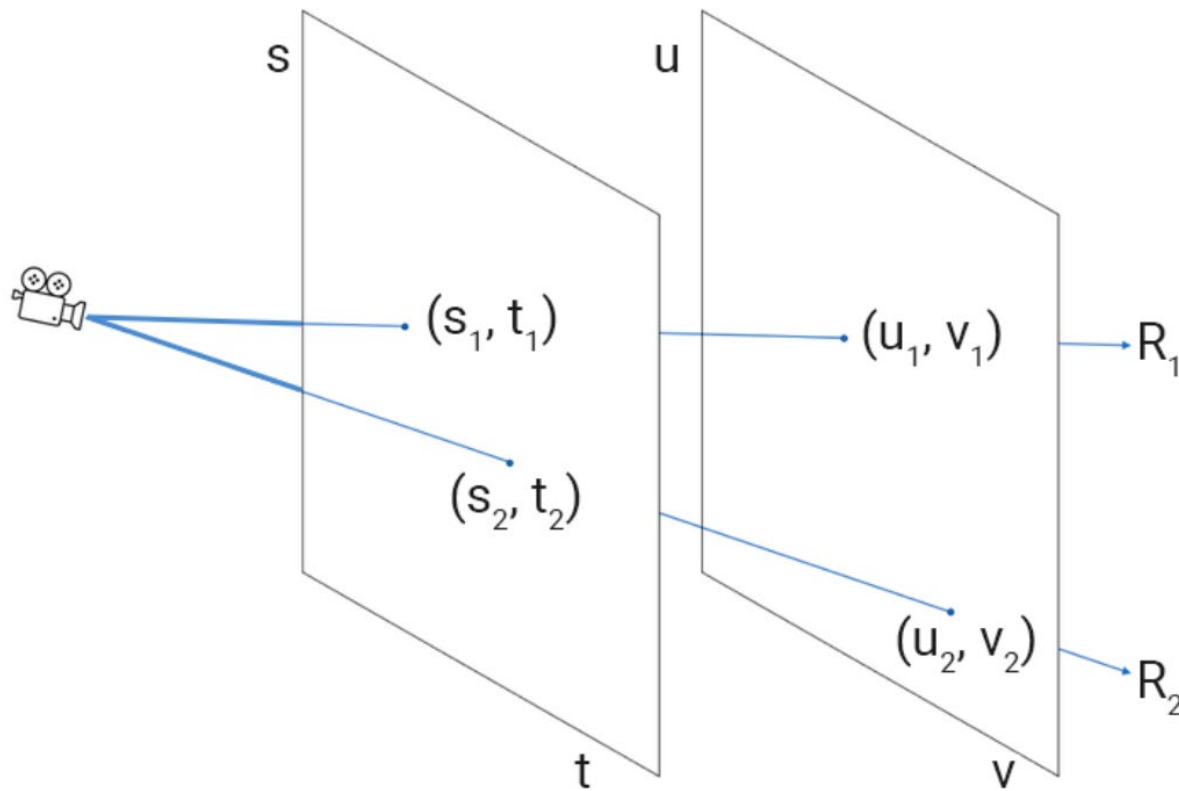
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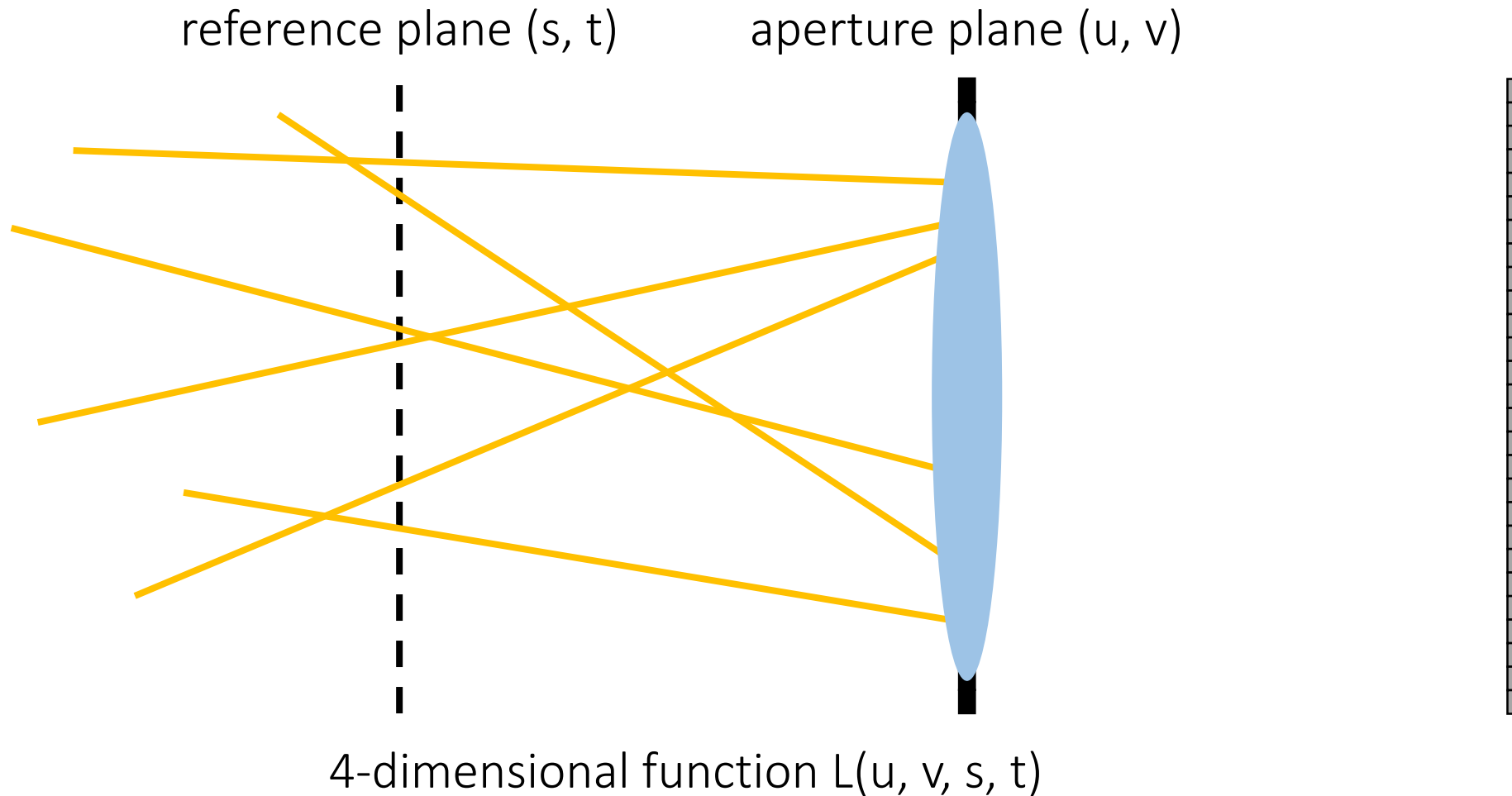


# 4D parameterization of light fields

- We can reparameterize the light field in 4D using just two planes:

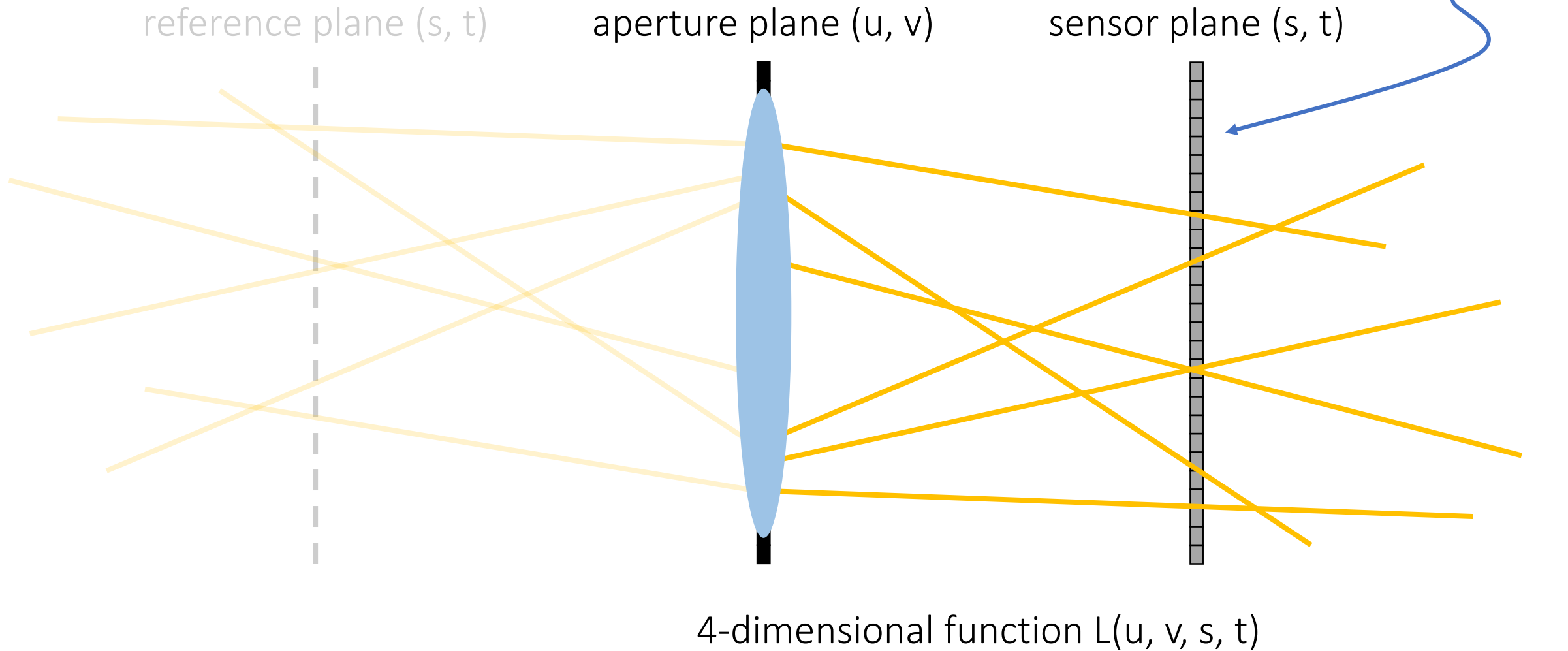


# 4D parameterization of light fields



Rays are parameterized based on their intersections with two planes.

# 4D parameterization of light fields

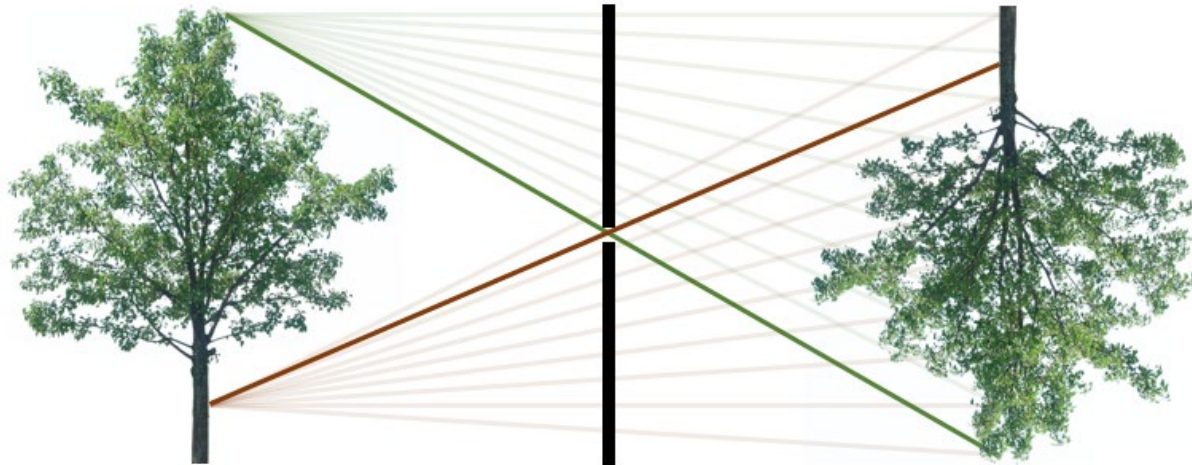


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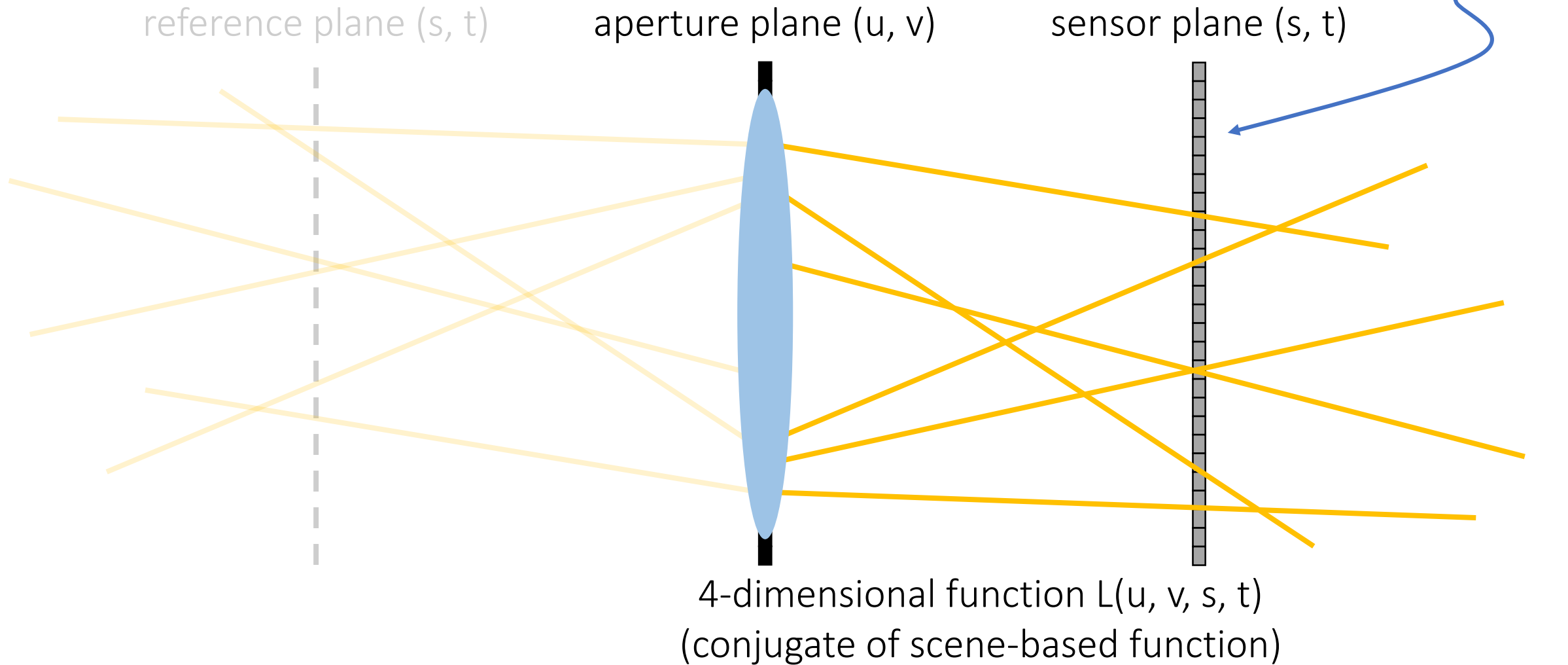


## \* Conjugate of a plane (in an optical imaging system)

- **Conjugate plane** of a given plane  $P$ : Plane  $P'$  such that points on  $P$  are imaged on  $P'$ .
- If an object is moved to the point occupied by its image, then the moved object's new image will appear at the point where the object originated → **The object and its image are interchangeable.**
- This comes from the principle of reversibility which states light rays will travel along the originating path if the light's direction is reversed (Helmholtz reciprocity).

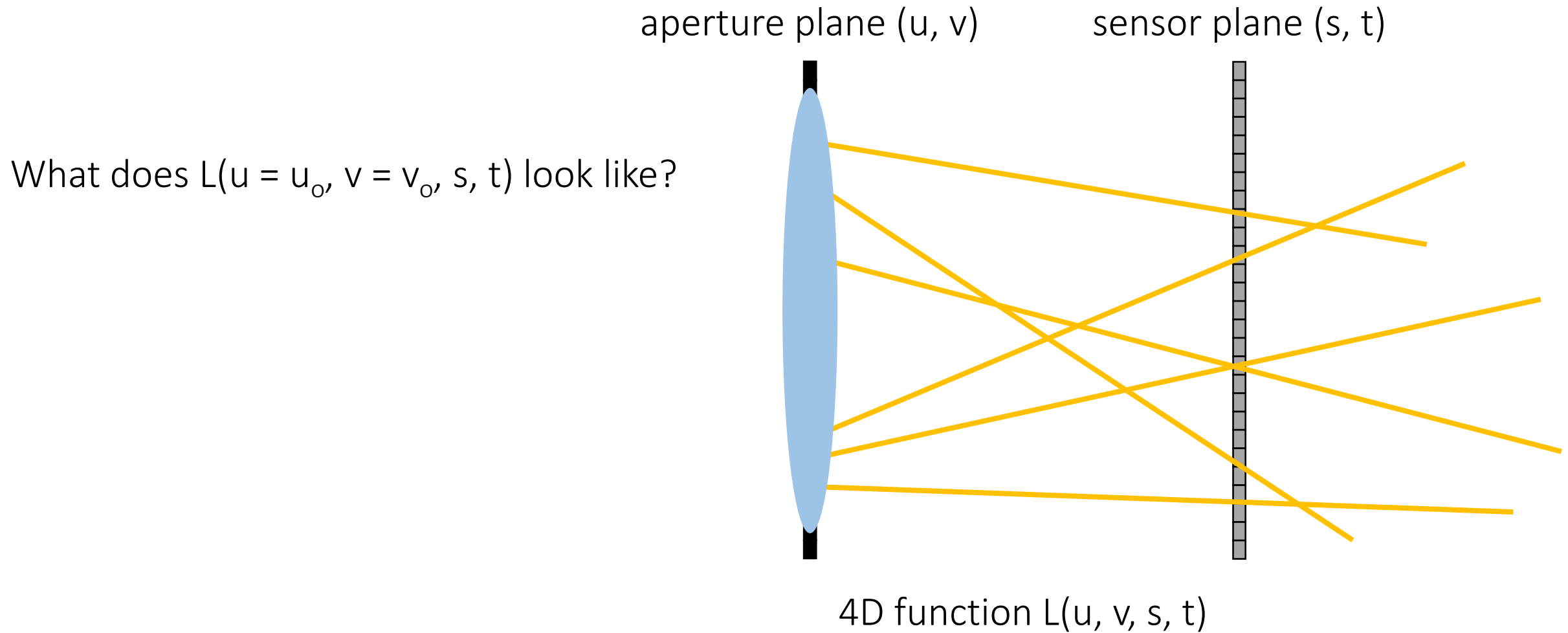


# 4D parameterization of light fields



Rays are parameterized based on their intersections with two planes.

# Understanding light fields



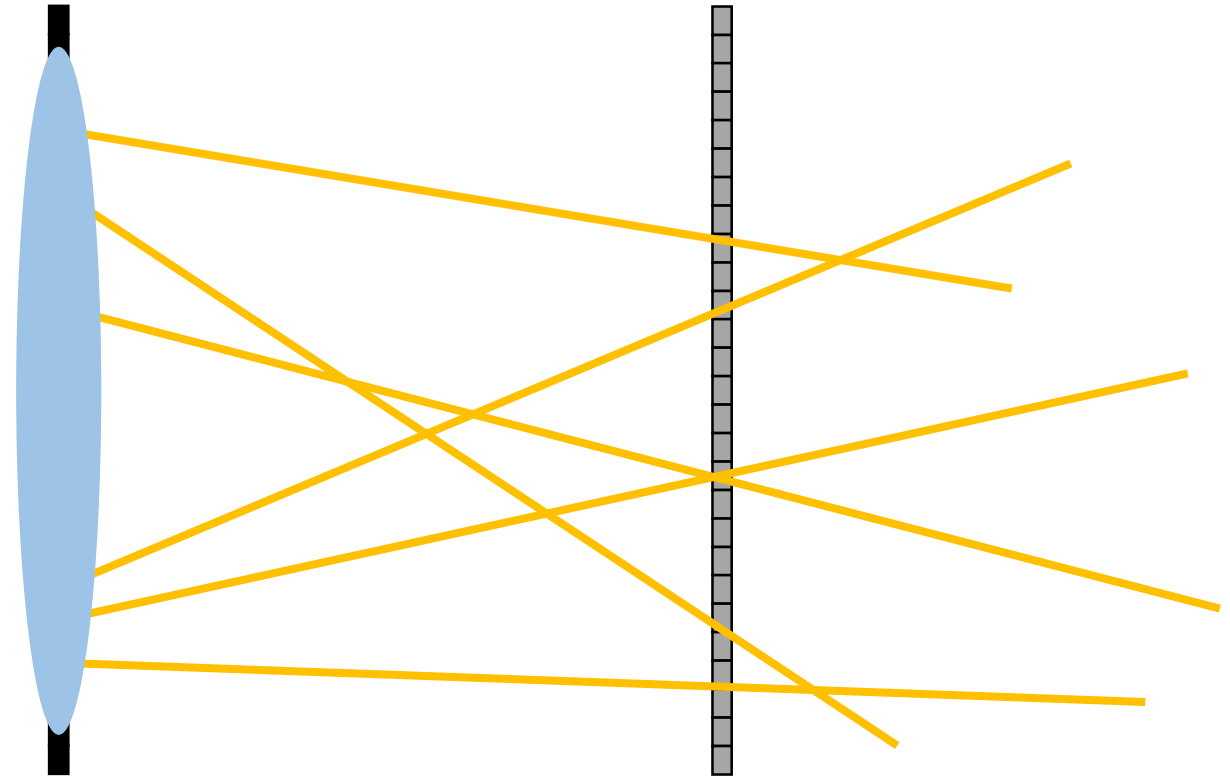
# Understanding light fields

What does  $L(u = u_o, v = v_o, s, t)$  look like?

- a pinhole image from a certain viewpoint

aperture plane (u, v)

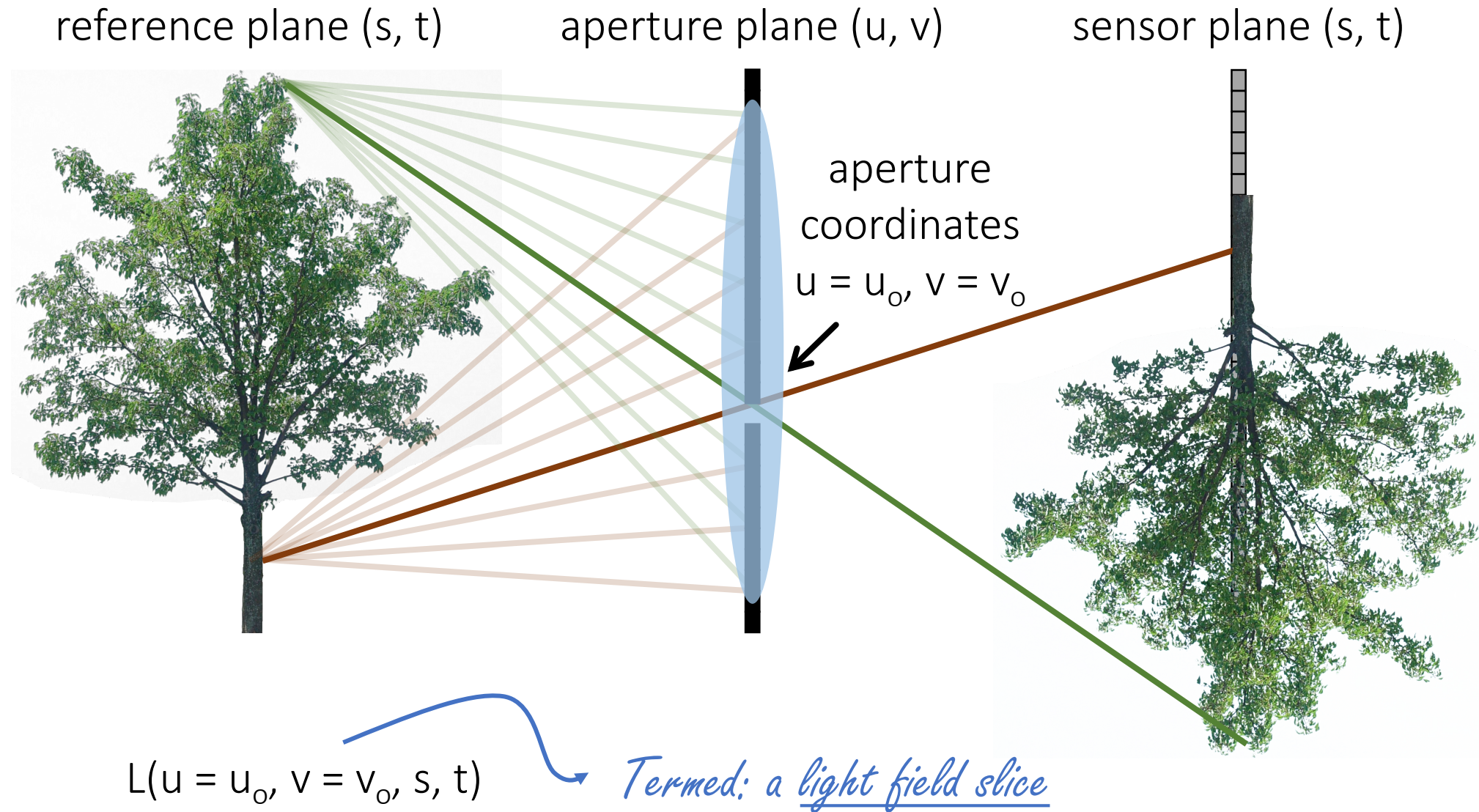
sensor plane (s, t)



4D function  $L(u, v, s, t)$



# Understanding light fields



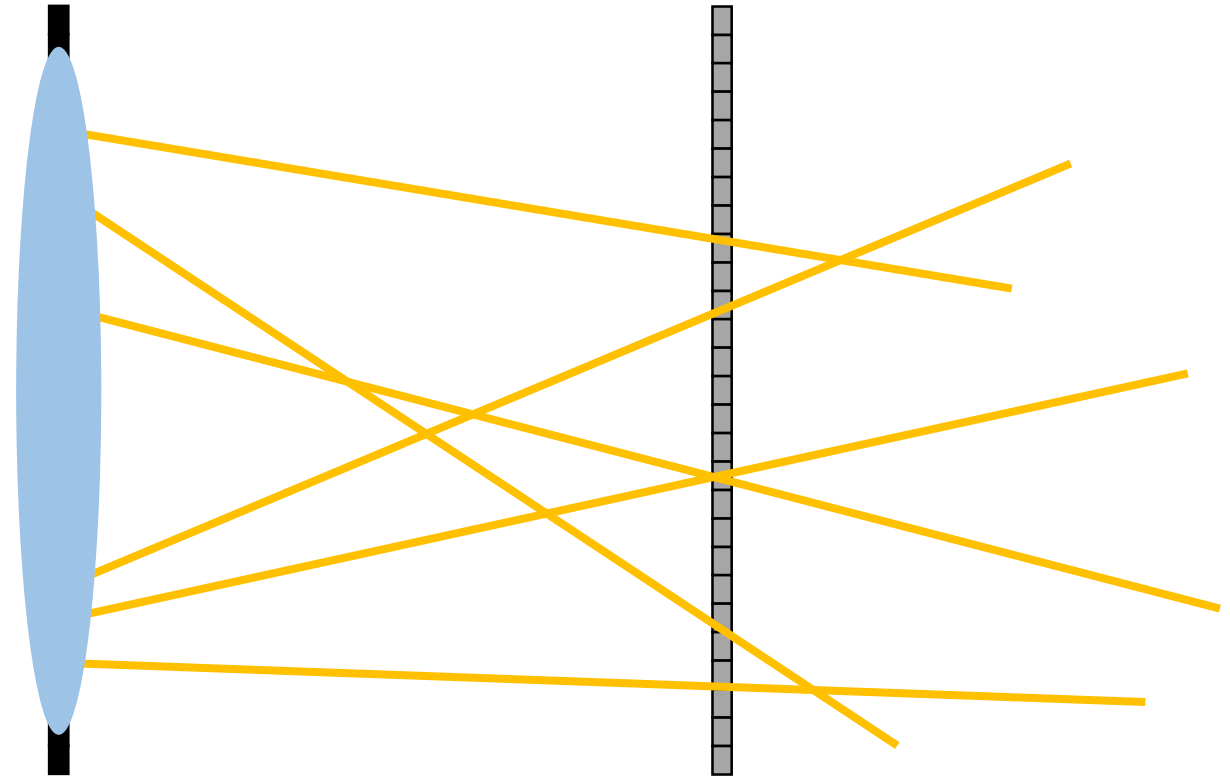
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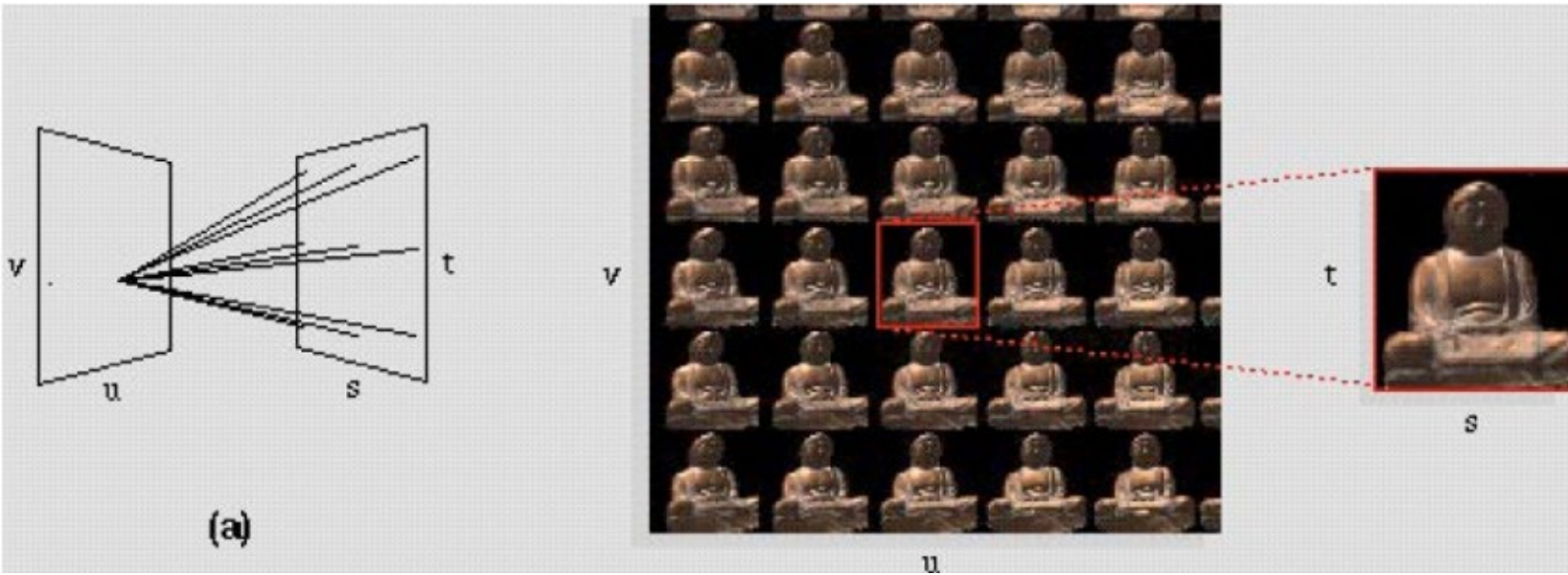
aperture plane (u, v)

sensor plane (s, t)



4D function  $L(u, v, s, t)$

# Understanding light fields



$L(u = u_o, v = v_o, s, t)$  is a pinhole image from a certain viewpoint

# Understanding light fields

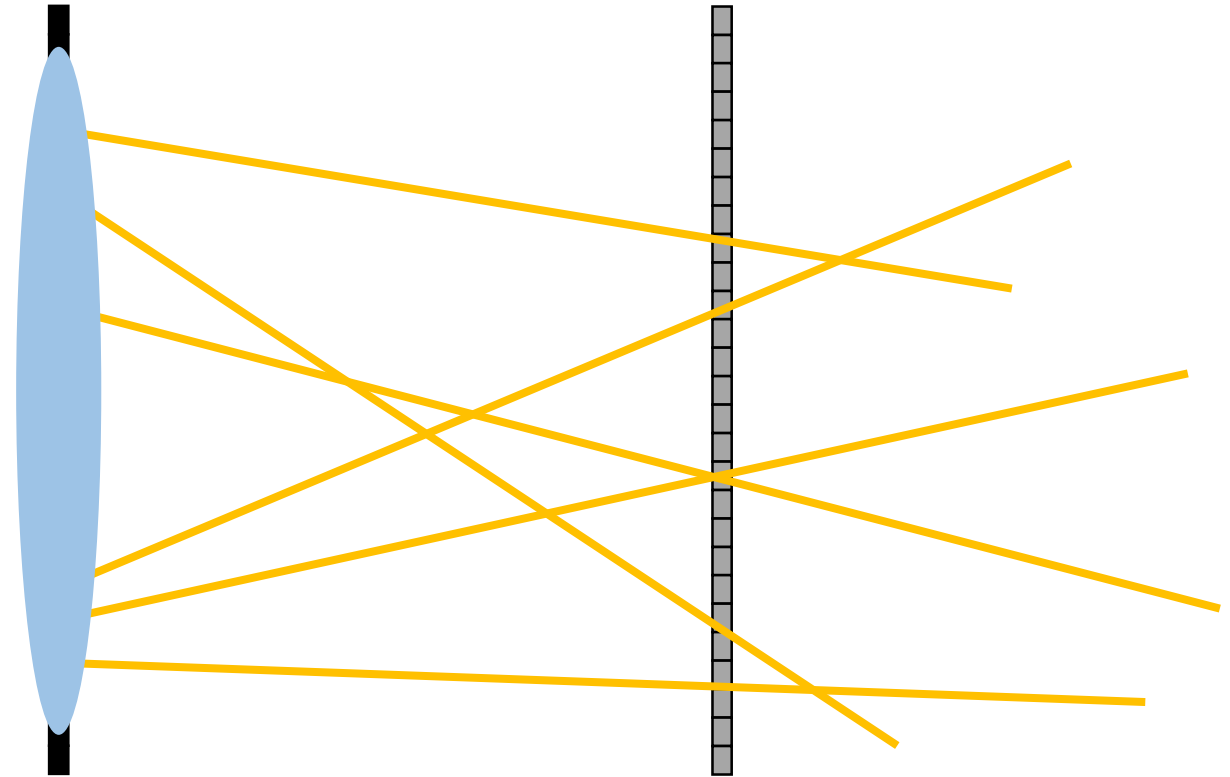
What does  $L(u = u_o, v = v_o, s, t)$  look like?

- a pinhole image from a certain viewpoint

What does  $L(u, v, s = s_o, t = t_o)$  look like?

aperture plane (u, v)

sensor plane (s, t)



4D function  $L(u, v, s, t)$



# Understanding light fields

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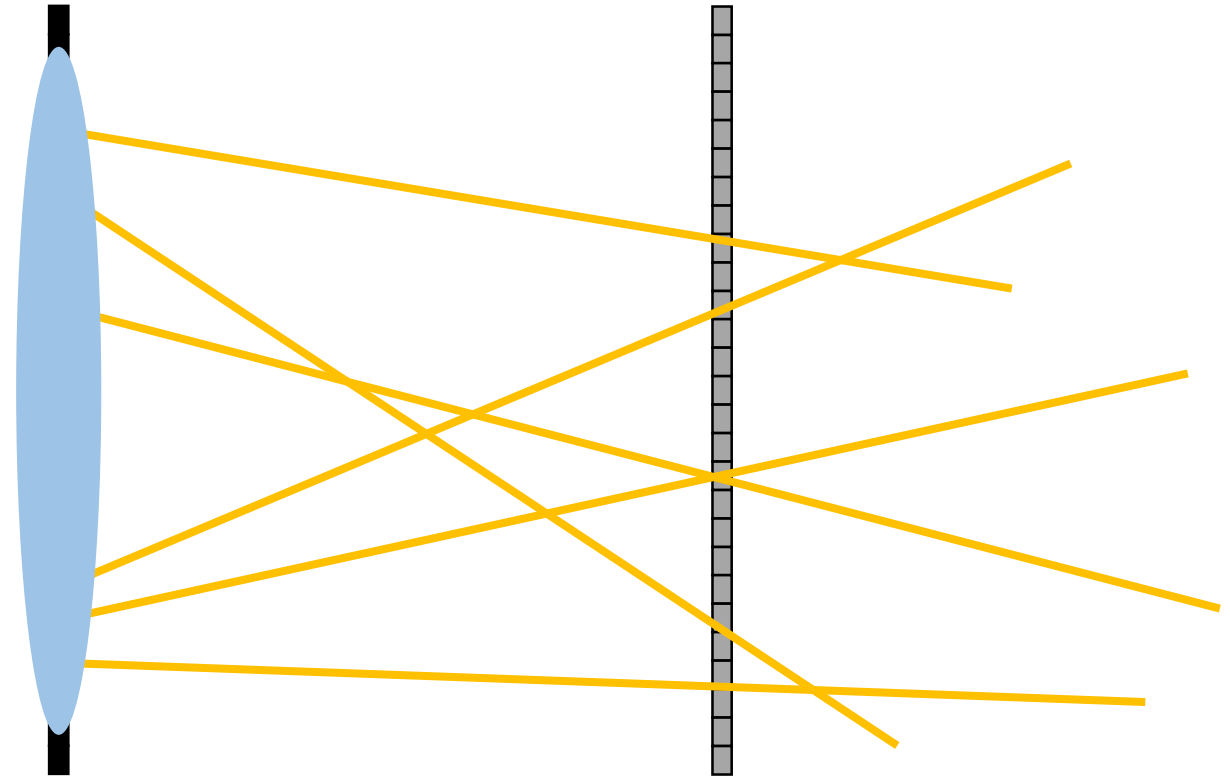
- a pinhole image from a certain viewpoint

What does  $L(u, v, s = s_o, t = t_o)$  look like?

- radiance emitted by a certain (in-focus) point at various directions

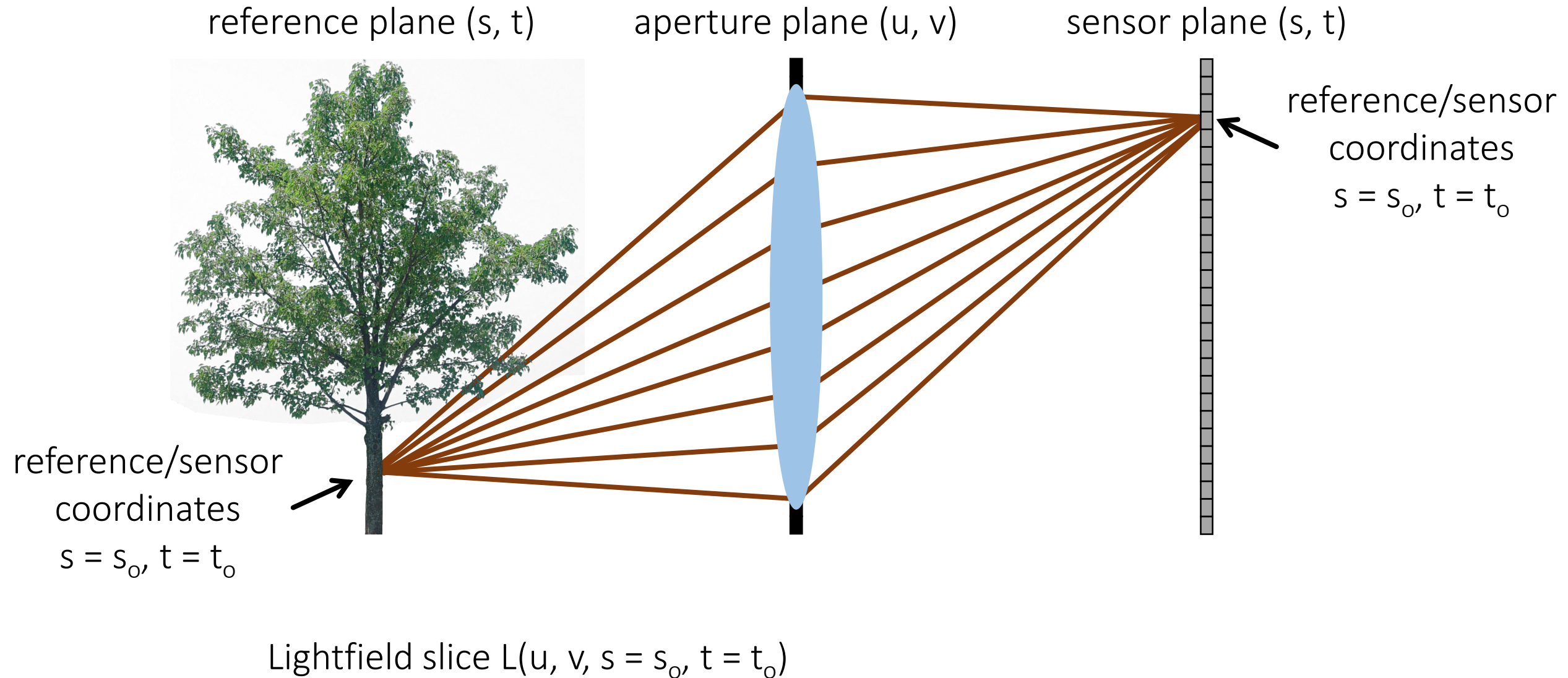
aperture plane (u, v)

sensor plane (s, t)



4D function  $L(u, v, s, t)$

# Understanding light fields



# Understanding light fields

What does  $L(u = u_o, v = v_o, s, t)$  look like?

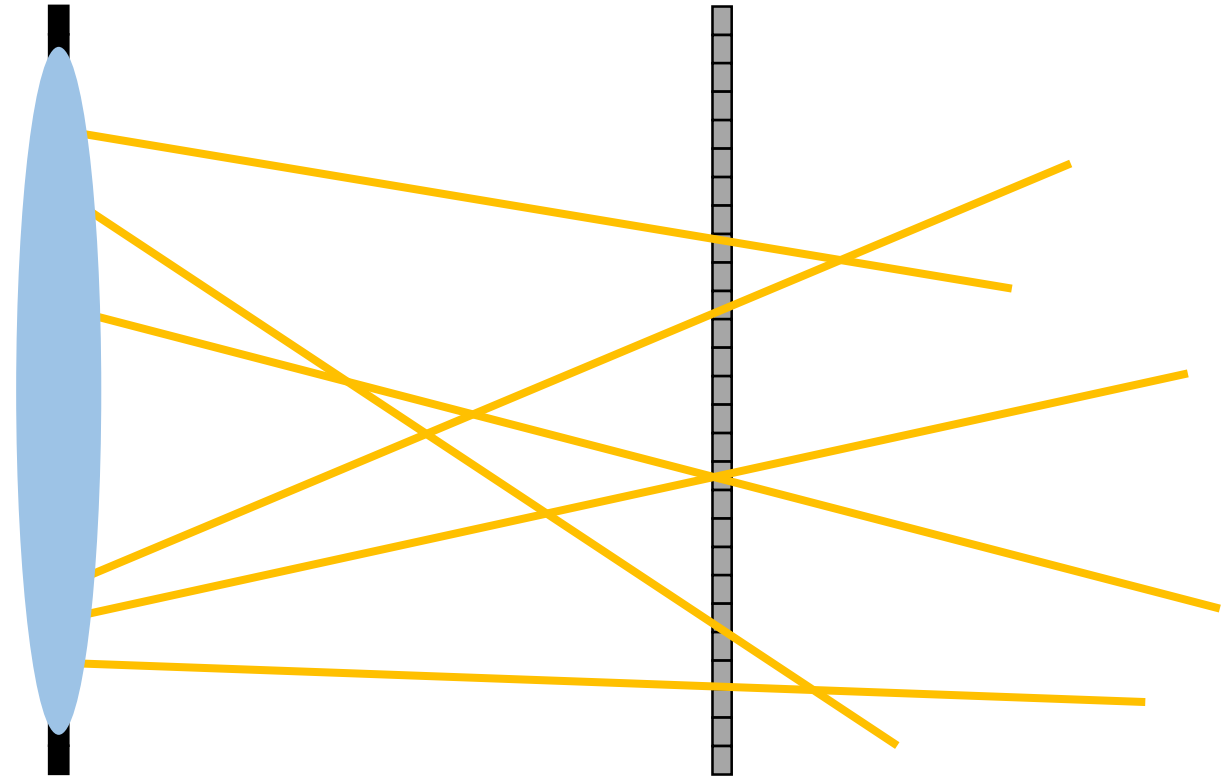
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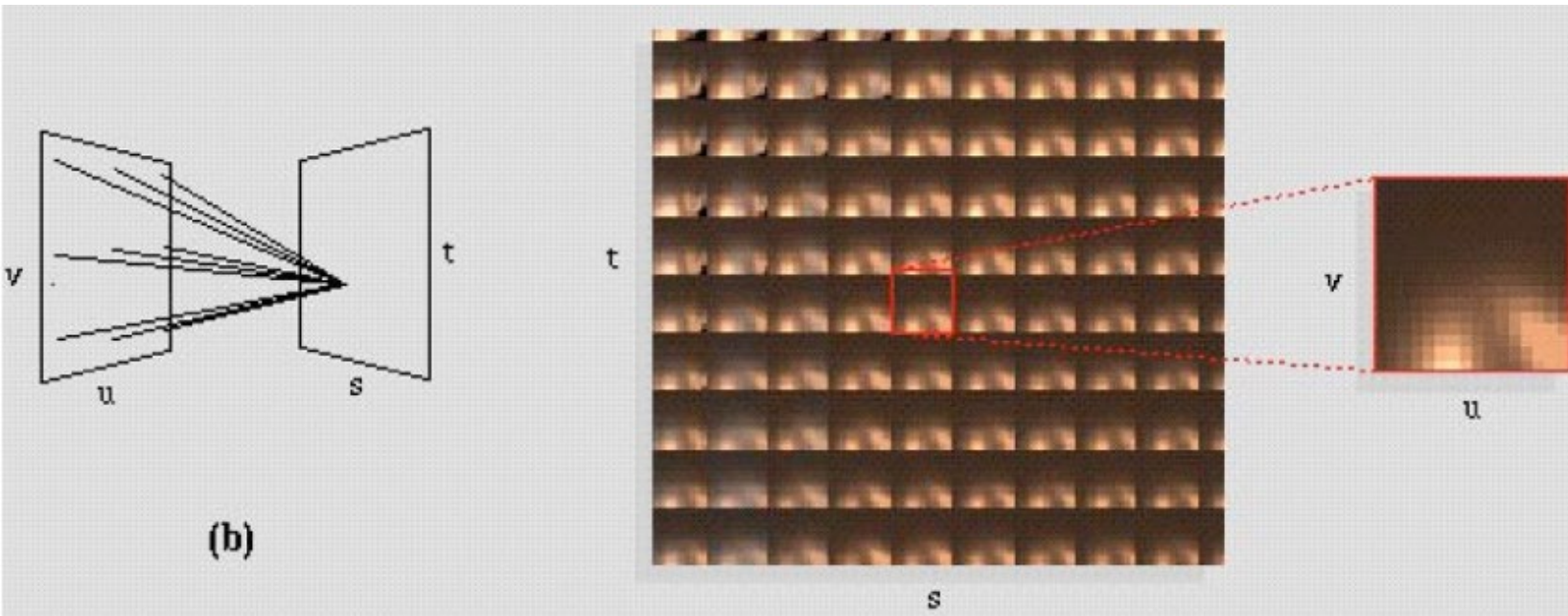
aperture plane (u, v)

sensor plane (s, t)



4D function  $L(u, v, s, t)$

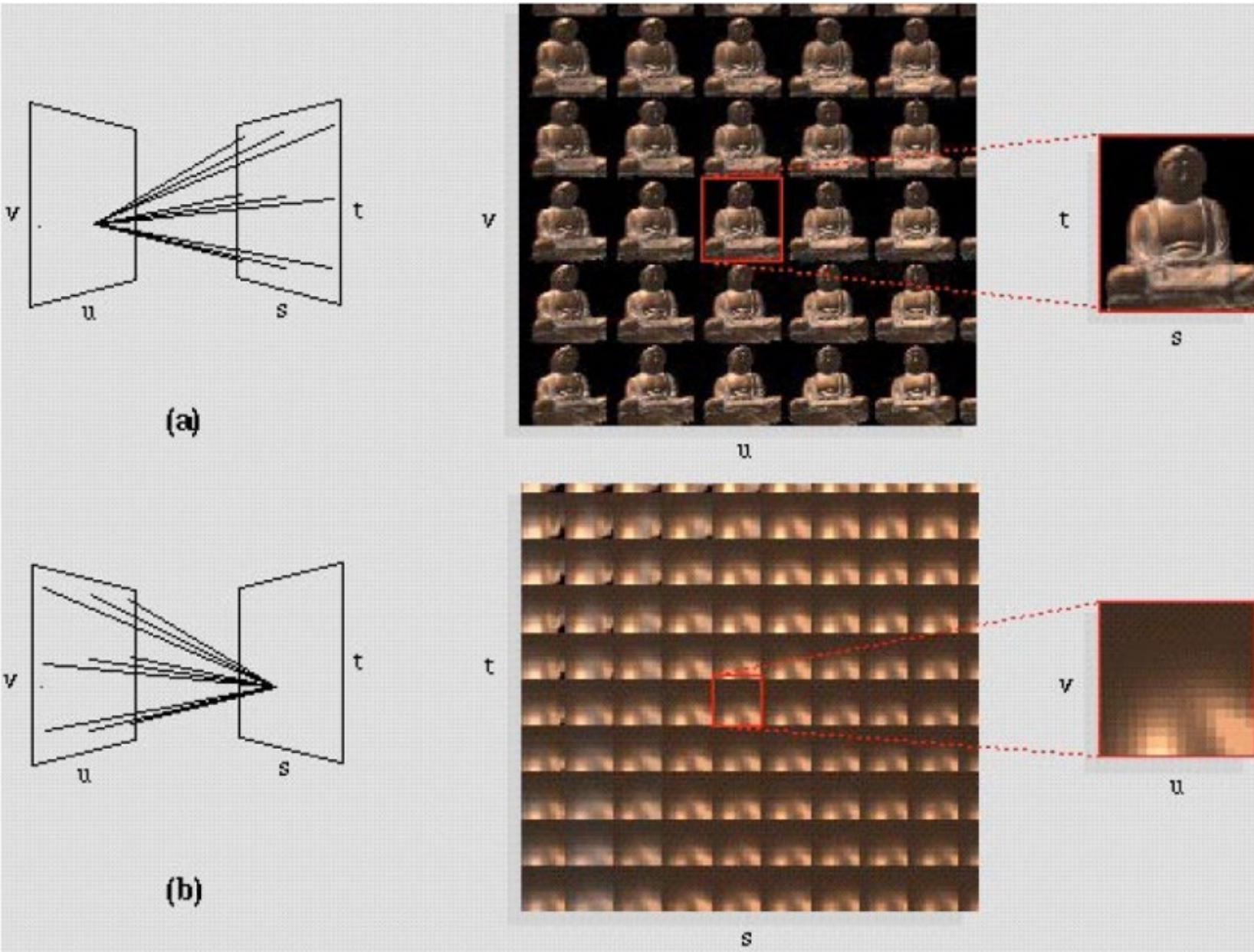
# Understanding light fields



$L(u, v, s = s_o, t = t_o)$  is the radiance emitted by a certain (in-focus) point at various directions



# Understanding light fields: 2 ways of visualizing them



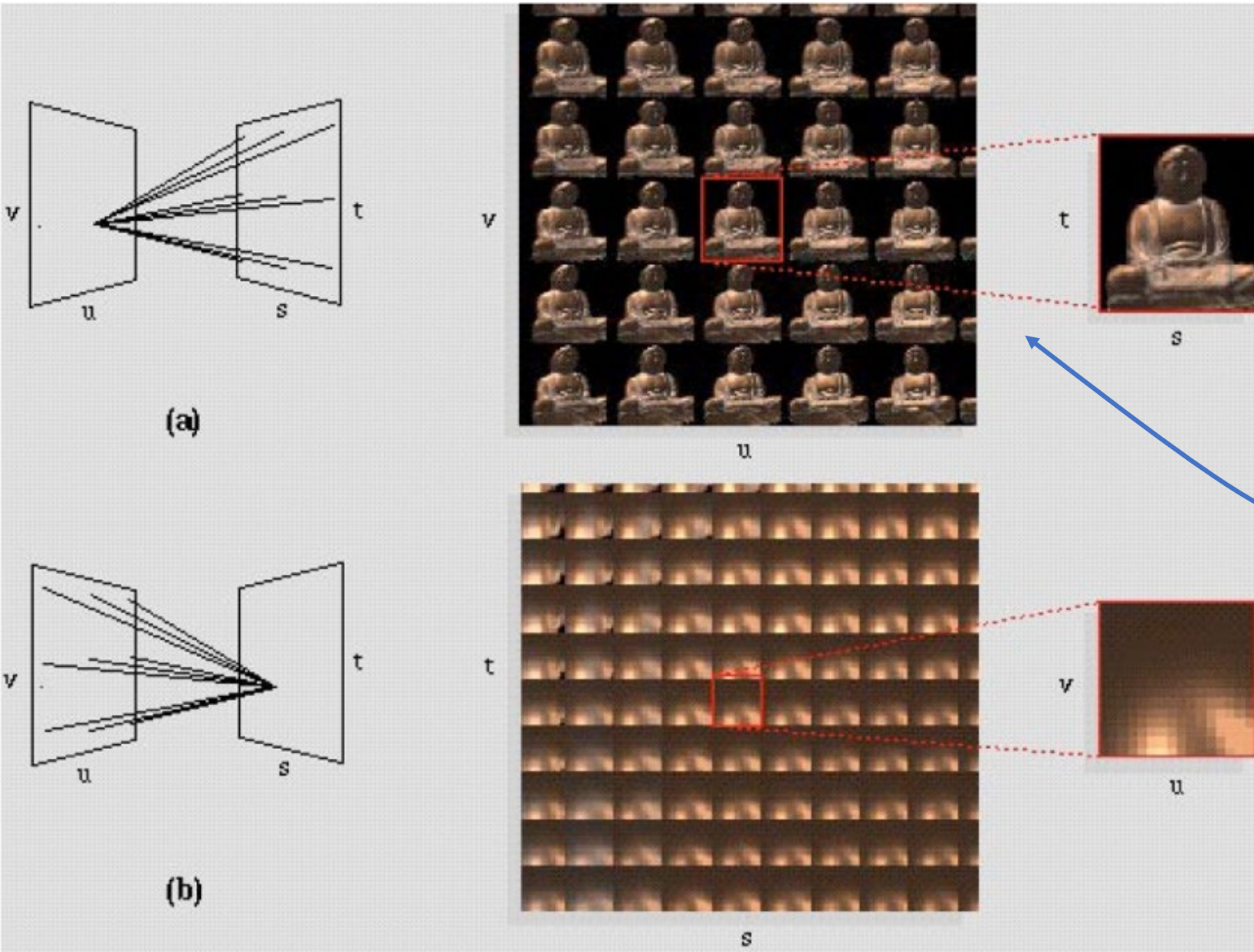
$L(u = u_o, v = v_o, s, t)$  is a pinhole image from a certain viewpoint

$L(u, v, s = s_o, t = t_o)$  is the radiance emitted by a certain (in-focus) point at various directions

Demo:

<http://lightfield.stanford.edu/lfs.html>

# Understanding light fields: 2 ways of visualizing them



$L(u = u_o, v = v_o, s, t)$  is a pinhole image from a certain viewpoint

*In a practical and straightforward way: A set of images from different viewpoints*

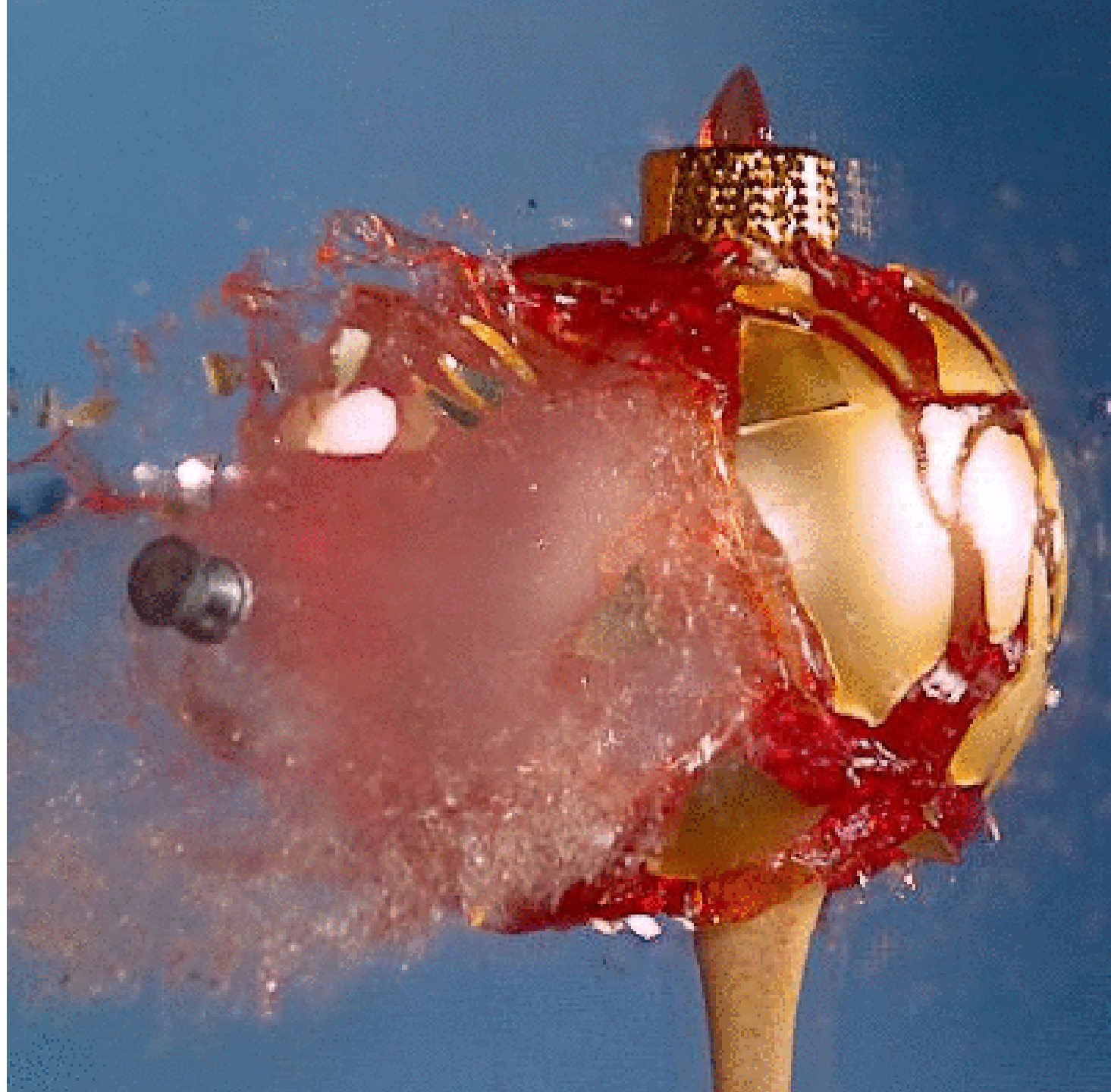
$L(u, v, s = s_o, t = t_o)$  is the radiance emitted by a certain (in-focus) point at various directions

Demo:

<http://lightfield.stanford.edu/lfs.html>



A '*new kind*'  
of pictures

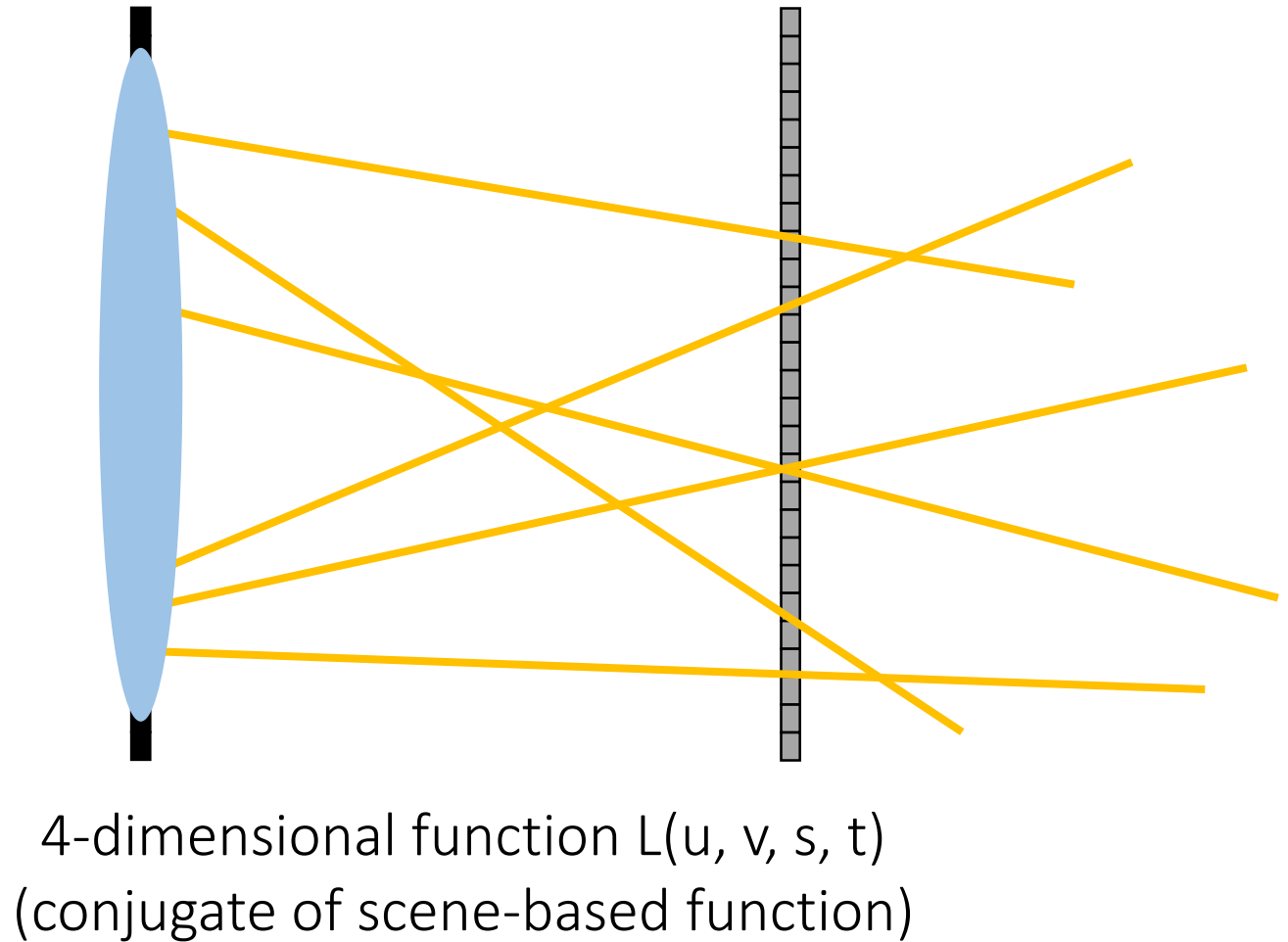


# 4D parameterization of light fields

*How can one capture the  
lightfield of a scene?*

aperture plane ( $u, v$ )

sensor plane ( $s, t$ )



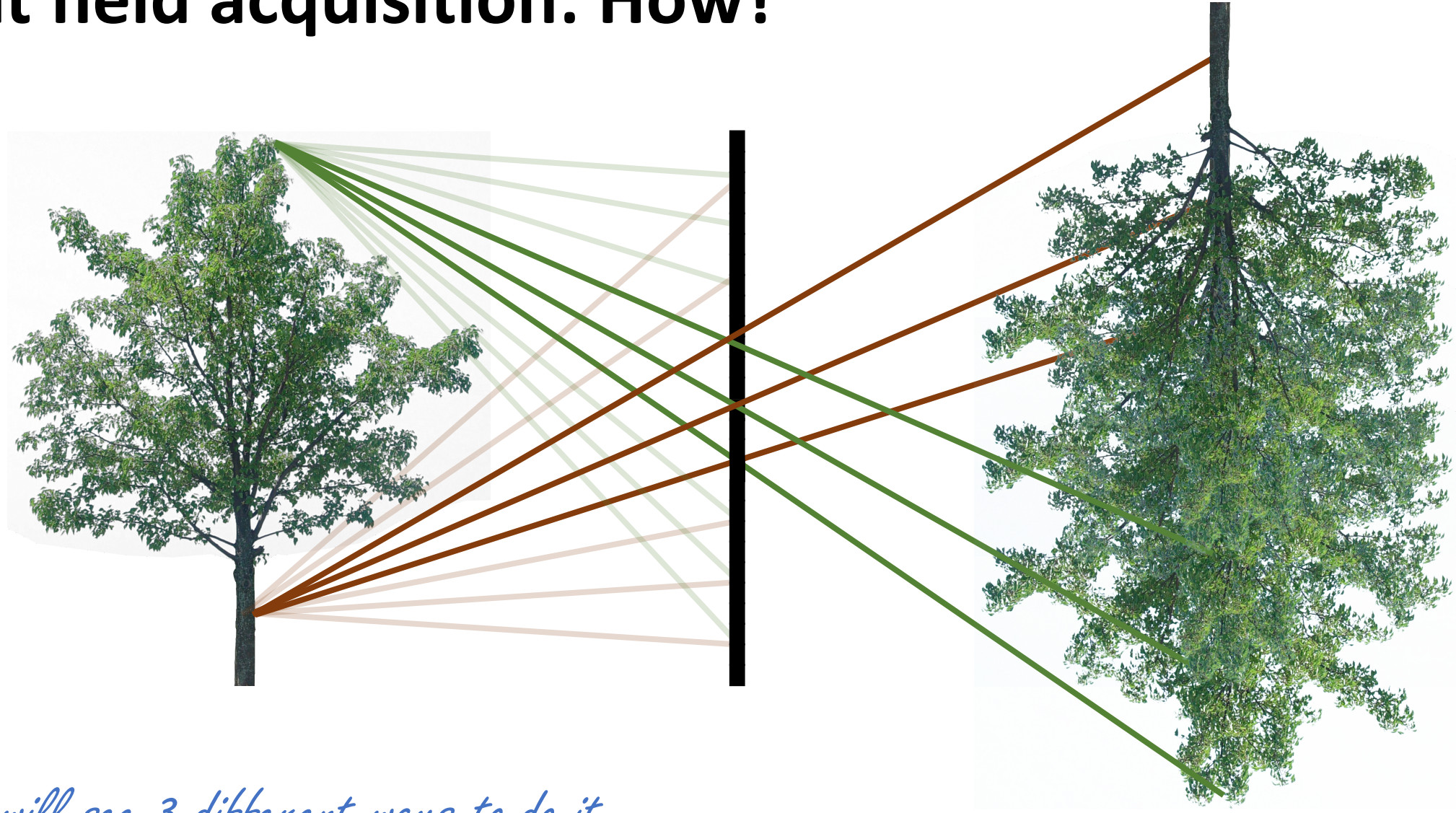
# Light Field Imaging

- ✓ • Concept
- ✓ • Parameterization
  - Acquisition
  - Rendering



# Light Field Acquisition

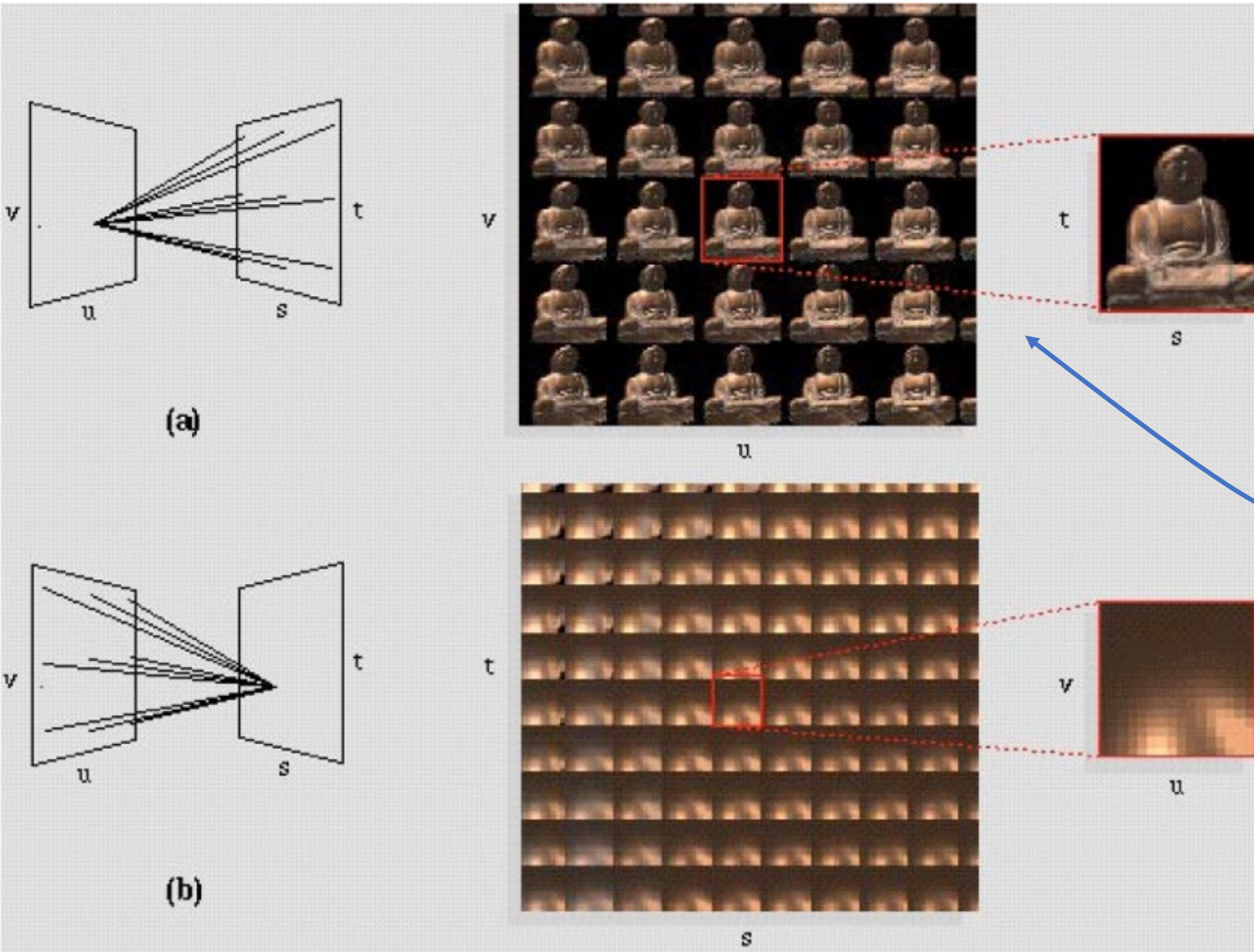
# Light field acquisition: How?



*We will see 3 different ways to do it.*



# Understanding light fields: 2 ways of visualizing them



$L(u = u_o, v = v_o, s, t)$  is a pinhole image from a certain viewpoint

*In a practical and straightforward way: A set of images from different viewpoints*

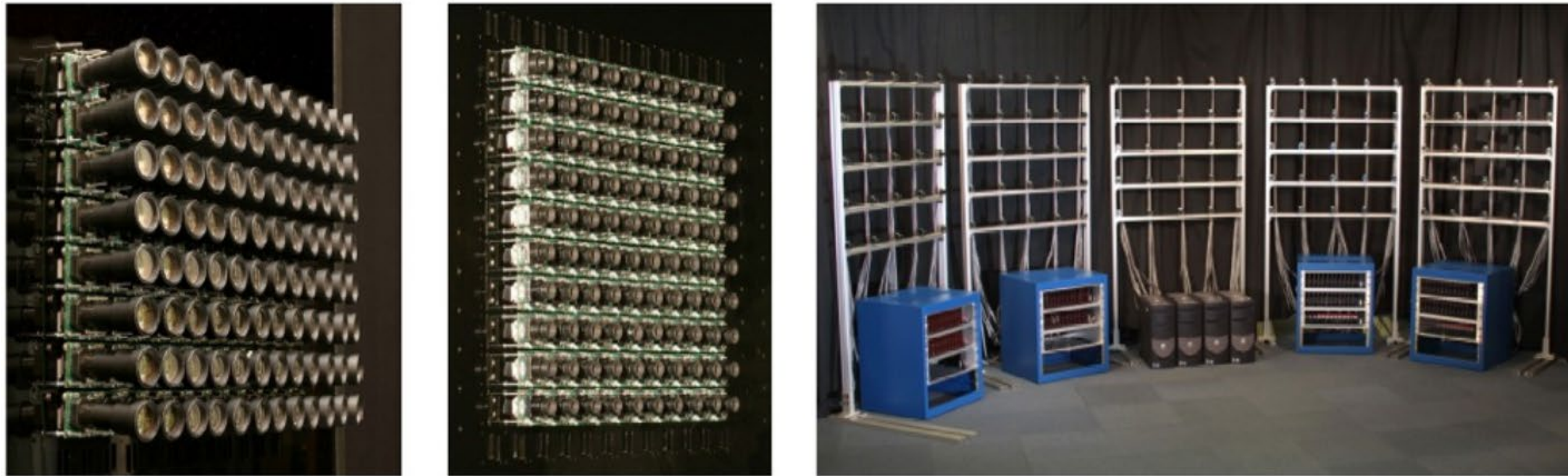
$L(u, v, s = s_o, t = t_o)$  is the radiance emitted by a certain (in-focus) point at various directions

Demo:

<http://lightfield.stanford.edu/lfs.html>

# Option 1: use multiple cameras

Stanford camera array

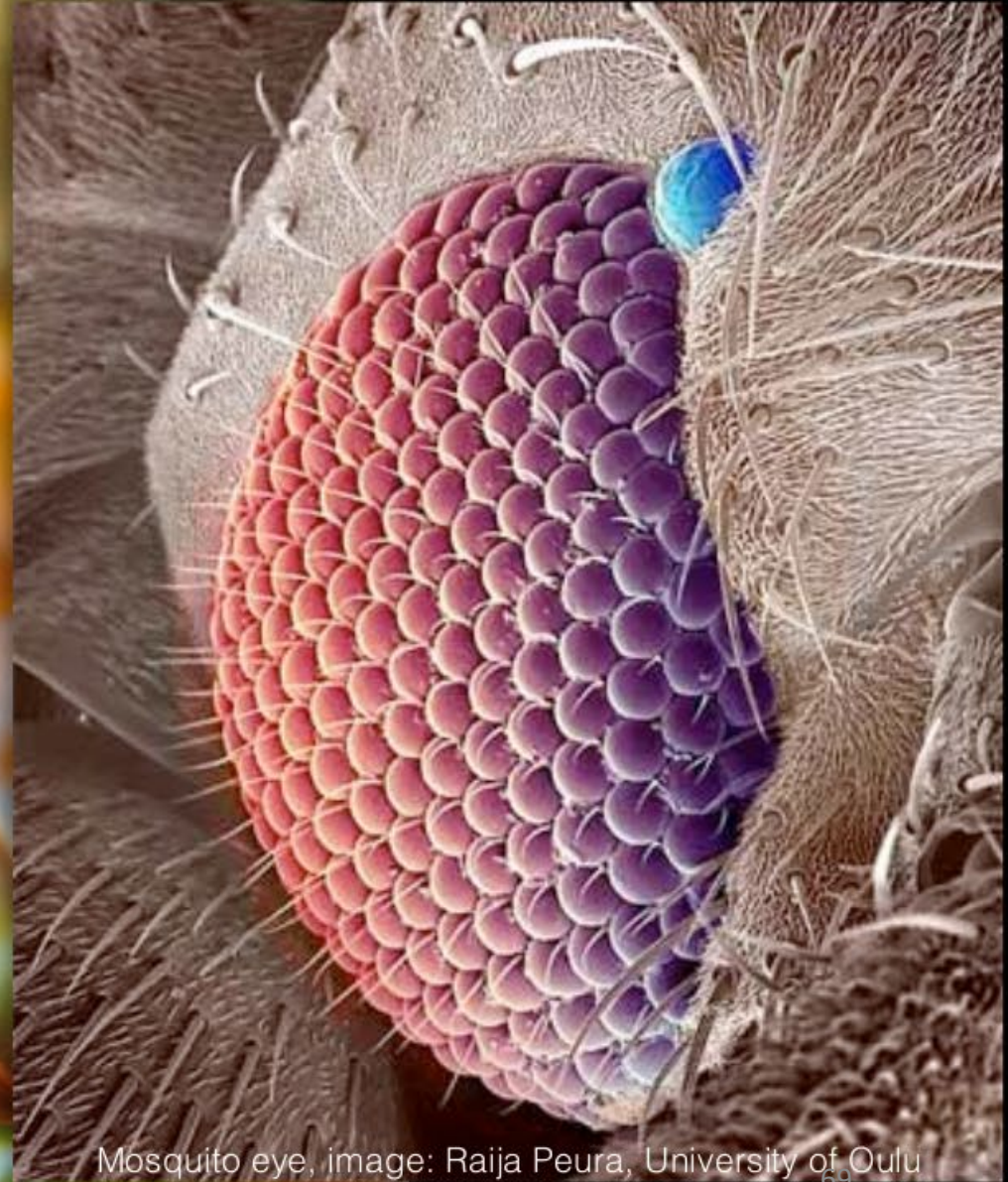




# Compound Eye vs Light Field



Image: National Geographic

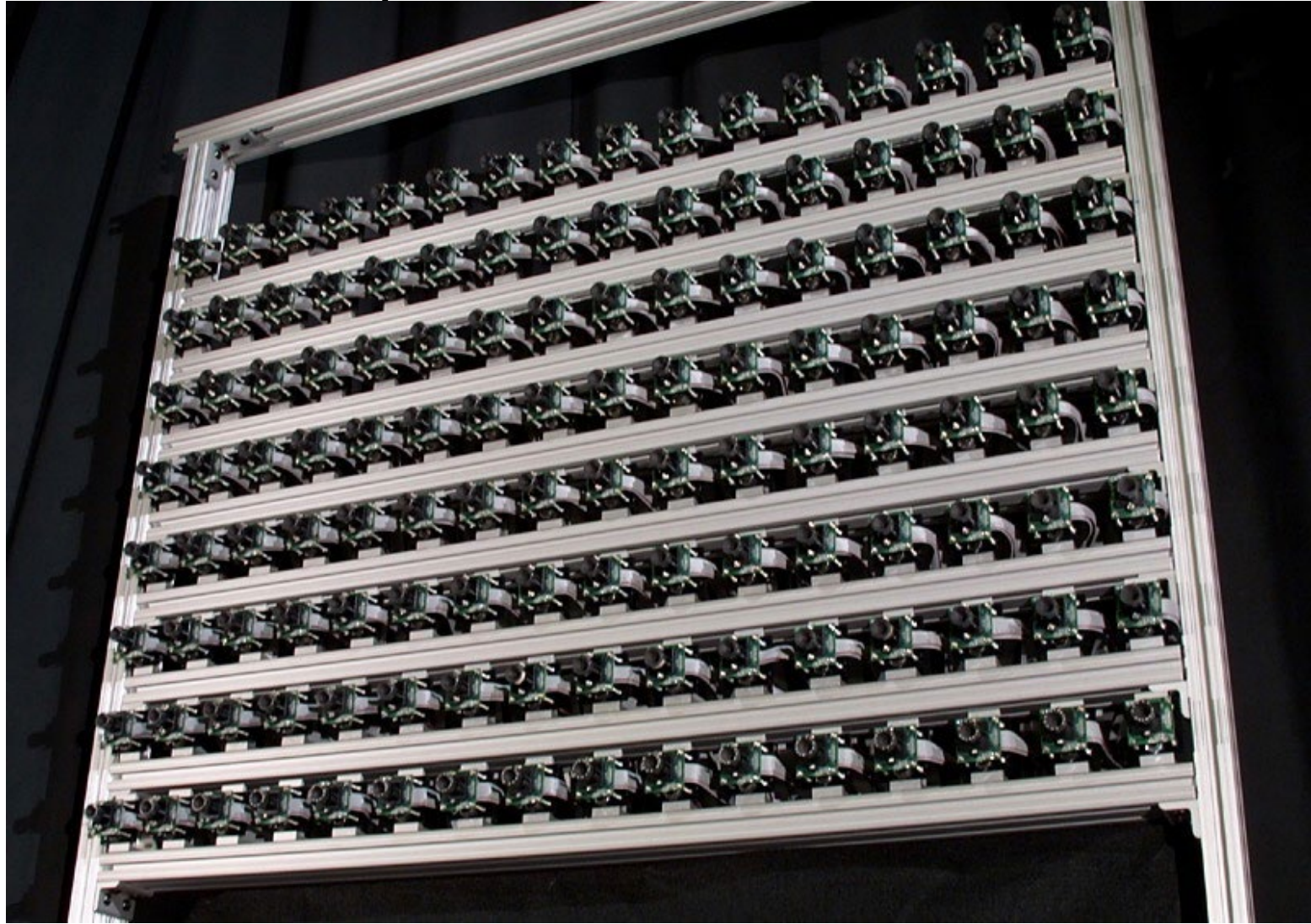


Mosquito eye, image: Raija Peura, University of Oulu



# Option 1: use multiple cameras

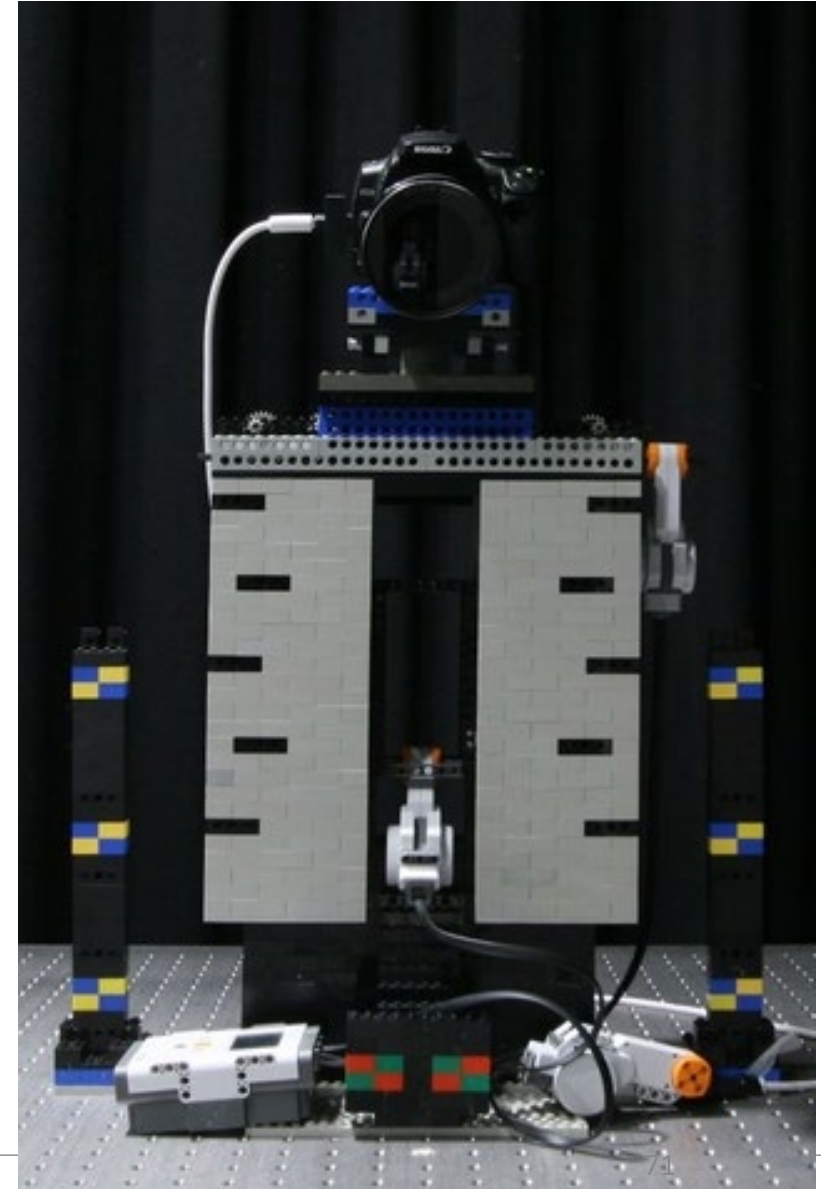
Problems?

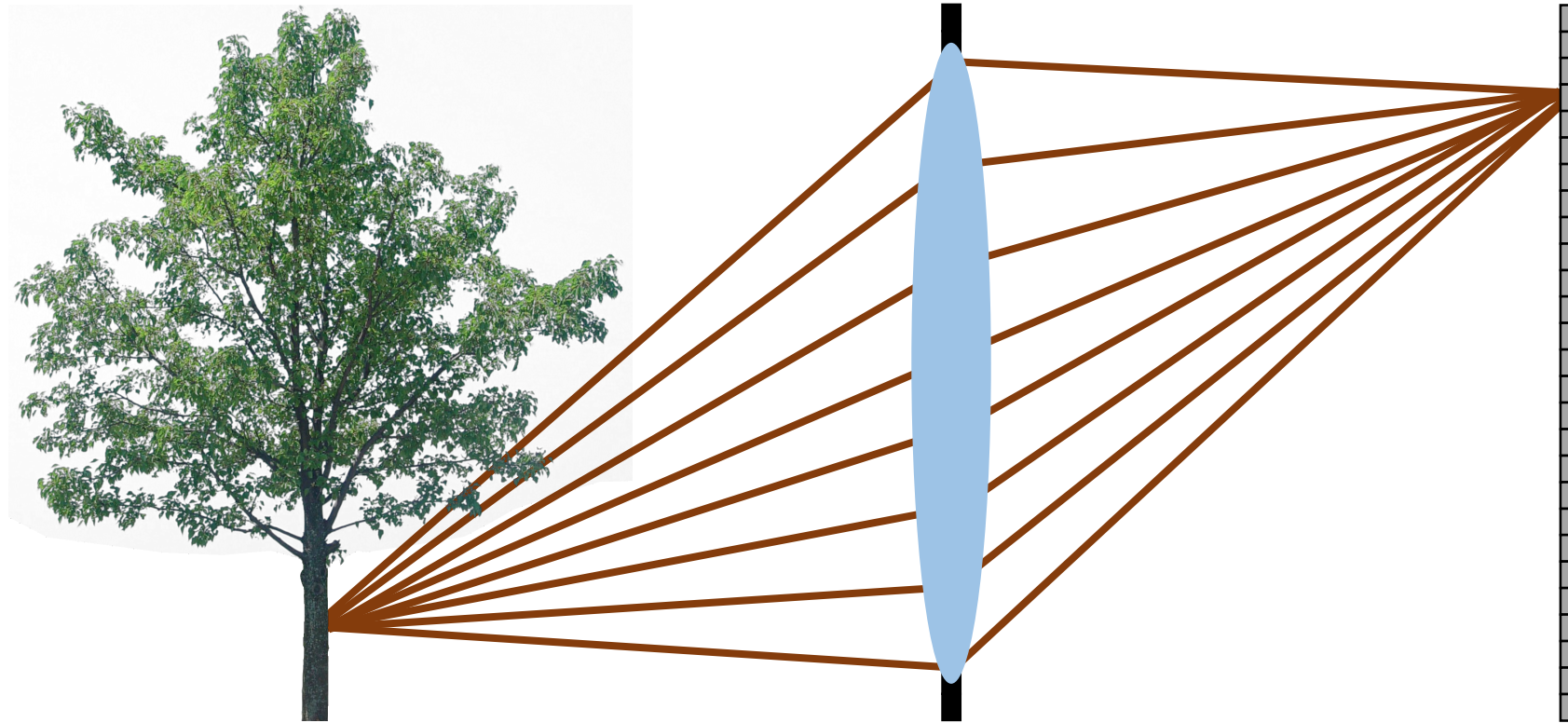


# Option 2: take multiple images with one camera

Single camera mounted on LEGO motor.

Problems?

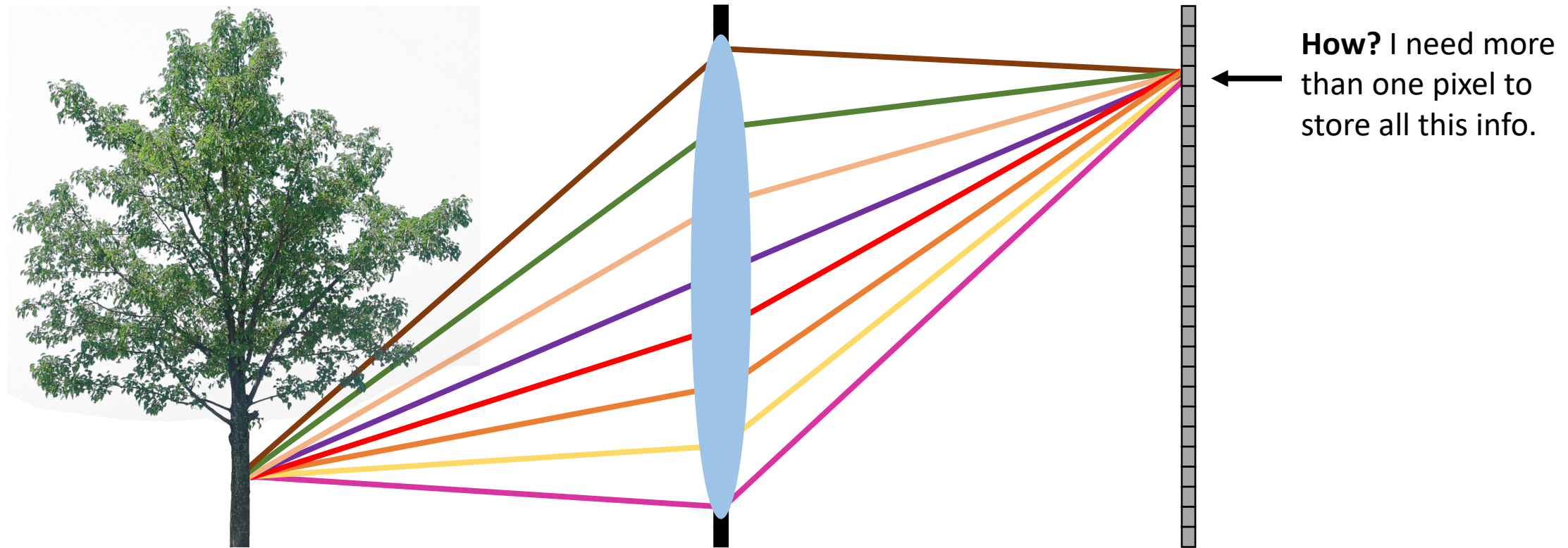




**Regular camera**: Integrates in each pixel of the sensor the angular information of each point in the scene

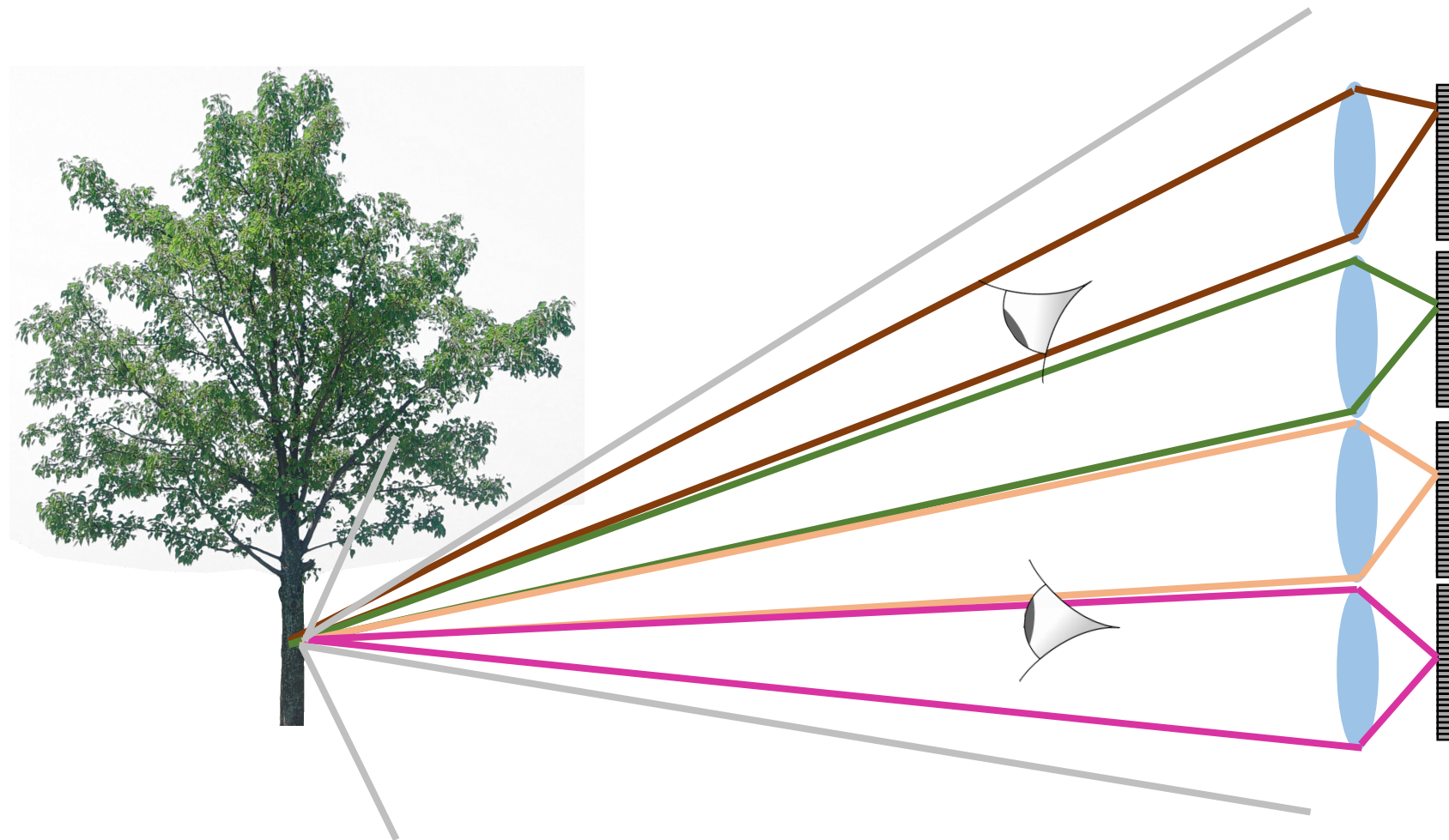
**Lightfield imaging**: Stores the angular information (i.e., different points of view) for each point in the scene.





**Regular camera:** Integrates in each pixel of the sensor the angular information of each point in the scene

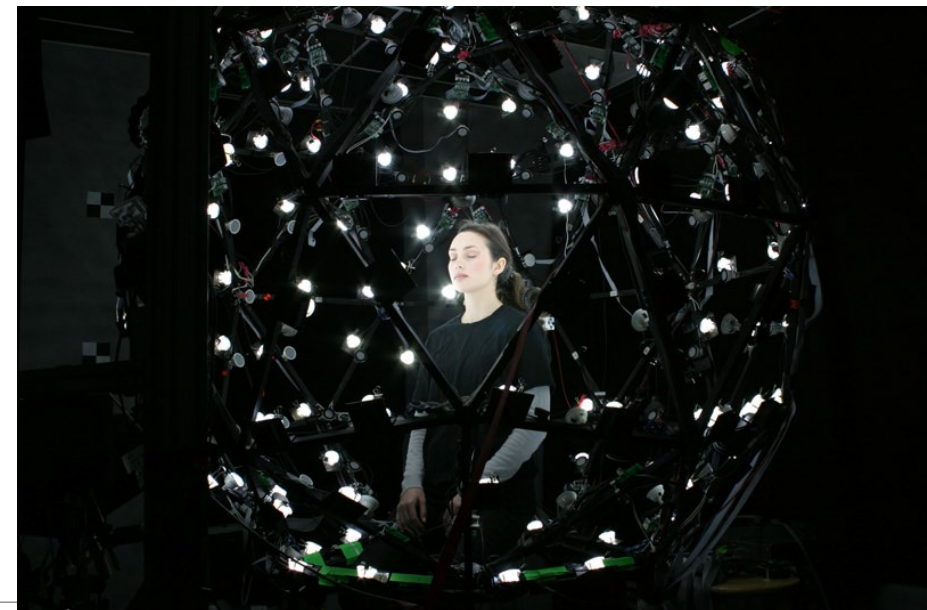
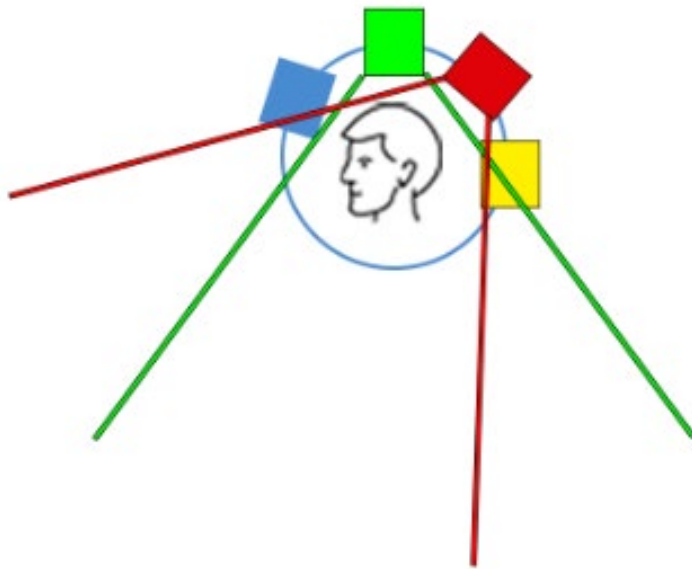
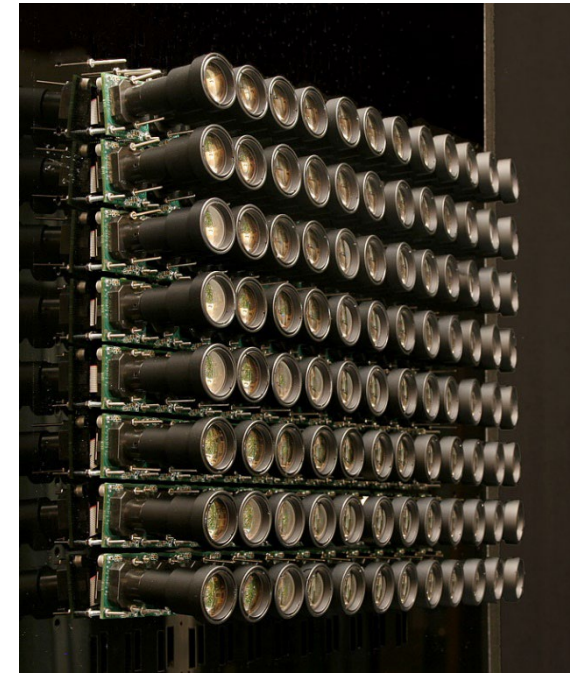
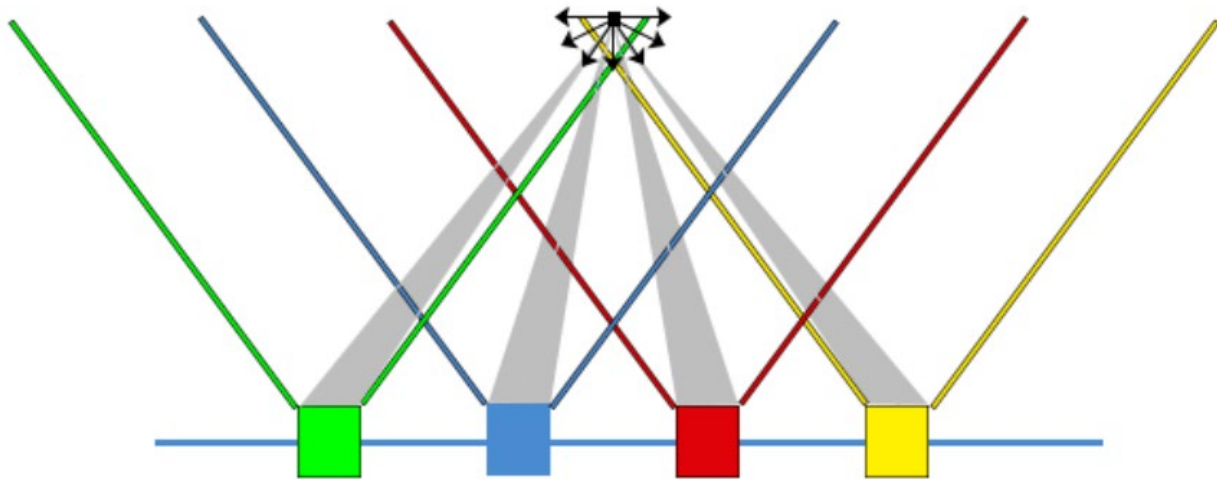
**Lightfield imaging:** Stores the angular information (i.e., different points of view) for each point in the scene.



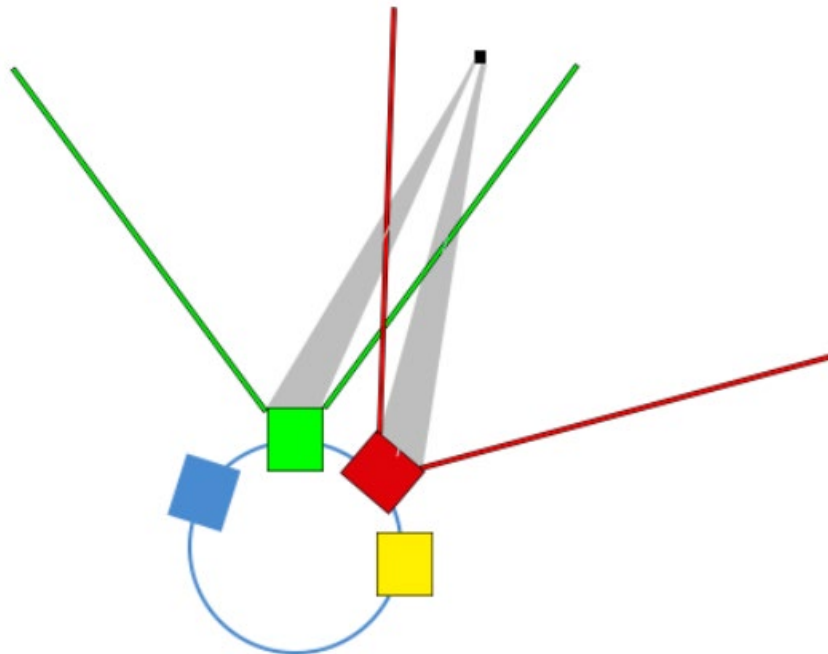
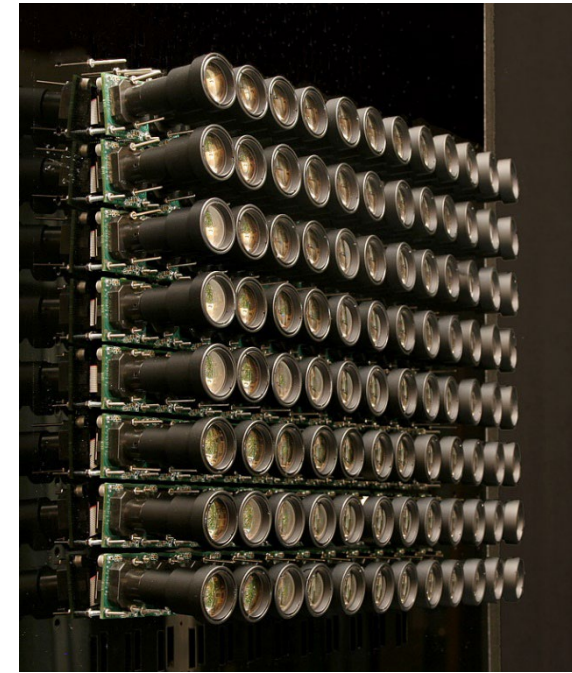
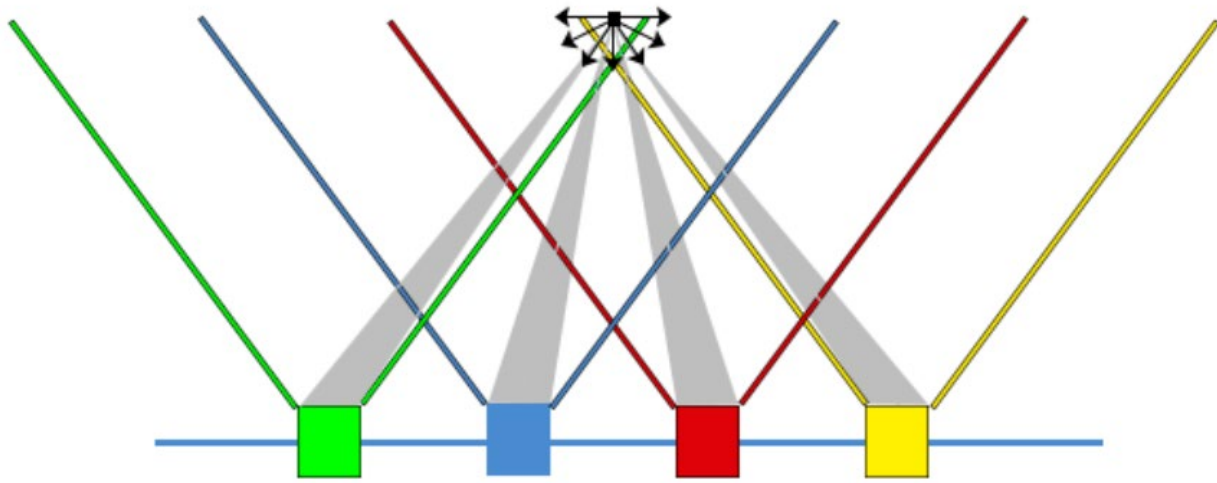
**Camera array:** Each will sample a subset of ray bundles

**Regular camera:** Integrates in each pixel of the sensor the angular information of each point in the scene

**Lightfield imaging:** Stores the angular information (i.e., different points of view) for each point in the scene.









“Lightfield”

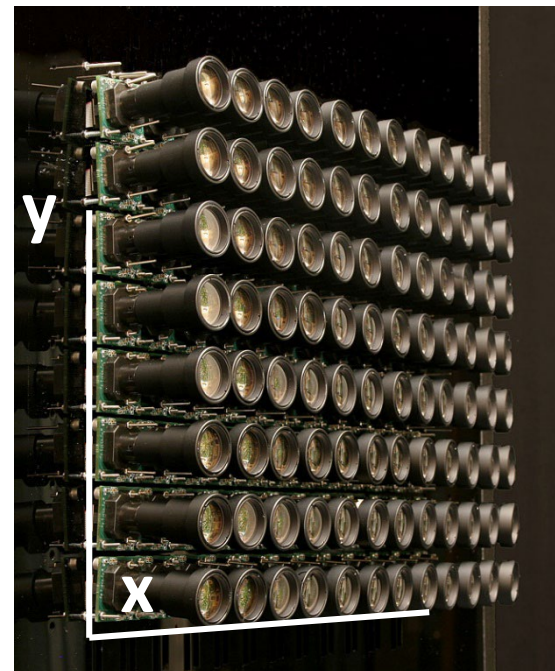
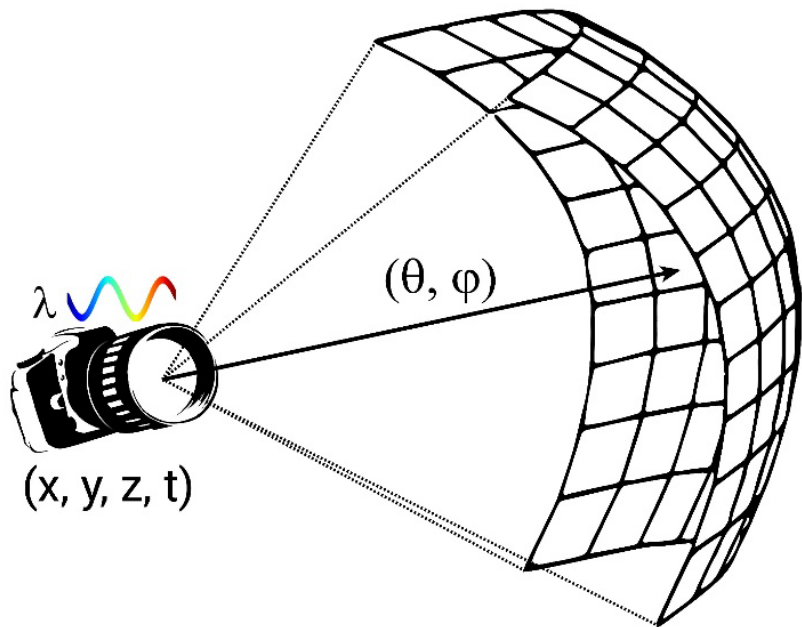


“Stereo only”



Source: <http://lightfield.stanford.edu/>





$(x, y)$   
 $(\theta, \Phi)$



$(x, y)$   
 $(\theta, \Phi)$



**Ideally both?**

Google 2020 – Immersive light field video with a layered mesh representation

<https://augmentedperception.github.io/deepviewvideo/>

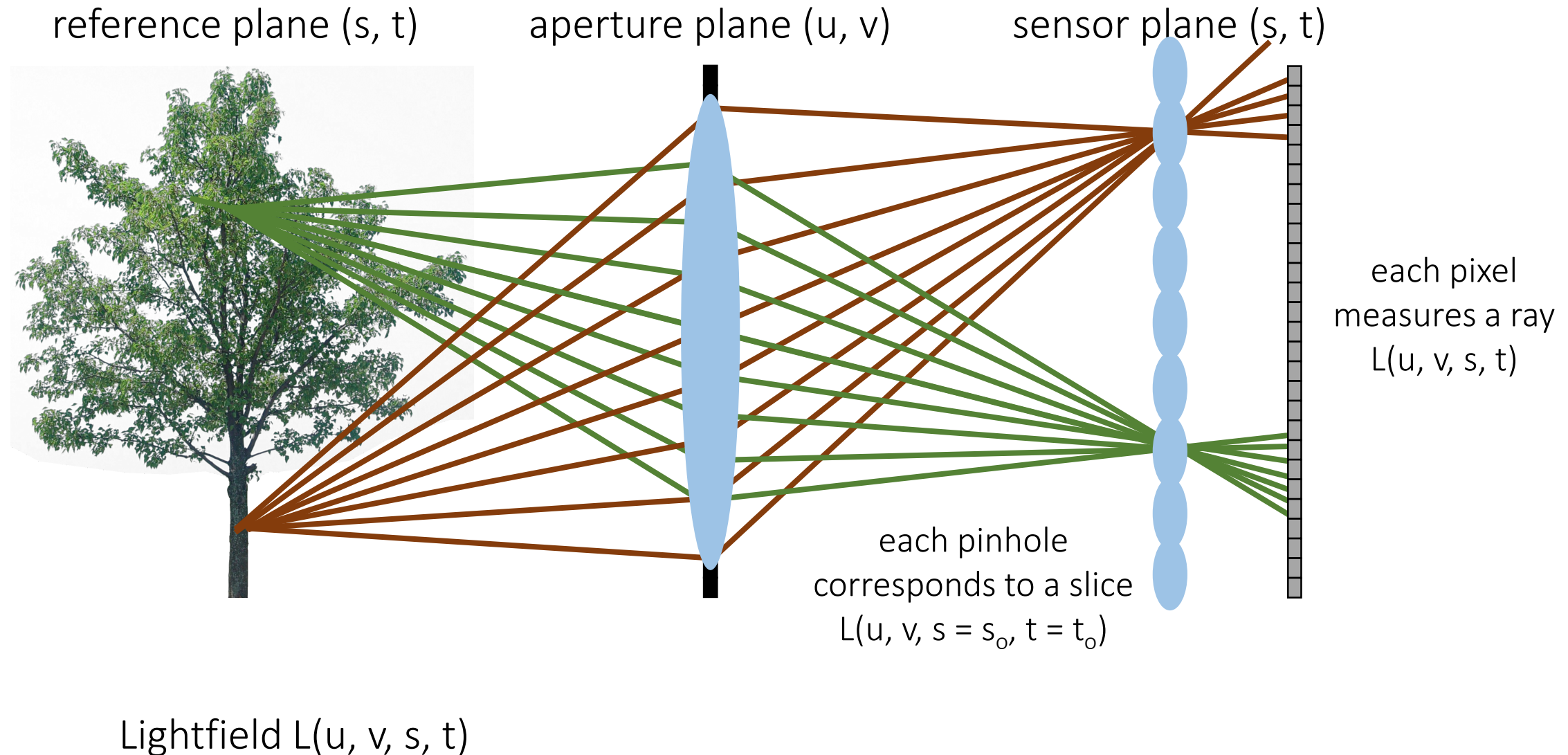
# Option 3: use a plenoptic camera

How to capture a 4D lightfield with a 2D sensor in a single shot?



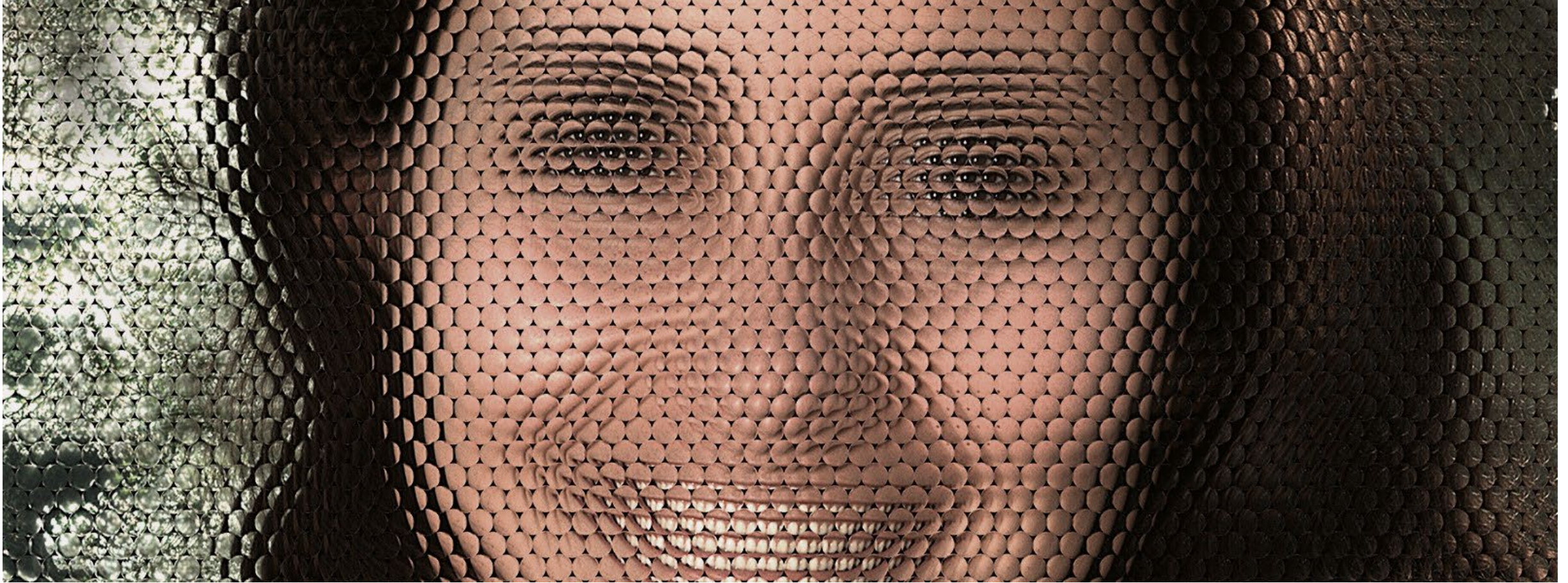


# Making a plenoptic camera





# Option 3: use a plenoptic camera



Source: Raytrix



# History of the plenoptic camera

164

SCIENTIFIC AMERICAN

August 19, 1911

## Integral Photography

A New Discovery by Professor Lippmann

By the French Correspondent of the Scientific American

PROF. LIPPMANN of Paris is working on a very remarkable new photographic method which he has termed "integral photography." The nature of this is best explained by reference to the accompanying diagram. It should be remarked in the first place that the process is designed to work without a camera. The sensitized plate is coated with a large number of very small globules of glass or other transparent material. In use each of these acts as a miniature eye, with a sensitive plate for its retina. The globules are very small, less than one-hundredth of an inch in diameter. In front each has a little blob which acts as the lens of the miniature eye. Suppose an object to be situated at  $M M'$ , as indicated in the diagram. Consider any globule. Rays from the points  $M M'$  will pass through the minute lens, and an image will be formed at the back of the globule upon the sensitized plate at  $M M'$ . Thus each globule will contain a minute picture of the objects placed in front of the plate. This picture is brought out by developing the plate and fixing it in the usual way. This of course yields a negative, which must be reversed by the use of a suitable reversing bath. The result is a transparency, in which all the minute pictures in the individual globules are combined into one image of the original object. To view it, a good light is placed behind the plate, so as to send light through all the globules. Suppose the eye viewing the transparency to be placed at  $O$ , the point which when the picture was taken, lay on a line joining one globule with the point  $M$  of the object. The process which now occurs will be the opposite of what happened when the picture was taken. In other words, the eye will receive from the globule a view of the image  $m$  of the point  $M$ . Other globules will send images of the point  $M'$ , etc., at the same time, so that the eye will see the different portions of the image coming from all the different globules. In this way a complete image of the object originally pre-

is changed in accordance with the direction in which the eye is looking at the globule.

After working out this remarkable principle, Prof. Lippmann presented it before the French Academy of Sciences and other scientific societies, where it awakened great interest. The realization of the process presents considerable technical difficulties. Owing to their small size, it was out of the question to shape each one of the globules separately. Prof. Lippmann tried using minute glass beads, such as are found in commerce and used for covering ornamental surfaces, such as picture post cards. He found, however, that these were too irregular in shape and size. Another idea which suggested itself was to stamp out a transparent sheet of collodion and gelatine, giving the surface the requisite shape. But the dies required were found to be too difficult to make in the laboratory, so that Prof. Lippmann has been forced to abandon his attempts in this direction for the time being. The fact is that the working out of a process of this kind is a technical

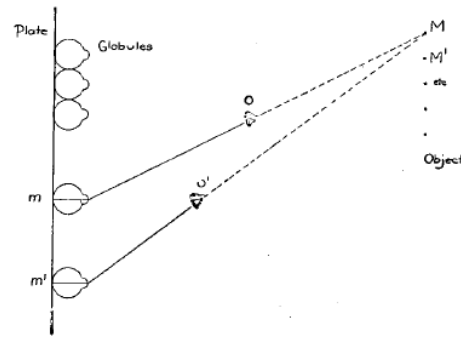


Diagram to illustrate the principle of Lippmann's integral photography.

on the plate, the portions of the plate illuminated by the several lenses being kept separate by an arrangement of black cardboard cells or partitions. One of our views represents the front of the camera with a set of twelve lenses, while the rear view shows the cells in the plate-holder. This latter carries a large plate of such size as to receive all the twelve pictures, and is placed in the camera as usual. The plate-holder, however, is made with a slide in the back as well as in the front, for a purpose which will appear presently. After the picture is taken, the negative is developed as usual, and is reversed either by means of a reversing bath, or by copying it, so that a transparent positive is obtained. The twelve pictures are not quite alike, for each lens forms its image from a somewhat different point of view. The transparency is put back in the holder, both slides of which are now opened, while light is sent through from behind. The observer looks in through the lenses with both eyes, when he sees a single view in relief of the object photographed. On moving the head from side to side, or up and down, the same effect is observed as would be under similar circumstances when looking at the real object, that is to say, objects which cover one another when looked at from one point, are seen to separate when viewed from another. Even with this simple apparatus the effect is very pleasing.

### Proposed Aeronautic Map

AT a recent meeting of the Academie des Sciences, M. Lallemand, chief of the French Government Survey Department, presented the project which the Aeronautic Commission intends to carry out establishing an aeronautic map. The proposed map is of 1:200,000 size, and is drawn up after a provisory model made by the Aero Club. Each plate of the aeronautic map will be a sheet taking in one degree in latitude and longitude and there will be twenty-

First conceptualized by Gabriel Lippmann, who called it *integral photography*.

- Original article appeared in French journal three years earlier, what is shown here is an American re-print.

August 19, 1911

Notice the date: more than a century ago.

# History of the plenoptic camera

## PARALLAX PANORAMAGRAMS MADE WITH A LARGE DIAMETER LENS

BY HERBERT E. IVES

Parallax panoramagrams—pictures showing stereoscopic relief through a wide range of distances and angles of observation—have heretofore been made by any one of several kinds of camera, in which the object, the photographic lens, the opaque line grating and the sensitive plate are given a relative motion during the progress of the exposure. Details of these cameras, and of the characteristics of the pictures made by them may be obtained by reference to an earlier publication.<sup>1</sup> In all cases, whether separation of the grating and plate is depended on to cause the development of the panoramic strips by parallax, or a relative motion is given to plate and grating, a common characteristic is the motion of the lens with respect to the object, whereby different points of view are successively projected upon the grating and plate. This note is for the purpose of describing a method in which the moving lens is dispensed with, making possible a parallax panoramagram camera containing no moving parts whatever.

OPTICAL TECHNOLOGY

VOLUME 37, NUMBER 7

JULY 1970

## Autostereoscopy and Integral Photography

Yu.A. Dudnikov

UDC 778.39 + 778.4

A brief description of autostereoscopic techniques for obtaining three-dimensional images is presented; the advantages and faults of integral photography are analyzed. A comparison is made between holography and integral photography, and the prospects of this latter method are indicated.

The viewing of stereo-photographs without the aid of special optics (hence the term autostereoscopy), is a long-sought goal. As of now, the recently developed holography technique produces stereo images with the largest amount of information presently possible. However, its practical utilization is difficult since it requires coherent illumination.

The concept of holography, i. e., fixation of an image by photographing the pattern of interference of waves incident on the subject and reflected by it [1], derives from the color (non-stereoscopic) photography of G. Lippmann (1894), in which interference nodes and antinodes of the light waves incident on the photographic plate and reflected by its back mirror surface are fixed within the light-sensitive layer.

Autostereoscopic methods which do not require coherent illumination, such as the parallax stereogram of Berthier (1896) [2] and the parallax panoramagram of Kanolt (1915) [3], were

successful in their time, and in improved form are being used even today.

The parallax stereogram is a replica from a normal stereoscopic pair, made optically through a line pattern, i. e., through a screen of alternating vertical transparent and opaque lines. The lenses of the objective of the printing device superimpose the images of the two stereoscopic negatives on one another, but the screen subdivides them into narrow alternating strips. One must view the combined image through the same screen, set up in such a way that the right eye sees the strips belonging to the left negative and the left eye those of the right negative (Fig. 1). The image thus produced is half as bright as that obtained with the same illumination in a stereoscope, and contains less visual information, since only half of the area of each frame is viewed (and this without magnification). The stereoscopic effect is satisfactory but in order not to lose it, both the photograph and ones head must be perfectly stationary.

Reappeared under different forms and names throughout the century.

- The left paper is from 1930, the right one from 1970.



# History of the plenoptic camera

## Single Lens Stereo with a Plenoptic Camera

Edward H. Adelson and John Y.A. Wang

*Abstract*—Ordinary cameras gather light across the area of their lens aperture, and the light striking a given subregion of the aperture is structured somewhat differently than the light striking an adjacent subregion. By analyzing this optical structure, one can infer the depths of objects in the scene, i.e., one can achieve “single lens stereo.” We describe a novel camera for performing this analysis. It incorporates a single main lens along with a lenticular array placed at the sensor plane. The resulting “plenoptic camera” provides information about how the scene would look when viewed from a continuum of possible viewpoints bounded by the main lens aperture. Deriving depth information is simpler than in a binocular stereo system because the correspondence problem is minimized. The camera extracts information about both horizontal and vertical parallax, which improves the reliability of the depth estimates.

### I. INTRODUCTION

“**E**VERY BODY in the light and shade fills the surrounding air with infinite images of itself; and these, by infinite pyramids diffused in the air, represent this body throughout space and on every side.” Leonardo da Vinci [1] uses

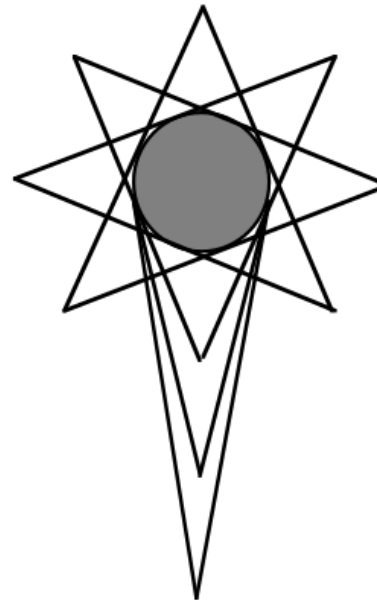


Fig. 1. Diagram from Leonardo's notebooks illustrating the fact that the light rays leaving an object's surface may be considered to form a collection of cones (which Leonardo calls “pyramids”), each cone constituting an image that would be seen by a pinhole camera at a given location.

Re-introduced to computer vision and graphics in 1992, which was also when the term *plenoptic camera* was coined.

# History of the plenoptic camera

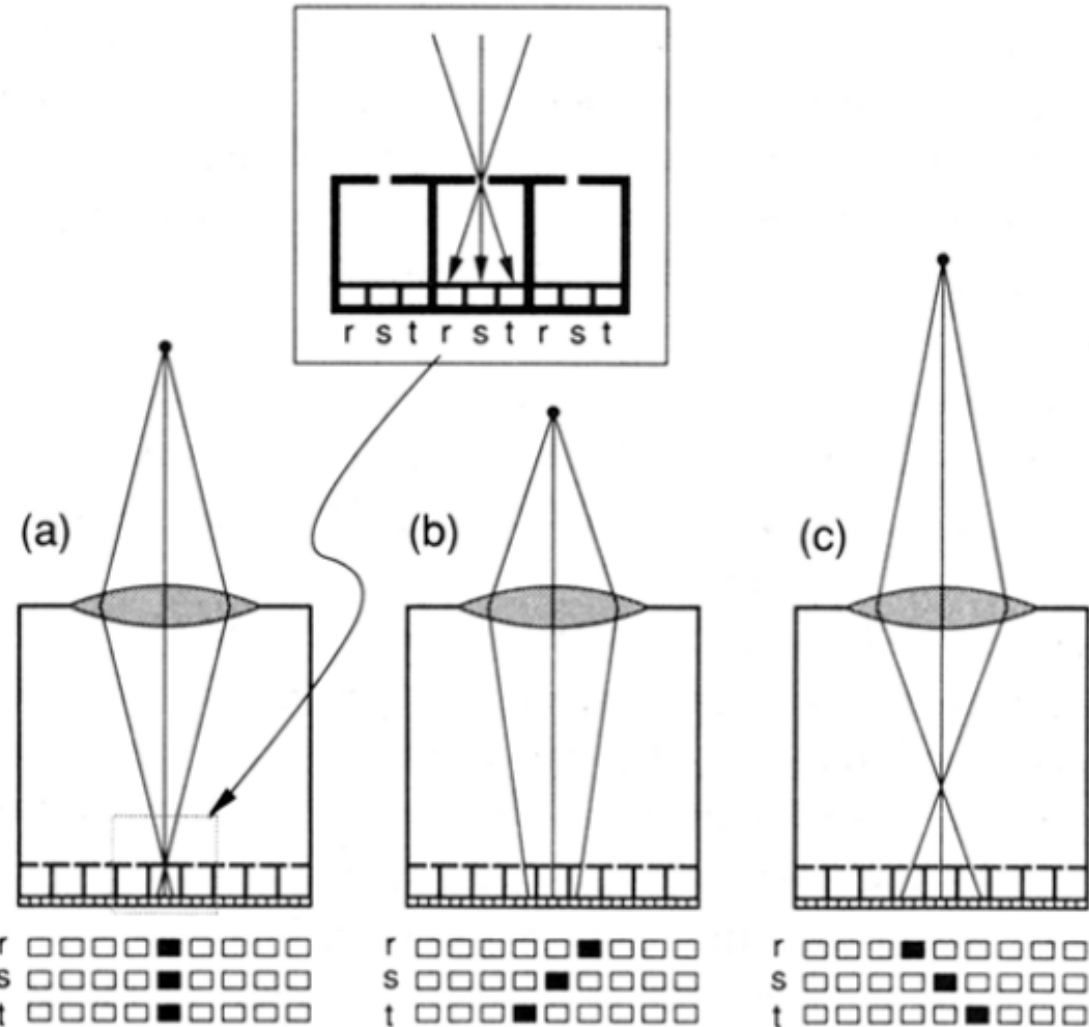


Figure from the 1992 paper, which shows the version of the plenoptic camera with pinholes instead of lenslets on the image plane.

Fig. 5. Array of miniature pinhole cameras placed at the image plane can be used to analyze the structure of the light striking each macropixel.

# History of the plenoptic camera

Stanford Tech Report CTSR 2005-02

## Light Field Photography with a Hand-held Plenoptic Camera

Ren Ng\*   Marc Levoy\*   Mathieu Brédif\*   Gene Duval†   Mark Horowitz\*   Pat Hanrahan\*

\*Stanford University

†Duval Design

### Abstract

This paper presents a camera that samples the 4D light field on its sensor in a single photographic exposure. This is achieved by inserting a microlens array between the sensor and main lens, creating a plenoptic camera. Each microlens measures not just the total amount of light deposited at that location, but how much light arrives along each ray. By re-sorting the measured rays of light to where they would have terminated in slightly different, synthetic cameras, we can compute sharp photographs focused at different depths. We show that a linear increase in the resolution of images under each microlens results in a linear increase in the sharpness of the refocused photographs. This property allows us to extend the depth of field of the camera without reducing the aperture, enabling shorter exposures and lower image noise. Especially in the macrophotography regime, we demonstrate that we can also compute synthetic photographs from a range of different viewpoints. These capabilities argue for a different strategy in designing photographic imaging systems.

To the photographer, the plenoptic camera operates exactly like an ordinary hand-held camera. We have used our prototype to take hundreds of light field photographs, and we present examples of portraits, high-speed action and macro close-ups.

**Keywords:** Digital photography, light field, microlens array, synthetic photography, refocusing.

Externally, our hand-held light field camera looks and operates exactly like a conventional camera: the viewfinder, focusing mechanism, length of exposure, etc. are identical. Internally, we augment the 2D photosensor by placing a microlens array in front of it, as proposed by Adelson and Wang [1992] in their work on the “plenoptic camera” (They did not build this device, but prototyped a non-portable version containing a relay lens.) Each microlens forms a tiny sharp image of the lens aperture, measuring the directional distribution of light at that microlens.

This paper explains the optical recipe of this camera in detail, and develops its theory of operation. We describe an implementation using a medium format digital camera and microlens array. Using this prototype, we have performed resolution experiments that corroborate the limits of refocusing predicted by the theory. Finally, we demonstrate examples of refocusing and view-point manipulation involving close-up macro subjects, human portraits, and high-speed action.

## 2 Related Work

The optical design of our camera is very similar to that of Adelson and Wang’s plenoptic camera [1992]. Compared to Adelson and Wang, our prototype contains two fewer lenses, which significantly shortens the optical path, resulting in a portable camera. These differences are explained in more detail Section 3.1 once sufficient technical background has been introduced. The other main differ-

Re-re-introduced to computer vision and graphics in 2005, when the term *light field photography* was also coined.

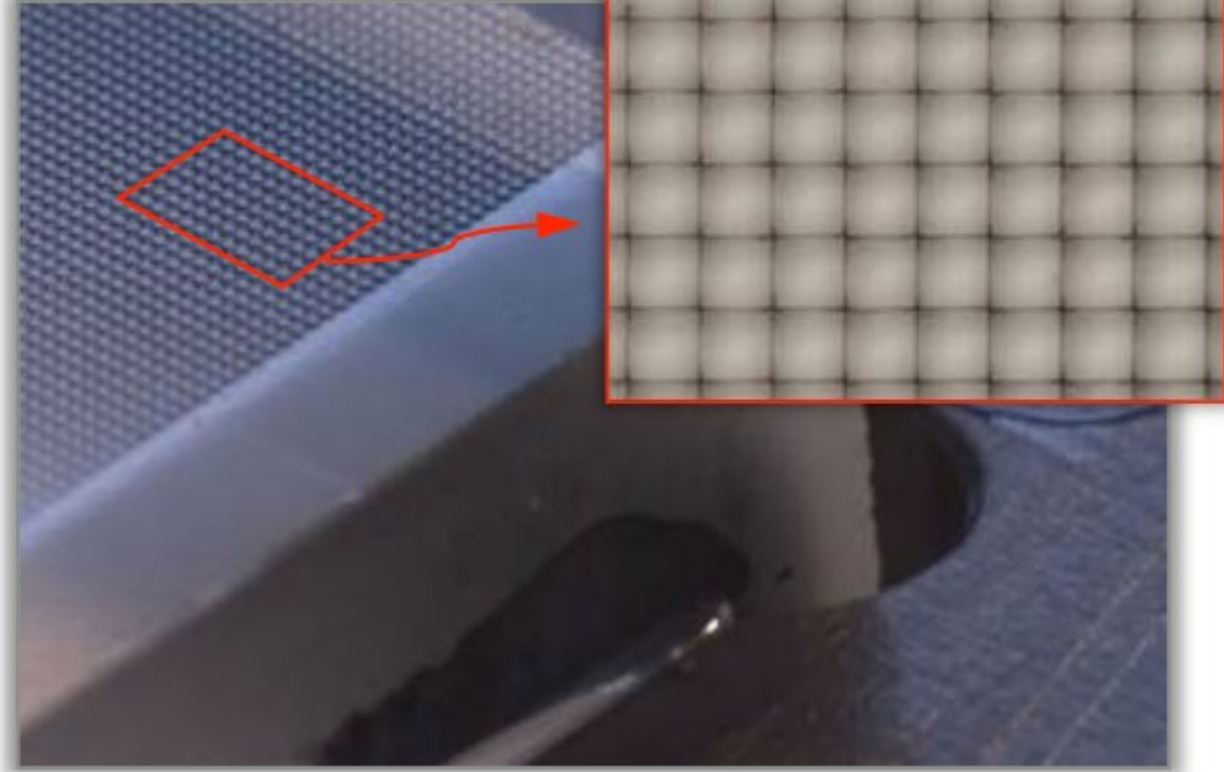


# Prototype plenoptic camera

- predecessor of Lytro
- resolution: 292x292px, 14x14 light field views



Kodak 16-megapixel sensor



125μ square-sided microlenses

# Commercial plenoptic camera

## Lens

The Lytro Light Field Camera starts with an 8X optical zoom, f/2 aperture lens. The aperture is constant across the zoom range allowing for unheard of light capture.

## Light Field Engine 1.0

The Light Field Engine replaces the supercomputer from the lab and processes the light ray data captured by the sensor.

The Light Field Engine travels with every living picture as it is shared, letting you refocus pictures right on the camera, on your desktop and online.

**Light Field Sensor**  
From a roomful of cameras to a micro-lens array specially adhered to a standard sensor, the Lytro's Light Field Sensor captures 11 million light rays.



# Commercial plenoptic camera

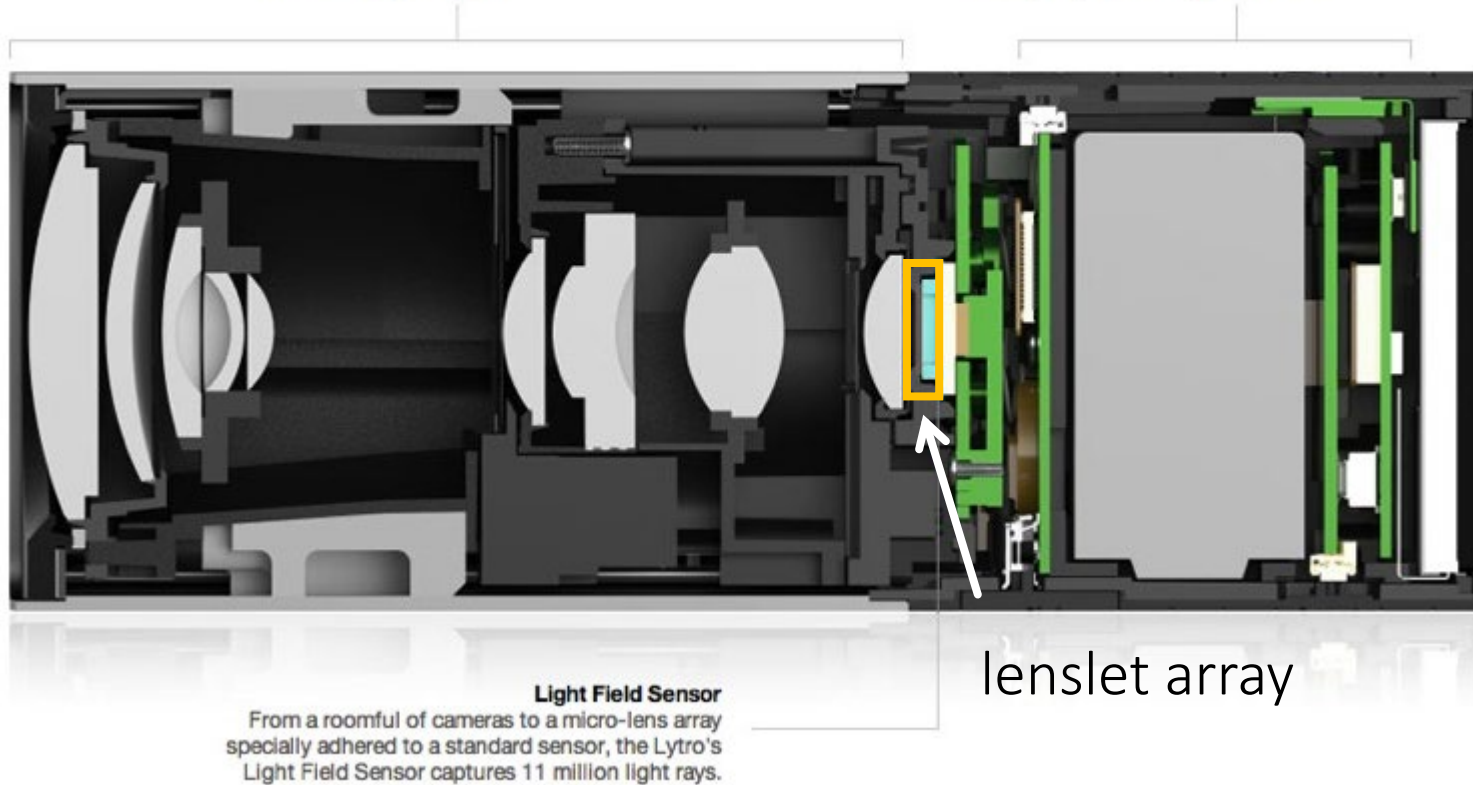
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lenslet array





- Stanford startup!

The image is a screenshot of the CrunchBase website showing the profile for the startup Lytro. The page has a dark header with the CrunchBase logo, a search icon, and a plus icon. Below the header, the Lytro profile is displayed with a blue background bar. The profile includes a 'FOLLOW' button, tabs for 'Info' and 'Section', and an 'ADD' button. The main content area shows the Lytro logo, a star rating of 23, and an eye icon with '25K'. The funding section states that Lytro has received \$90 Million in 3 Rounds from 6 Investors, with the most recent funding in 2014. The headquarters are listed as Mountain View, CA. The description states that Lytro is a light-field camera startup. The founders are listed as Ren Ng, and the categories are Photography, Hardware, and Hardware + Software. The website is listed as http://www.lytro.com. The page also features social media icons for Facebook and Twitter, and a help icon in the bottom right corner.

CrunchBase

Lytro

★ FOLLOW

Info Section

⊕ ADD

UPDATE ▾

**LYTRO**

★ 23

👁 25K

Funding Received  
**\$90 Million** in **3 Rounds** from **6 Investors**

Most Recent Funding  
**2014** / Undisclosed Amount and Round

Headquarters: **Mountain View, CA**

Description: Lytro, a light-field camera startup, markets their products to industrial and scientific applications.

Founders: **Ren Ng**

Categories: **Photography, Hardware, Hardware + Software**

Website: **http://www.lytro.com**

Facebook Twitter

www.crunchbase.com/organization/lytro

# Commercial plenoptic camera



newer version with  
higher resolution

# Industrial plenoptic cameras

Plenoptic cameras have become quite popular in lab and industrial settings.



- Much higher resolution, both spatial and angular, than commercial cameras.
- Support interchangeable lenses.
- Can do video.
- Very expensive.



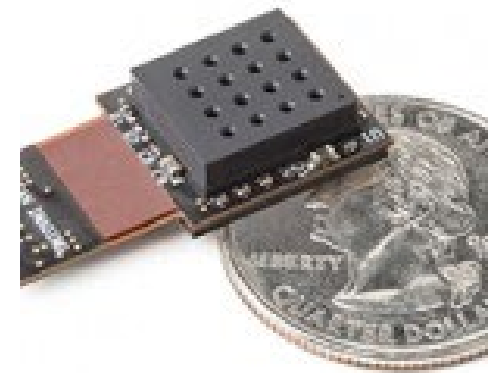
# Light field cameras proliferate...



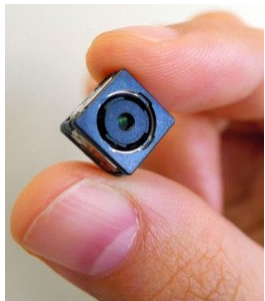
Raytrix™



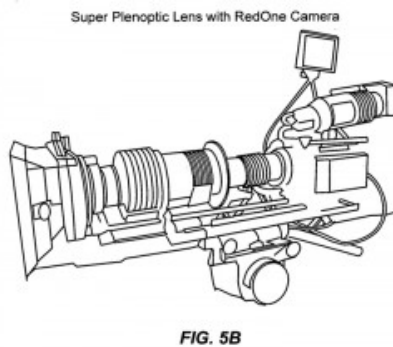
Lytro™



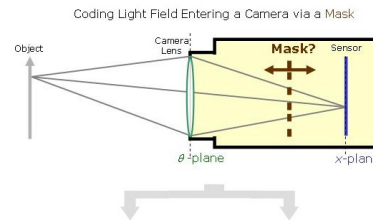
Pelican Imaging



Toshiba



Pixar



[Raskar et al. 2007]



Adobe



Cafadis

# Question

Given that the plenoptic camera design was known since 1908, why did it take a century for such a camera to be made commercially available?

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Given that the plenoptic camera design was known since 1908, why did it take a century for such a camera to be made commercially available?

- Difficult to manufacture good lenslet arrays at high resolution.
- Digital sensors **did not have sufficiently high resolution**.
- No known good use for them besides depth sensing.
  - The introduction of the handheld plenoptic camera was strongly influenced by the 1996 lightfield papers, which demonstrated all of the useful photographic operations one could replicate by having access to the entire lightfield.
- And of course, nobody thought of commercializing them.



# History of the plenoptic camera

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SCIENTIFIC AMERICAN

August 19, 1911

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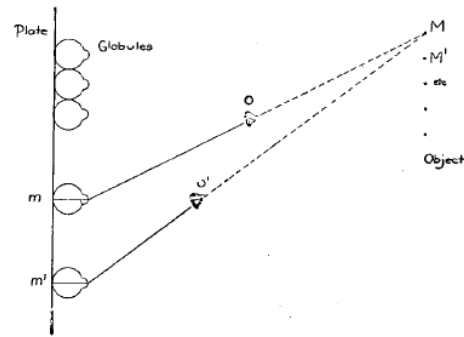
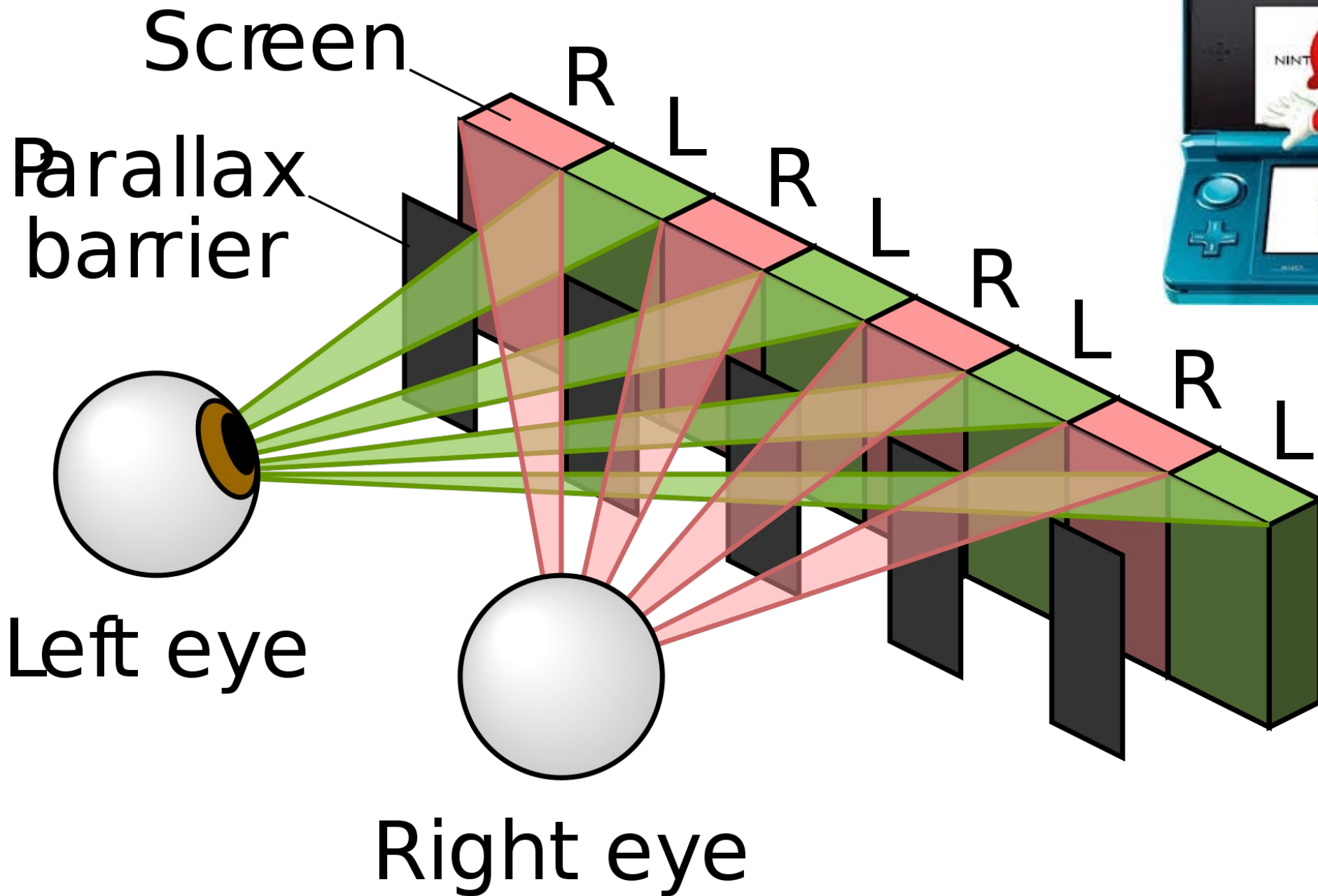


Diagram to illustrate the principle of Lippmann's integral photography.

### Proposed Aeronautic Map

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These concepts were not only used for acquisition, but also for display...

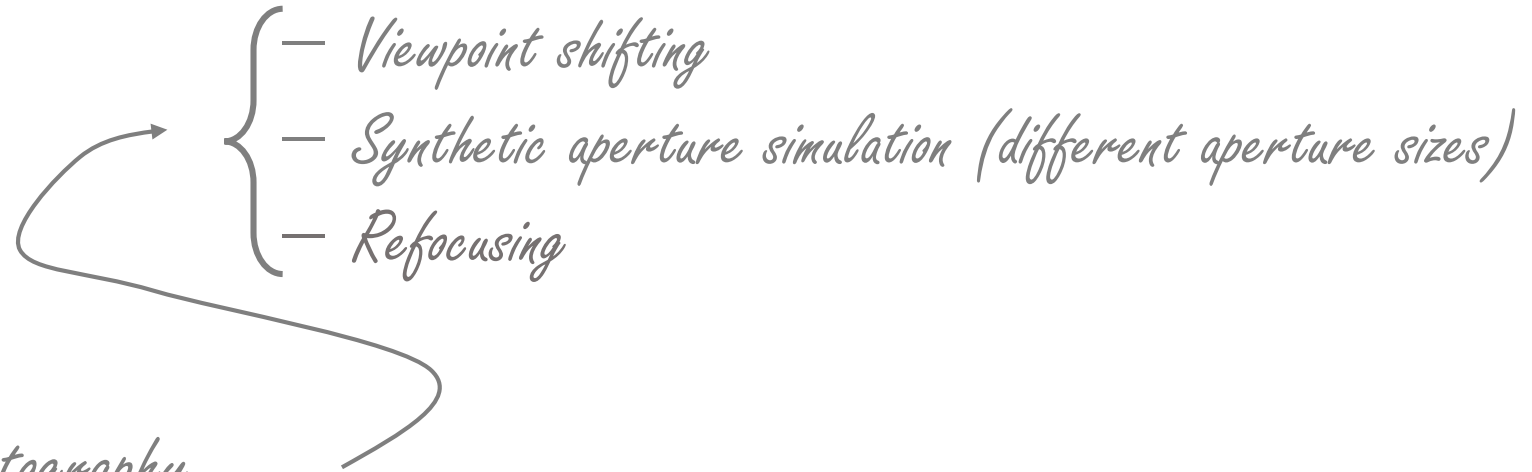


# Light Field Rendering

*(or how to generate images from light fields)*

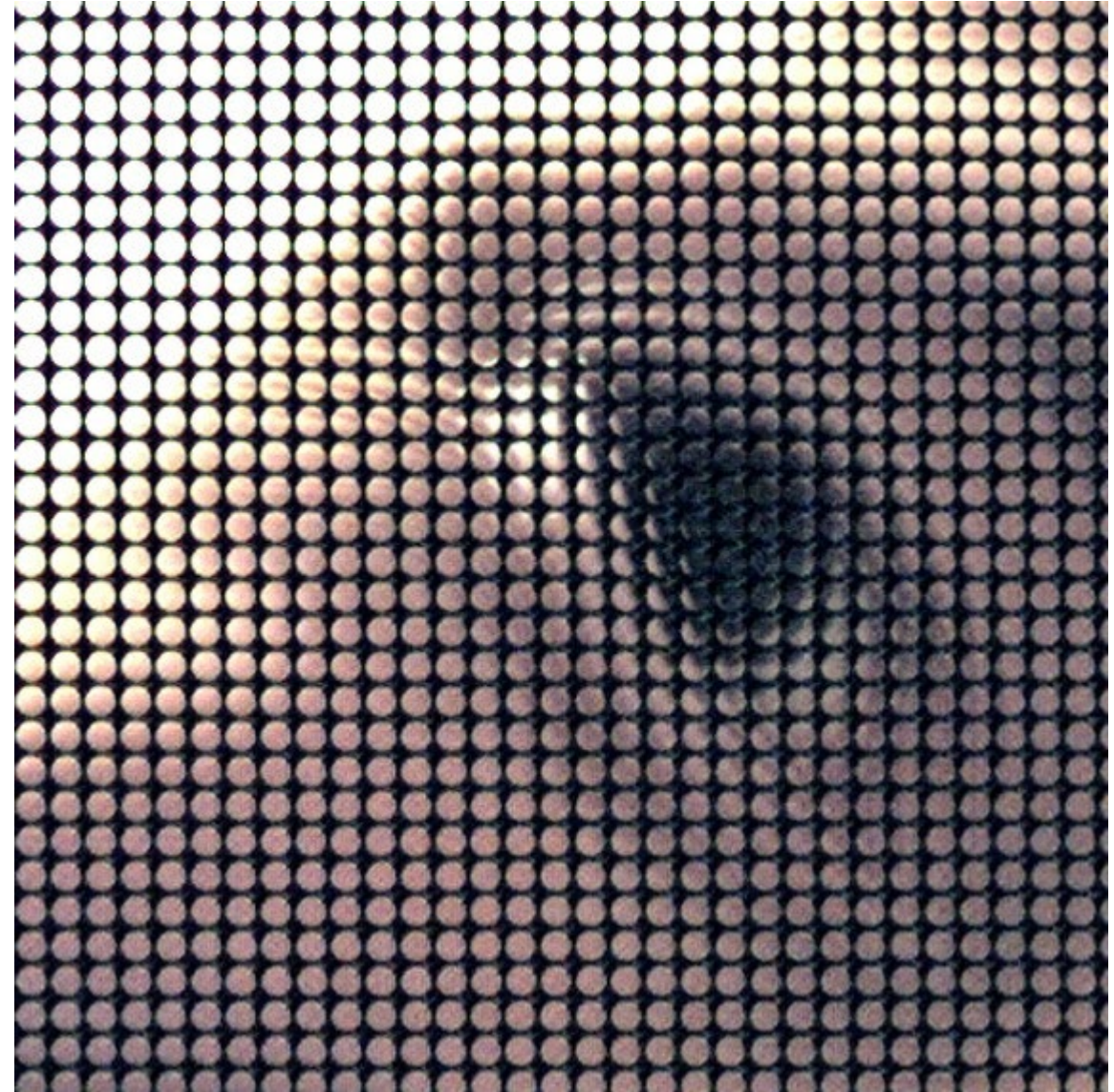
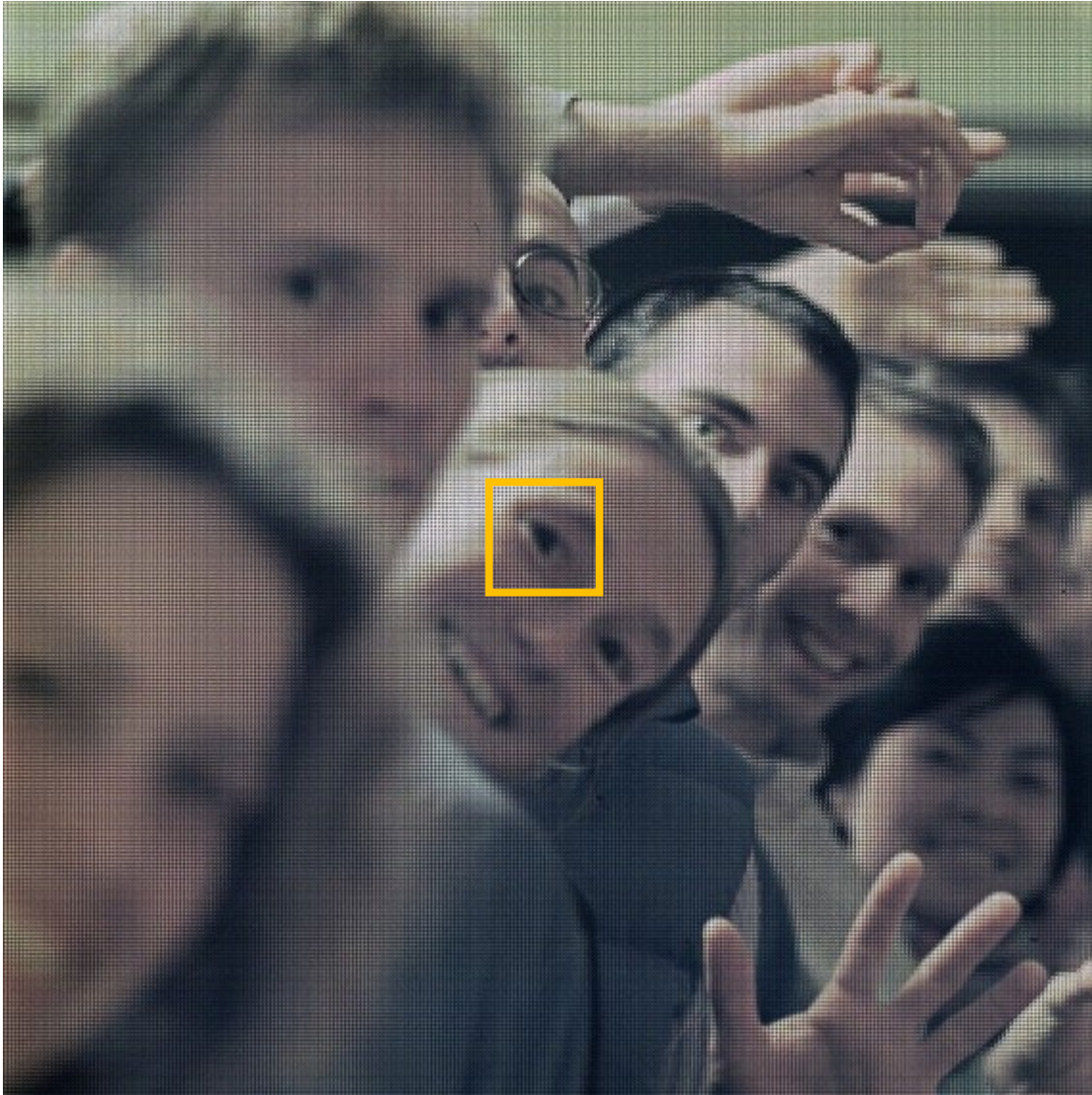
# Light Field Imaging

- ✓ • Concept
- ✓ • Parameterization
- ✓ • Acquisition
- Rendering – *Light field re-photography*





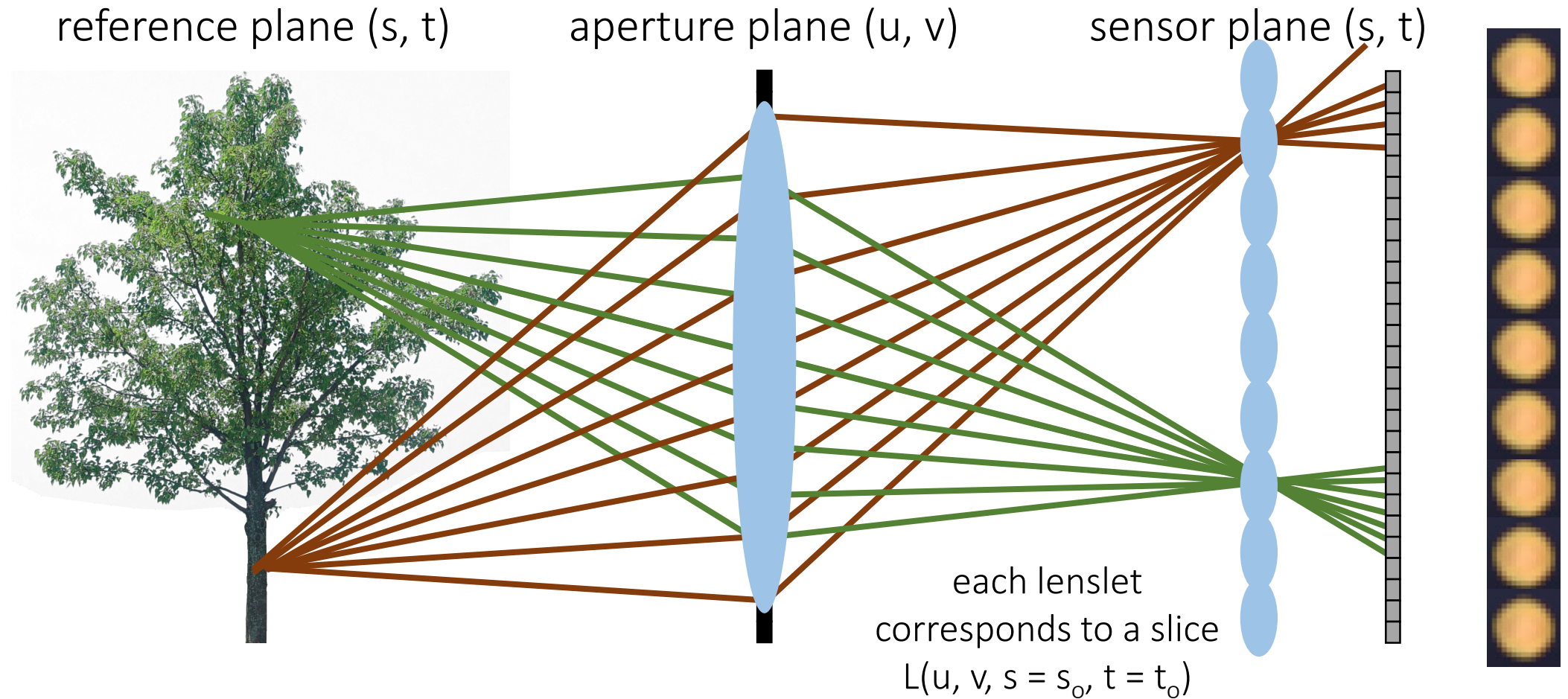
# A plenoptic “image”



What are these circles?



# A plenoptic camera

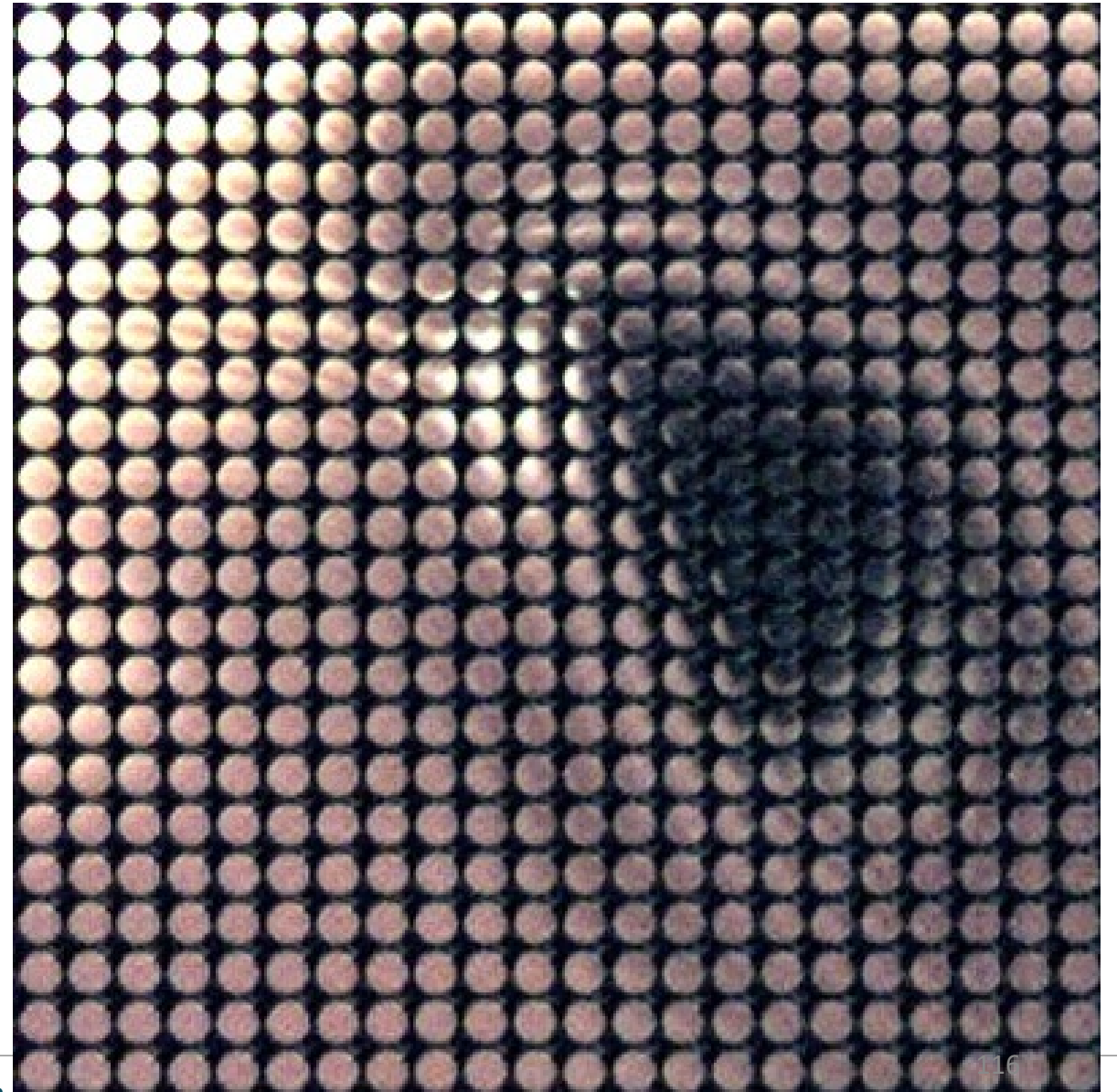


Lightfield  $L(u, v, s, t)$

# The plenoptic image

Which coordinates do I change when I move from one circle to another?

Which coordinates do I change when I move within each circle?



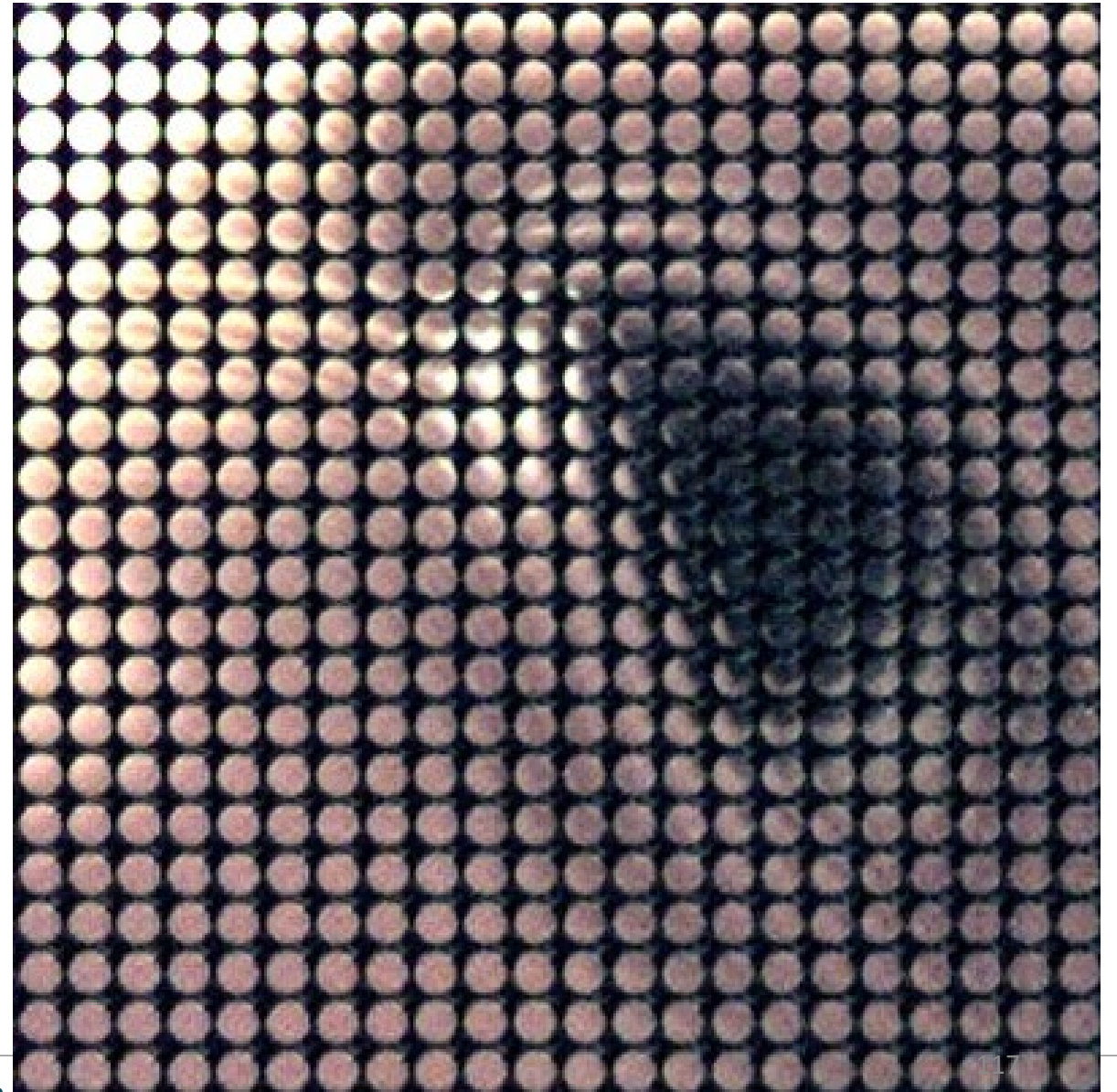
# The plenoptic image

Which coordinates do I change when I move from one circle to another?

- I change  $s, t$  (sensor plane) coordinates.

Which coordinates do I change when I move within each circle?

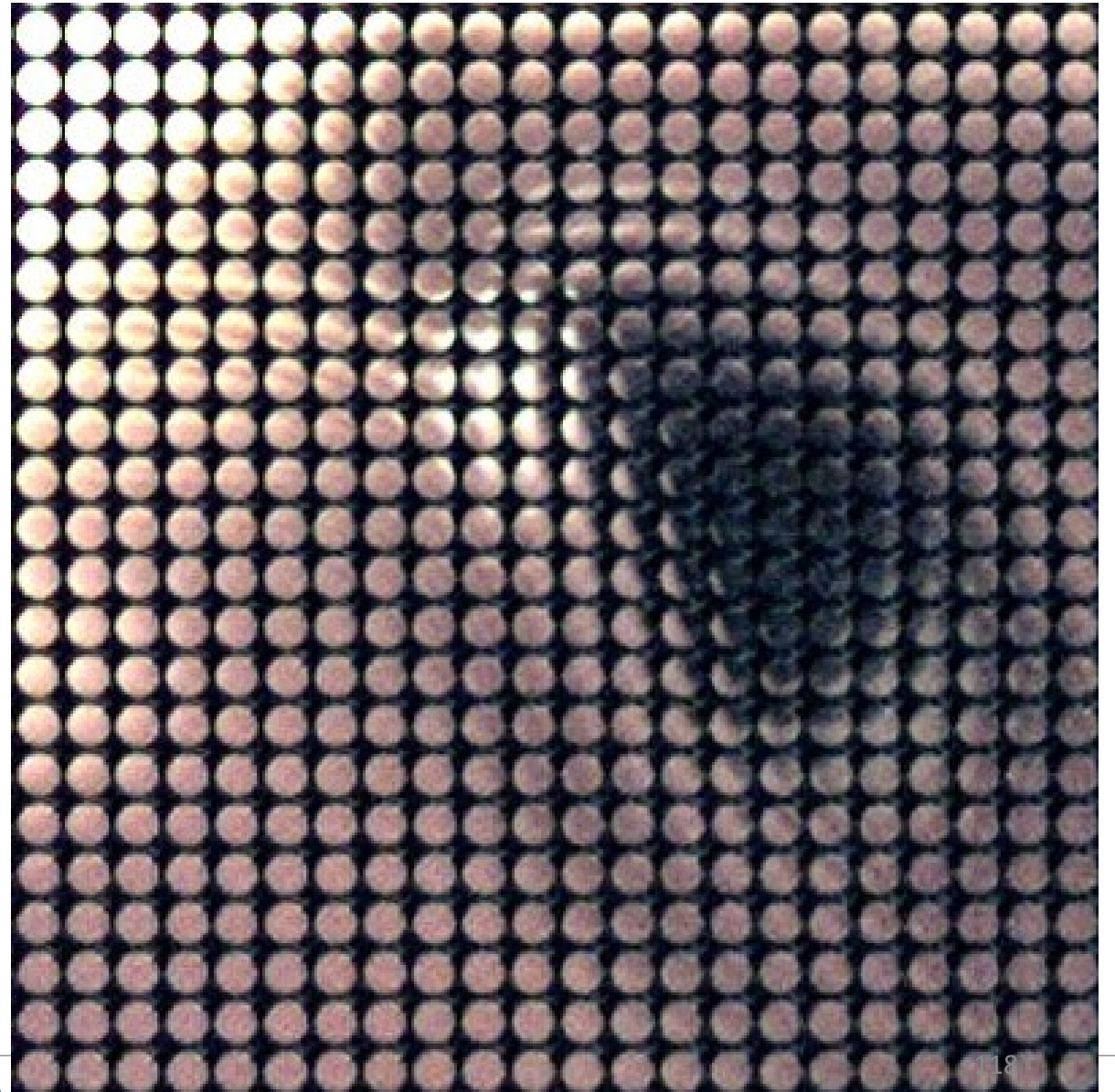
- I change  $u, v$  (aperture plane) coordinates.





# How do I...

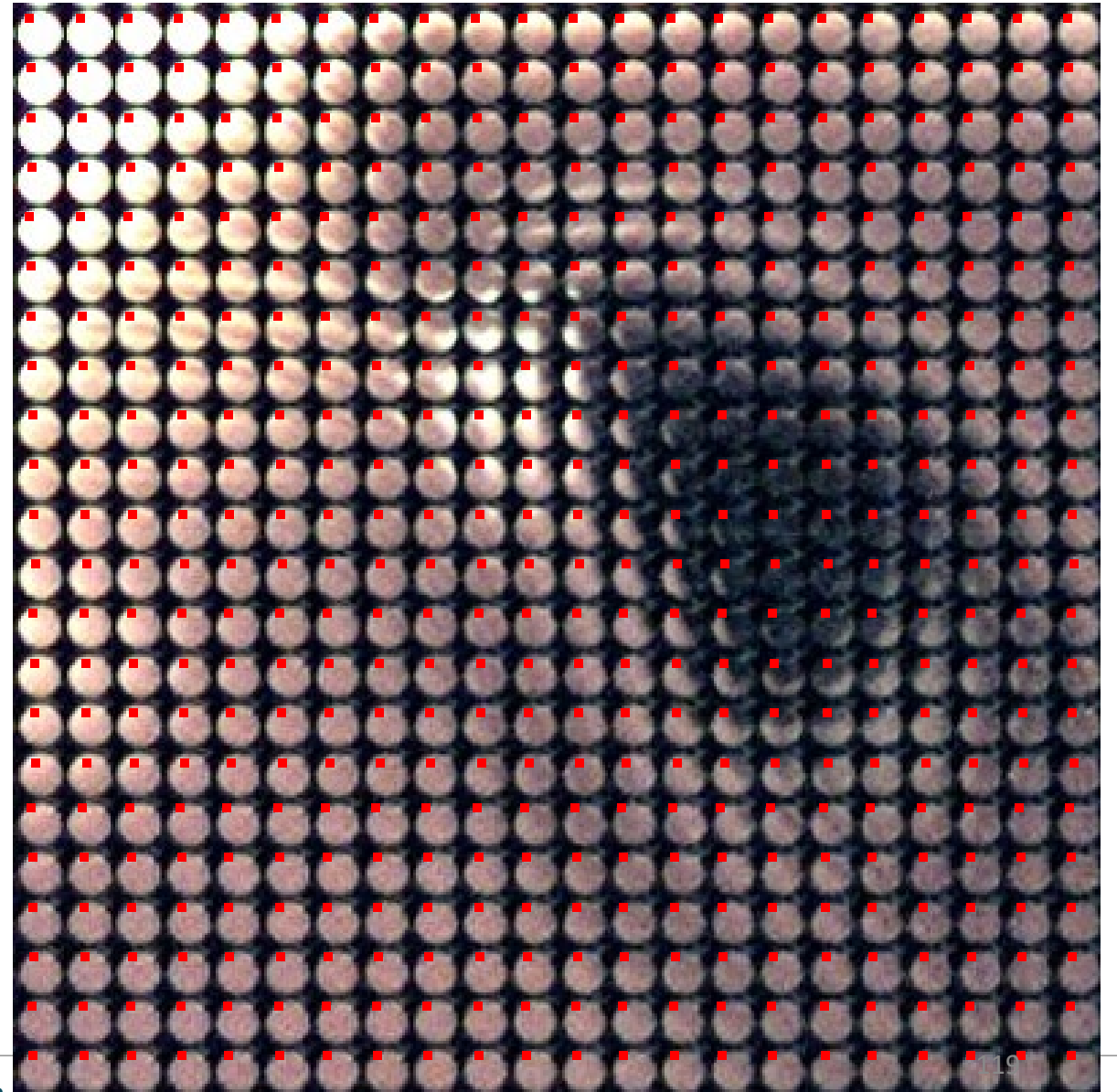
Simulate different viewpoints?



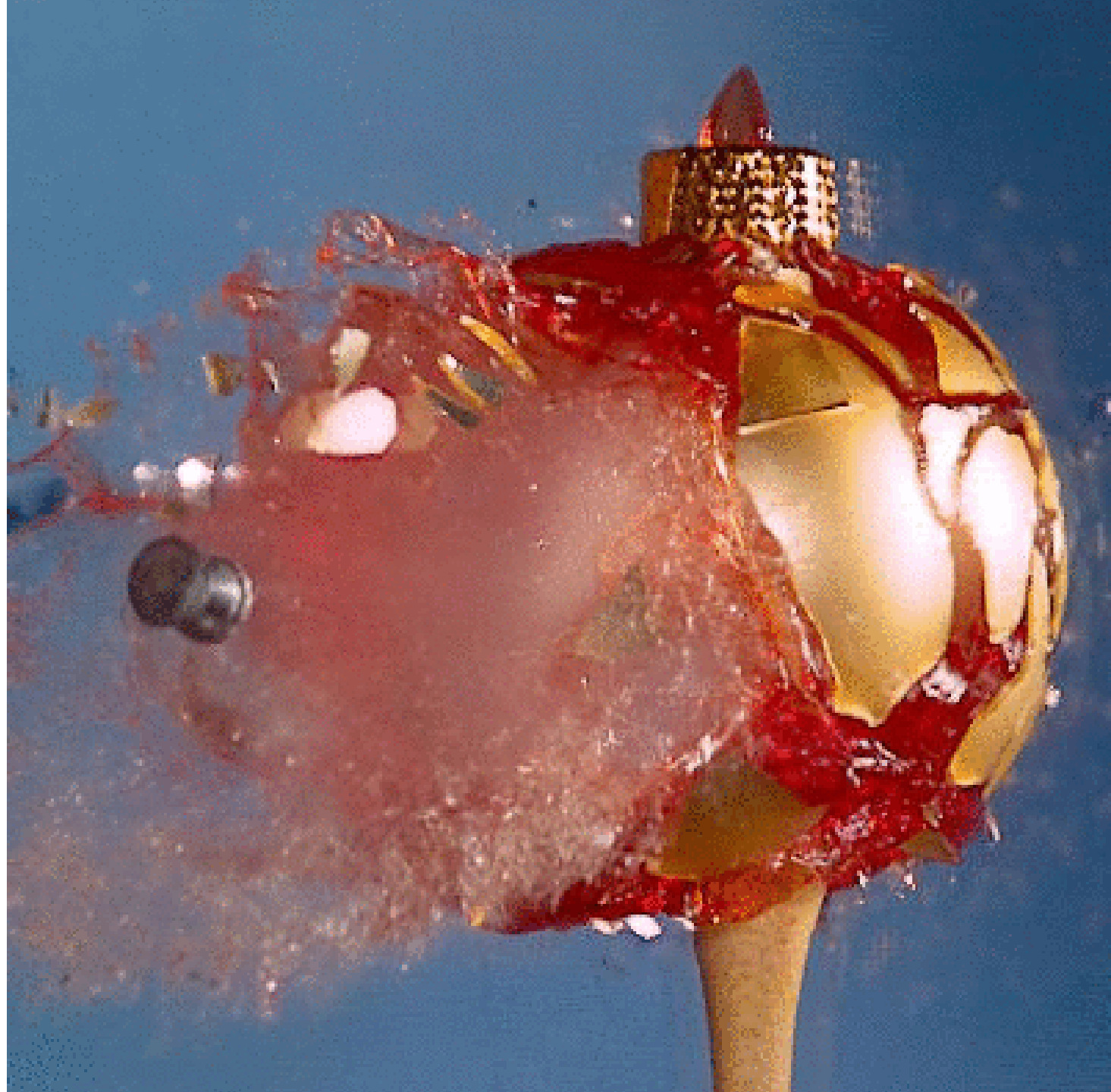
# How do I...

Simulate different viewpoints?

- Pick same pixel within each lenslet view



A '*new kind*'  
of pictures

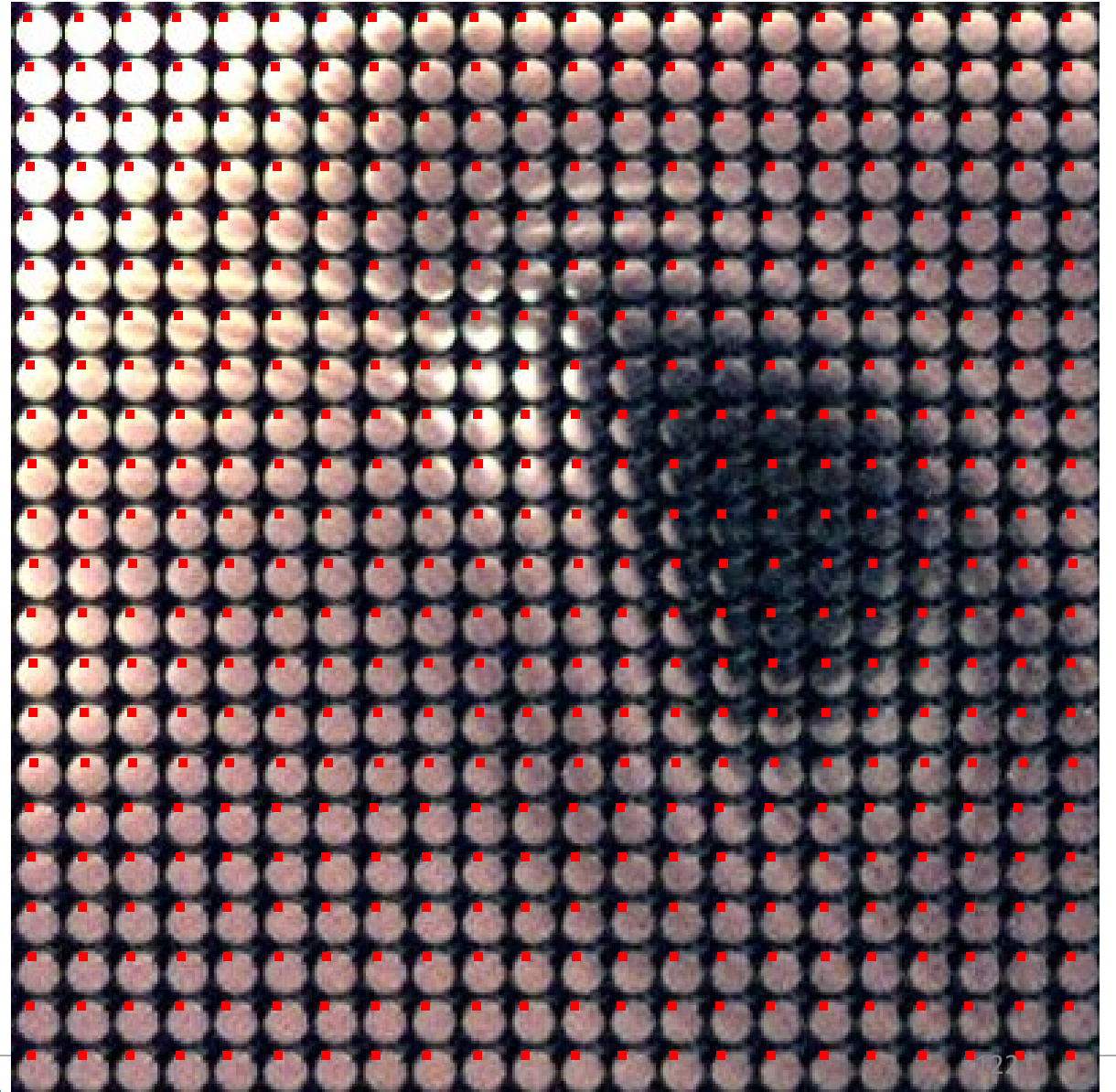


# How do I...

Simulate different viewpoints?

- Pick same pixel within each lenslet view

Simulate different aperture sizes?





# How do I...

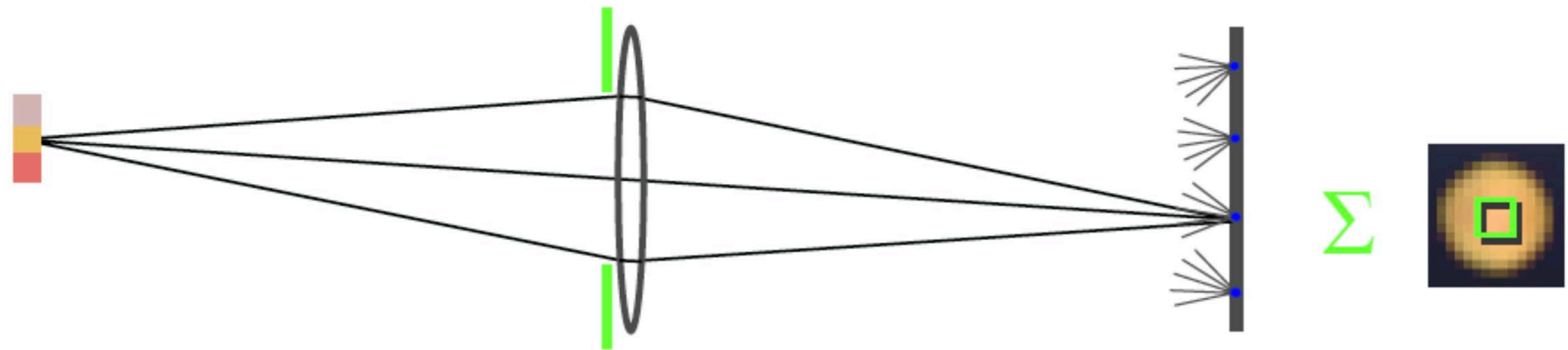
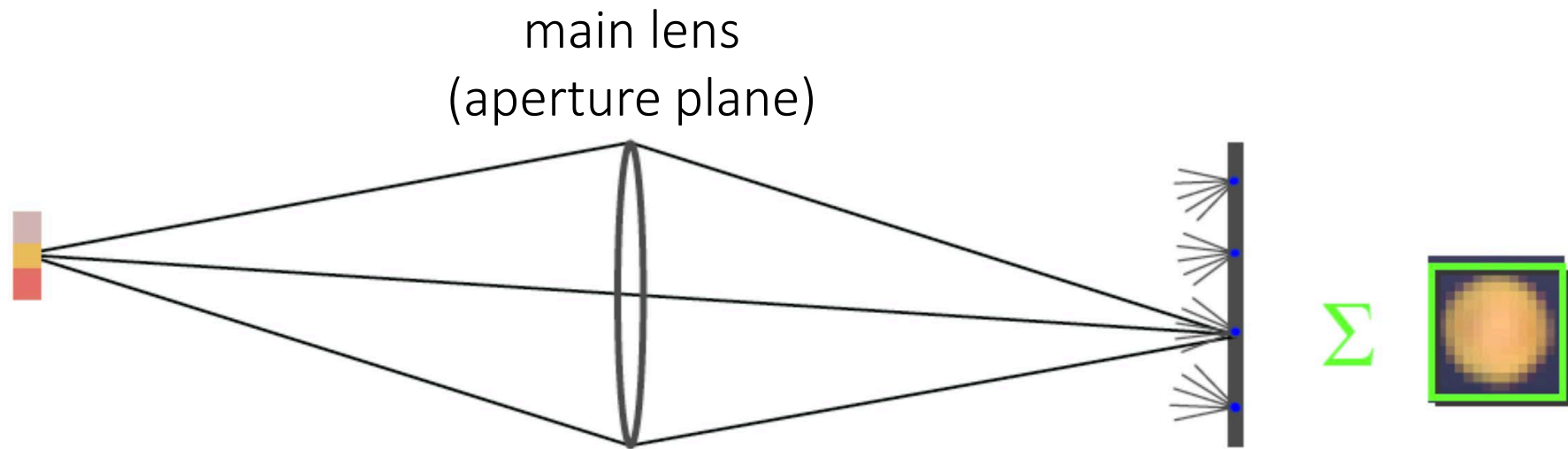
Simulate different viewpoints?

- Pick same pixel within each lenslet view

Simulate different aperture sizes?

- Sum more than one pixel within each lenslet view





**stopping down = summing only the central portion  
of each microlens**

# How do I...

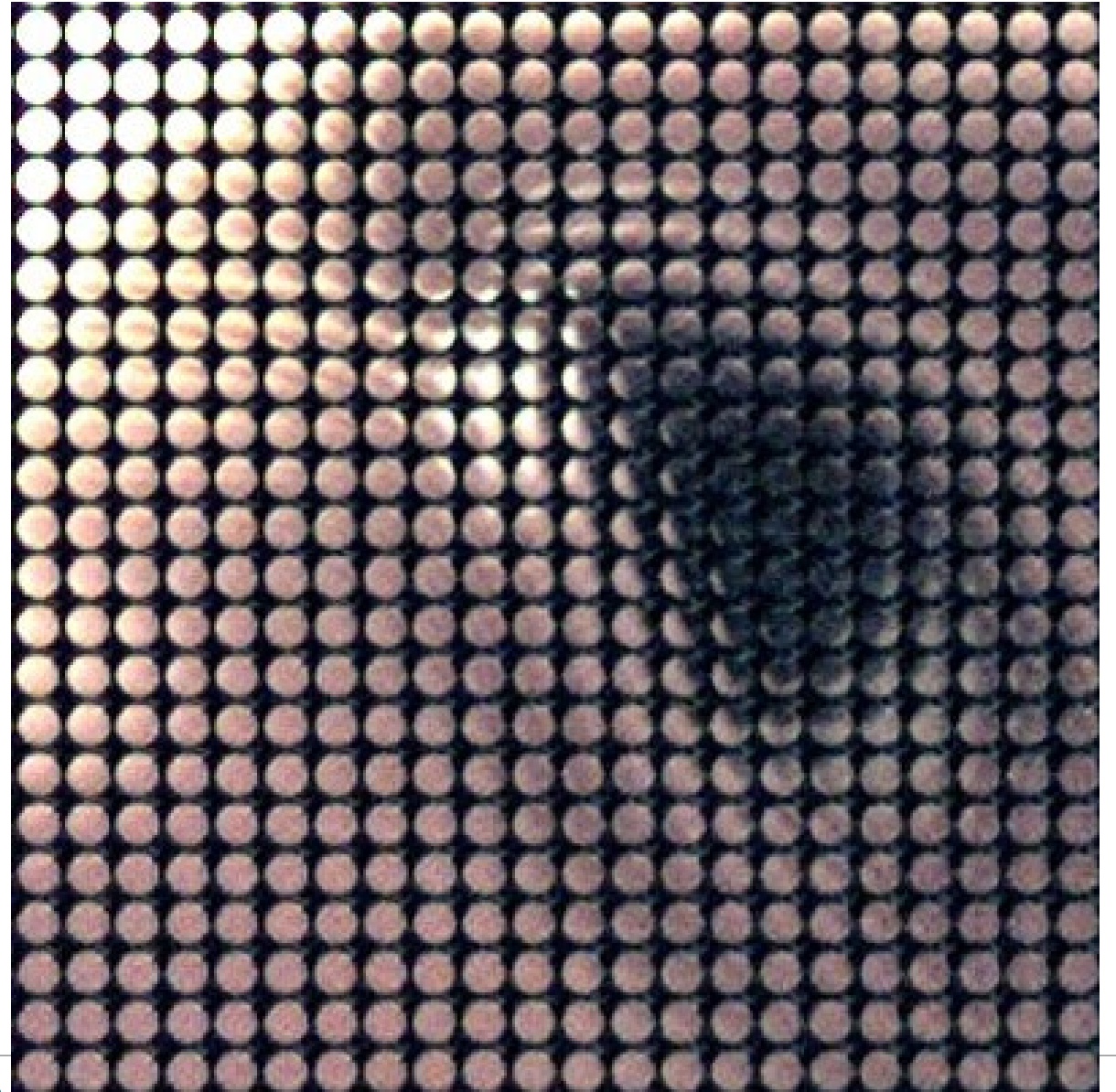
Simulate different viewpoints?

- Pick same pixel within each lenslet view

Simulate different aperture sizes?

- Sum more than one pixels within each lenslet view

Simulate focusing to different focus?



# *Not just view shifting...*

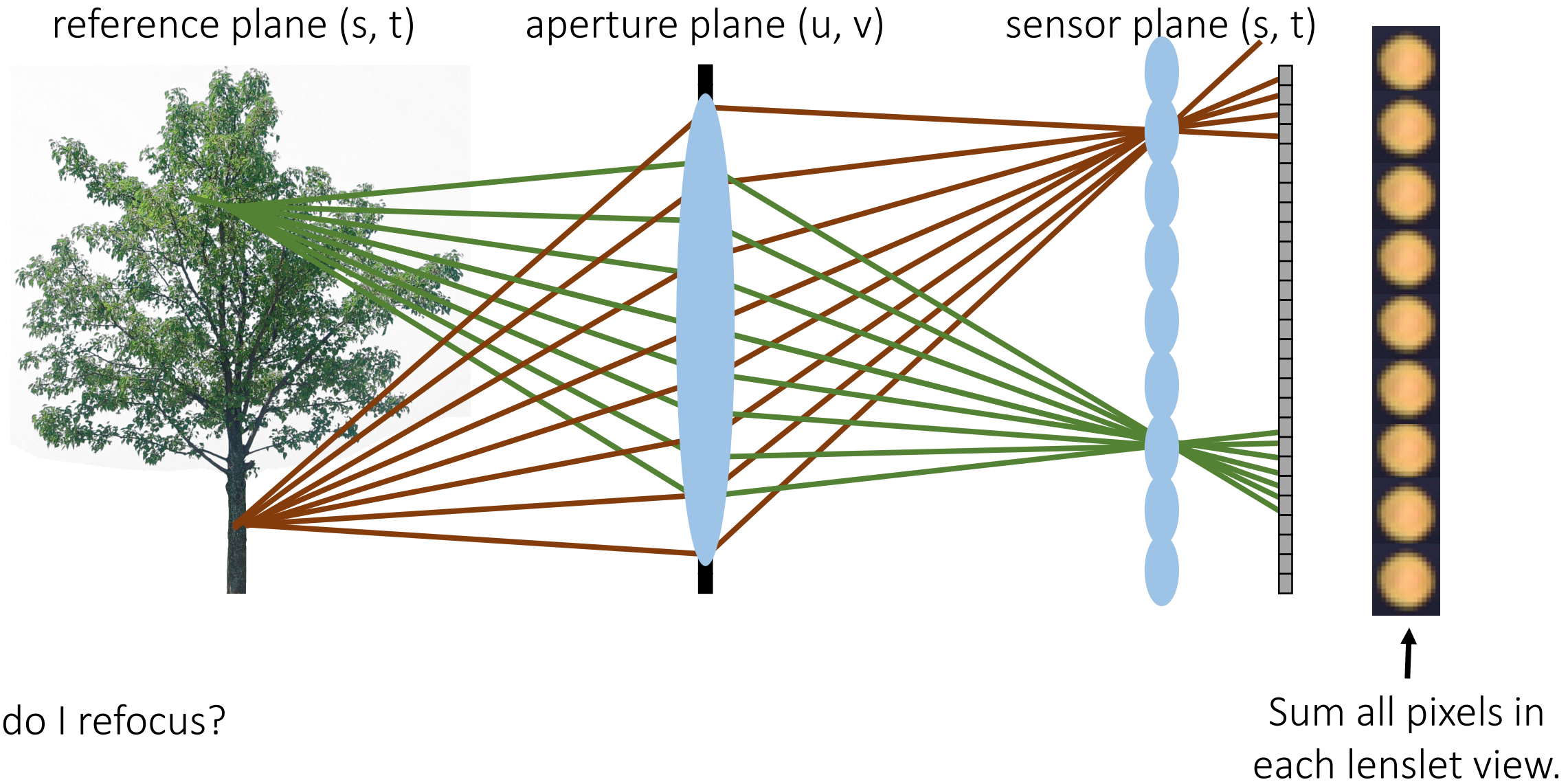
Source: LightField Forum



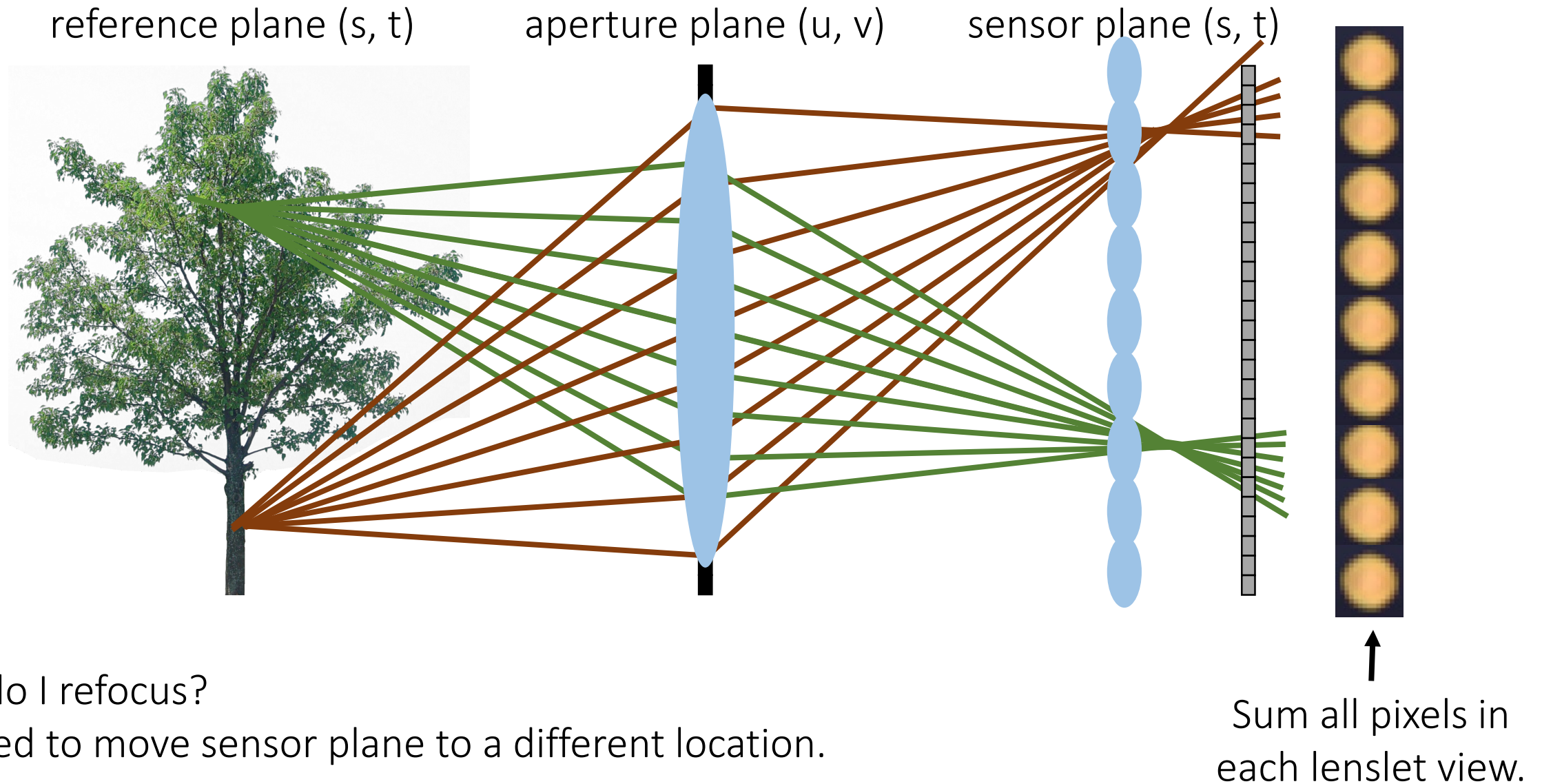
## Focus shifting



# Form lens image

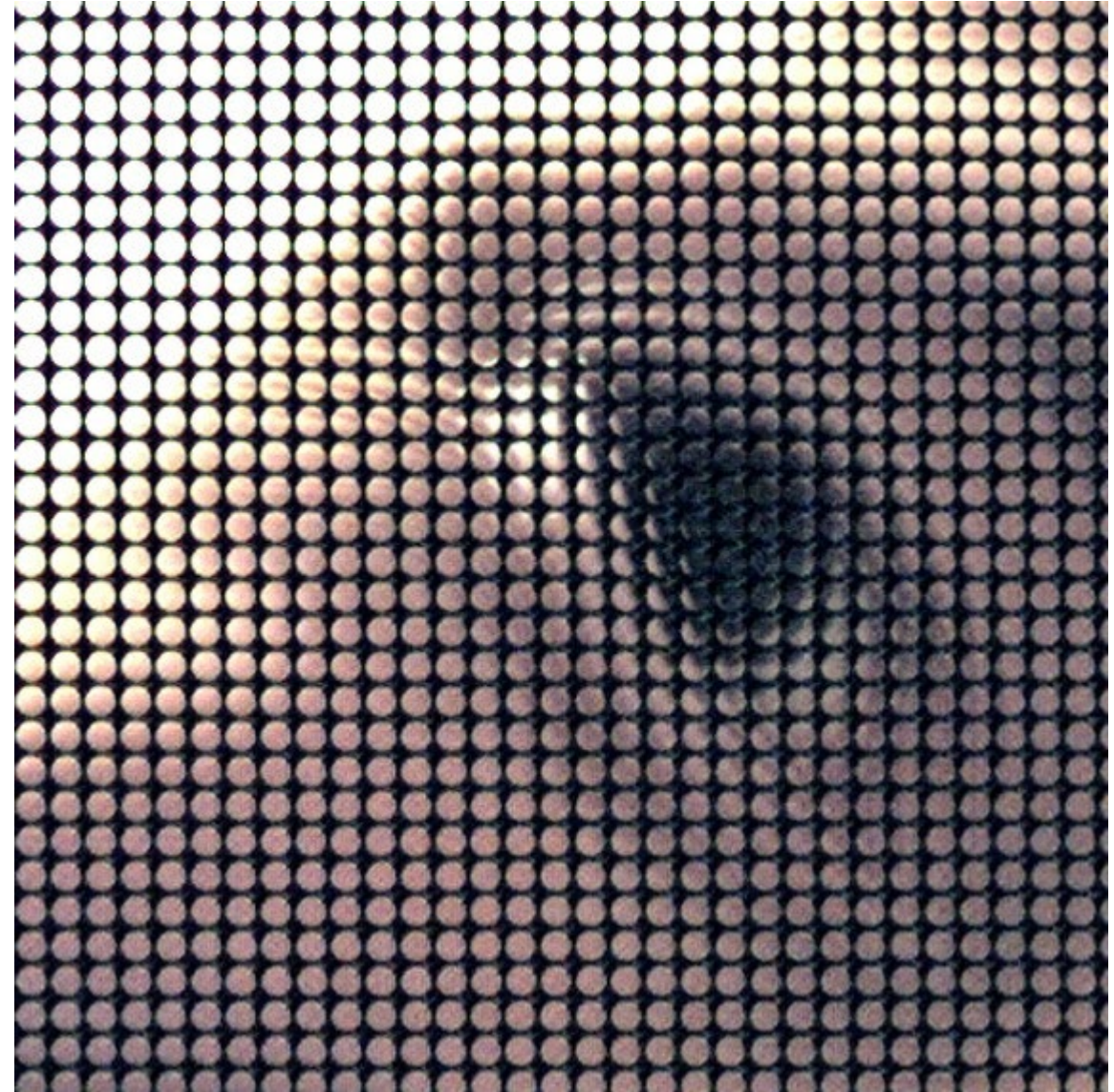
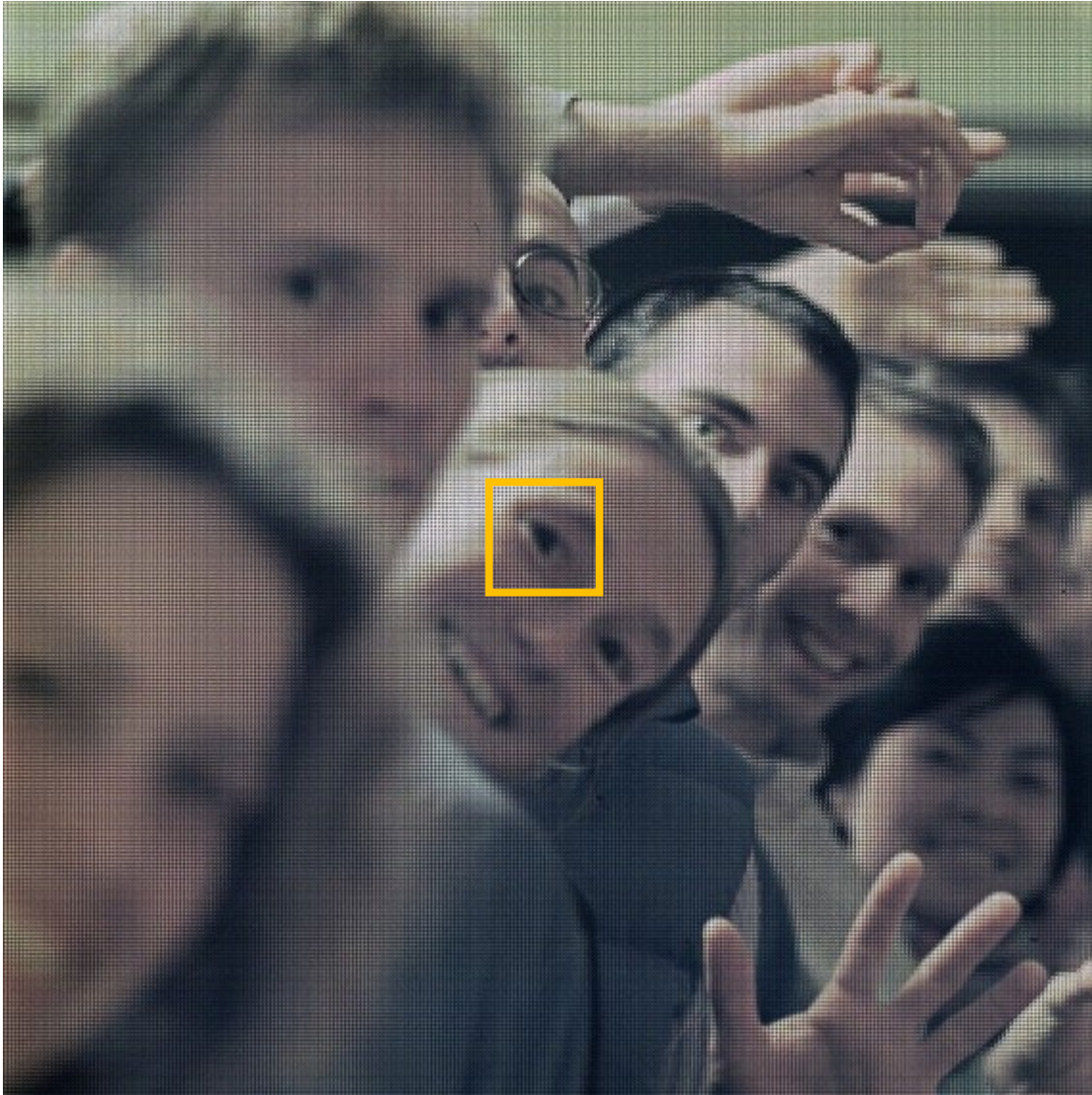


# Form lens image



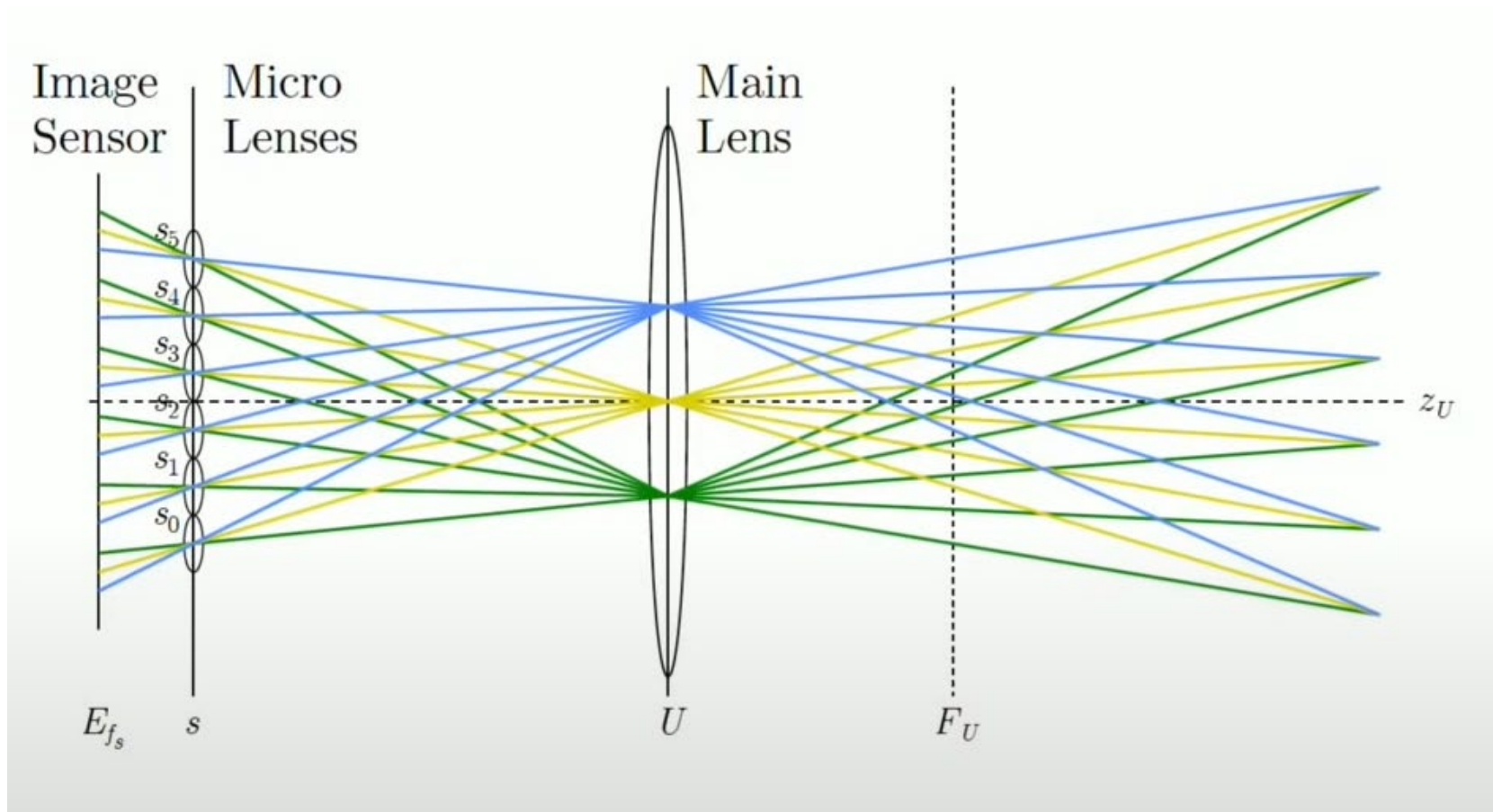


# A plenoptic “image”



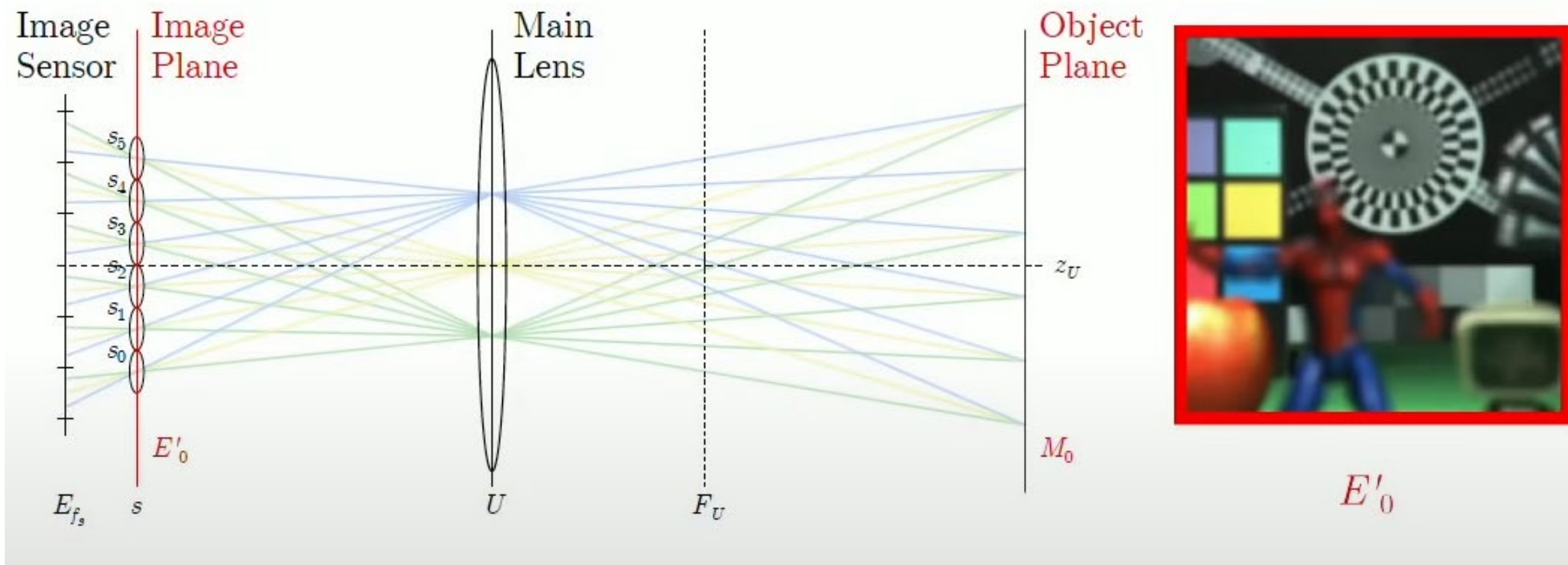
What are these circles?



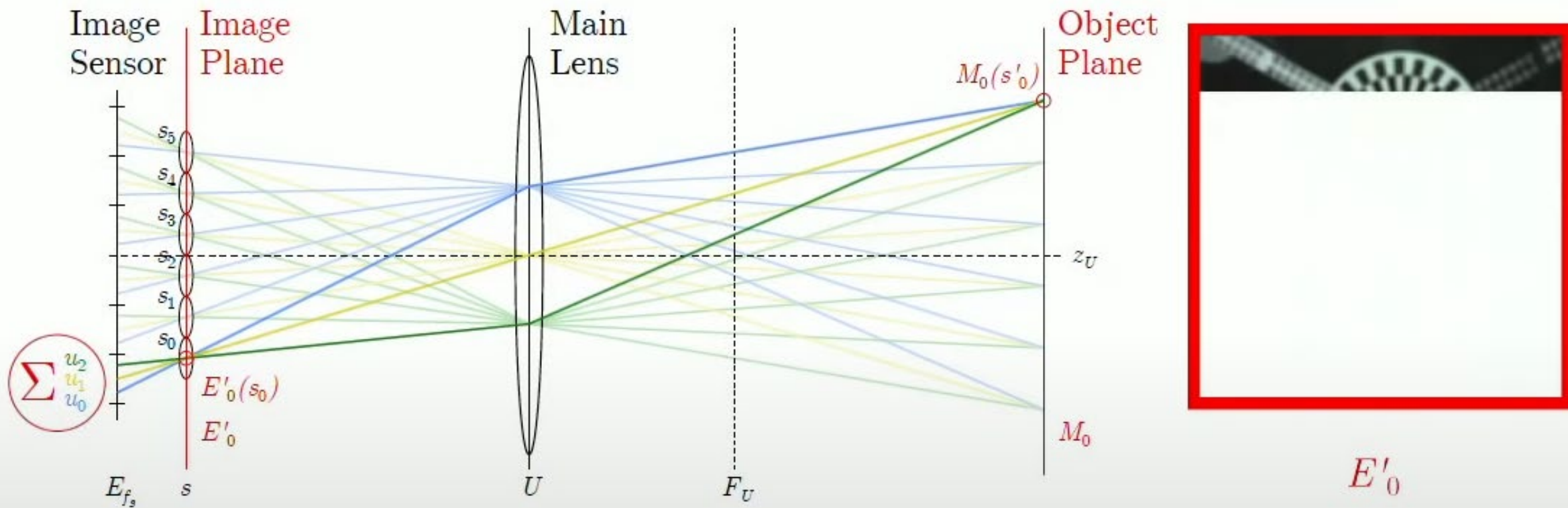


Source: <http://www.plenoptic.info/>

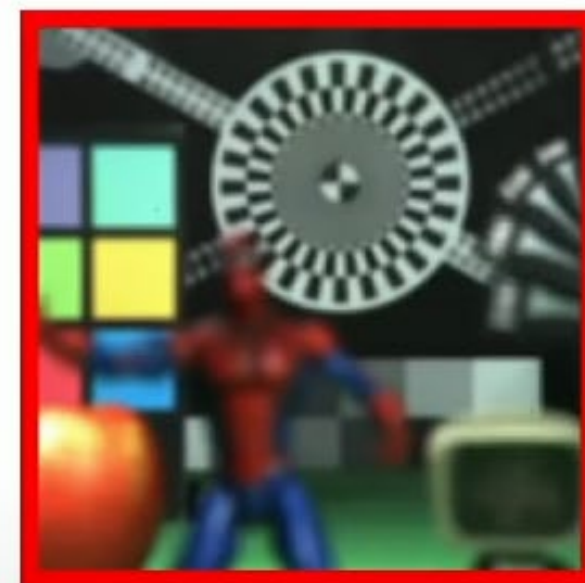
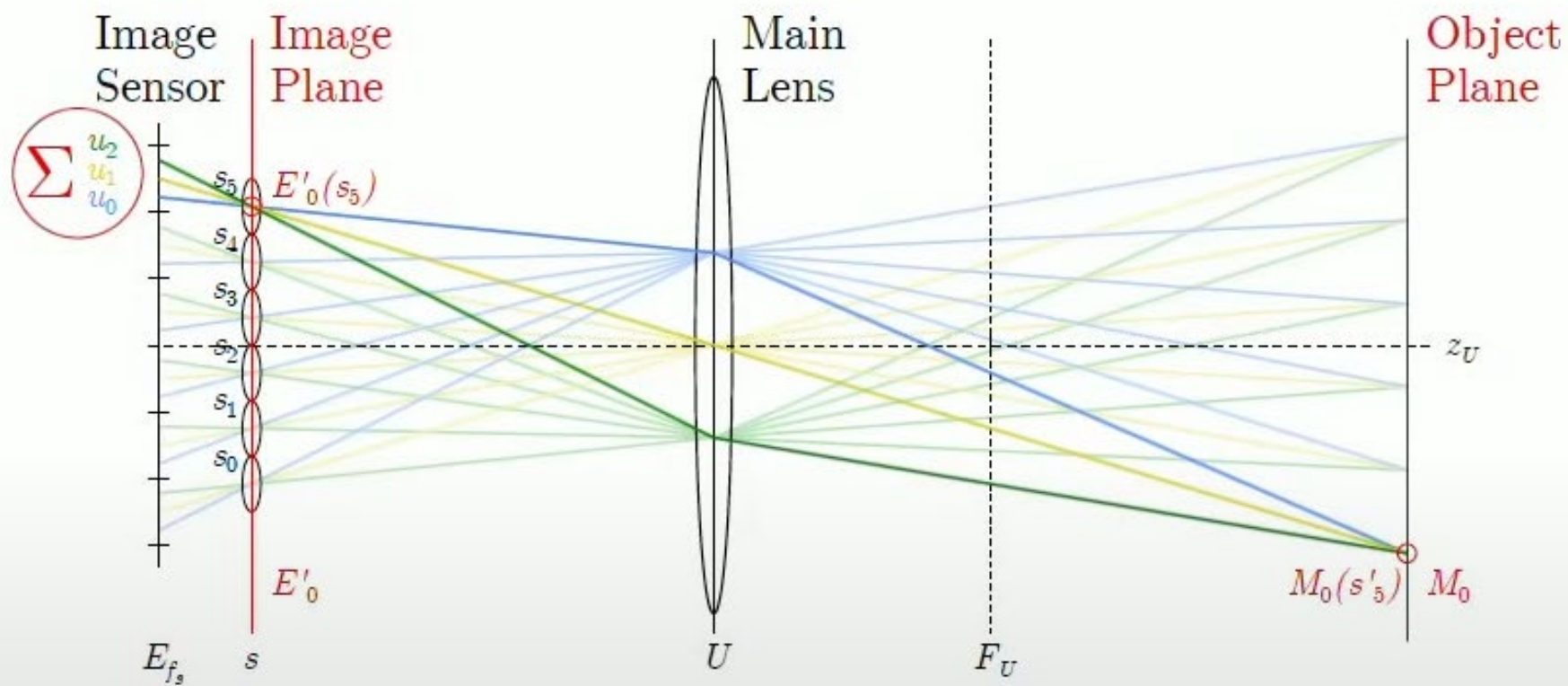




Source: <http://www.plenoptic.info/>



Source: <http://www.plenoptic.info/>

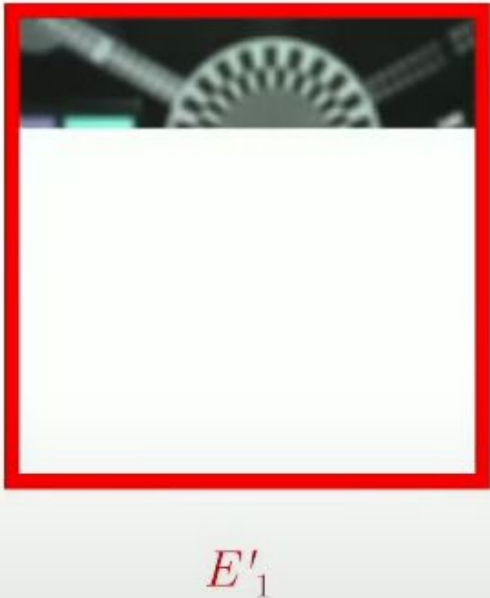
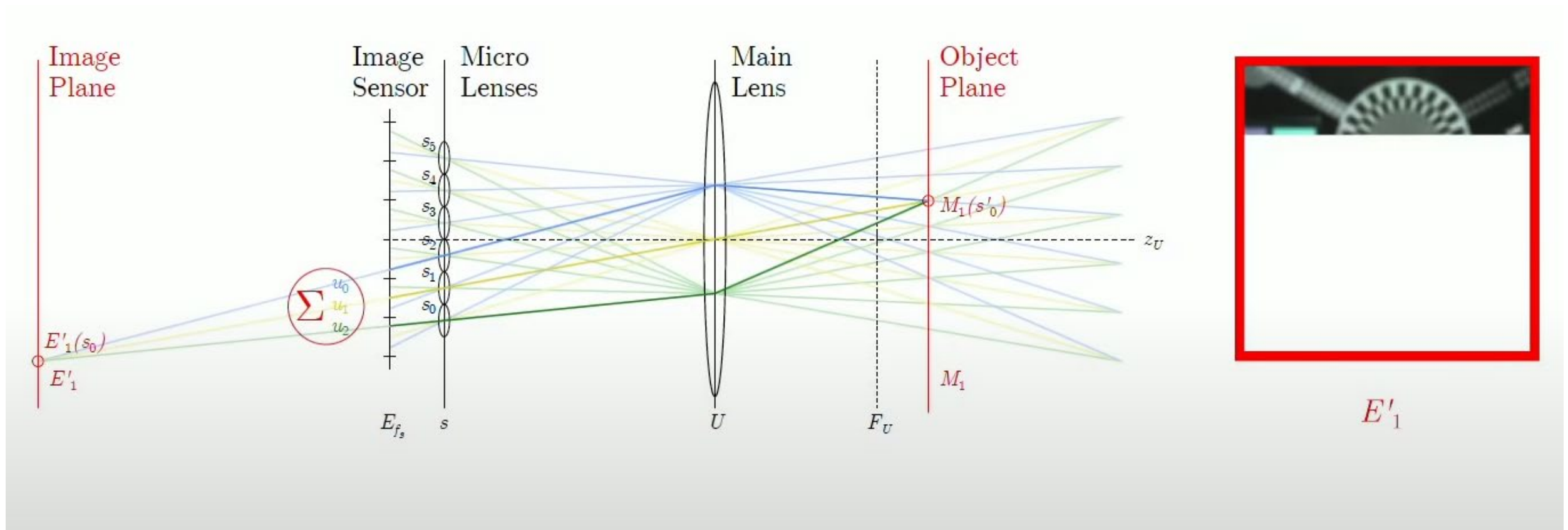


$E'_0$

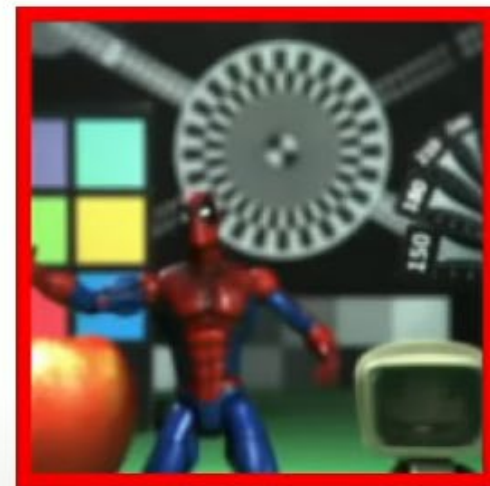
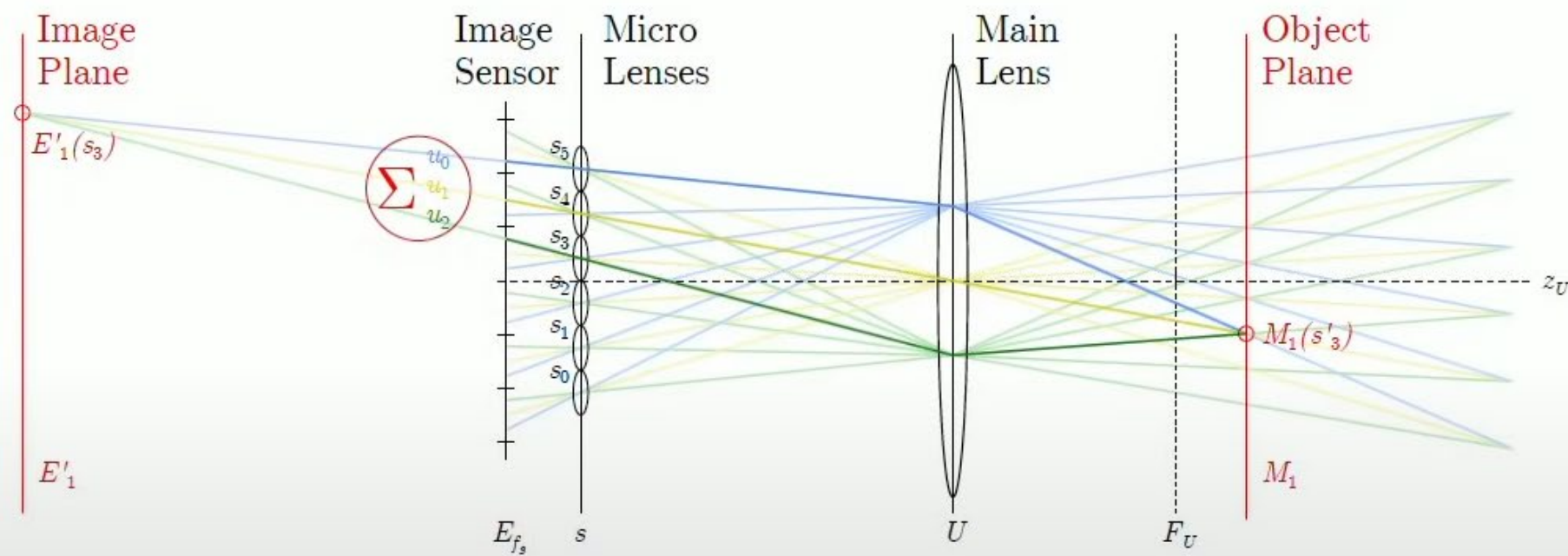
Source: <http://www.plenoptic.info/>







Source: <http://www.plenoptic.info/>



$E'_1$

Source: <http://www.plenoptic.info/>

# Refocusing



# Refocusing example





# Refocusing example



# Refocusing example



# Synthesizing images from a lightfield

Many more examples with demo: <http://lightfields.stanford.edu/>

# Three ways to measure a lightfield

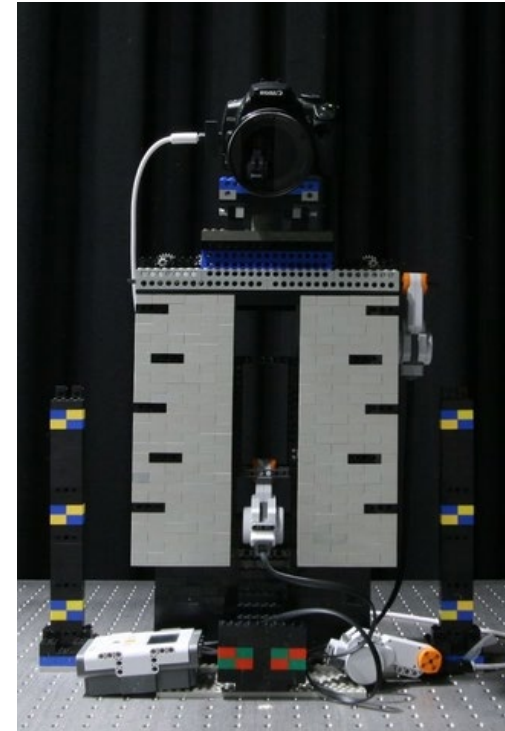
1) Use a plenoptic camera



2) Use a camera array



3) Use one camera multiple times

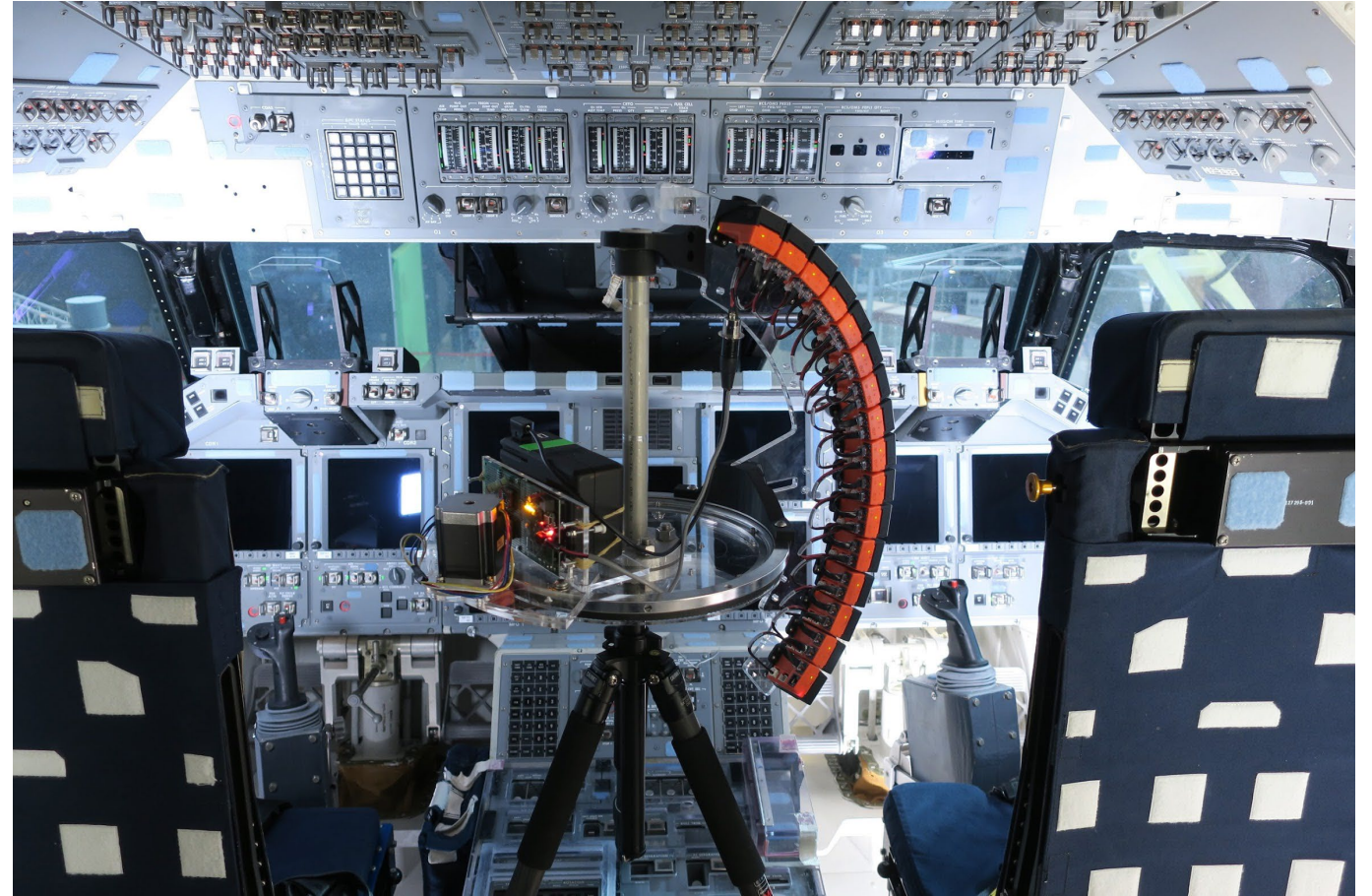


*Each has pros and cons.*



# Recent Advances

# Welcome to Light Fields (by Google)



For videos, paper, etc.: <https://augmentedperception.github.io/welcome-to-lightfields/>  
<https://augmentedperception.github.io/deepviewvideo/>





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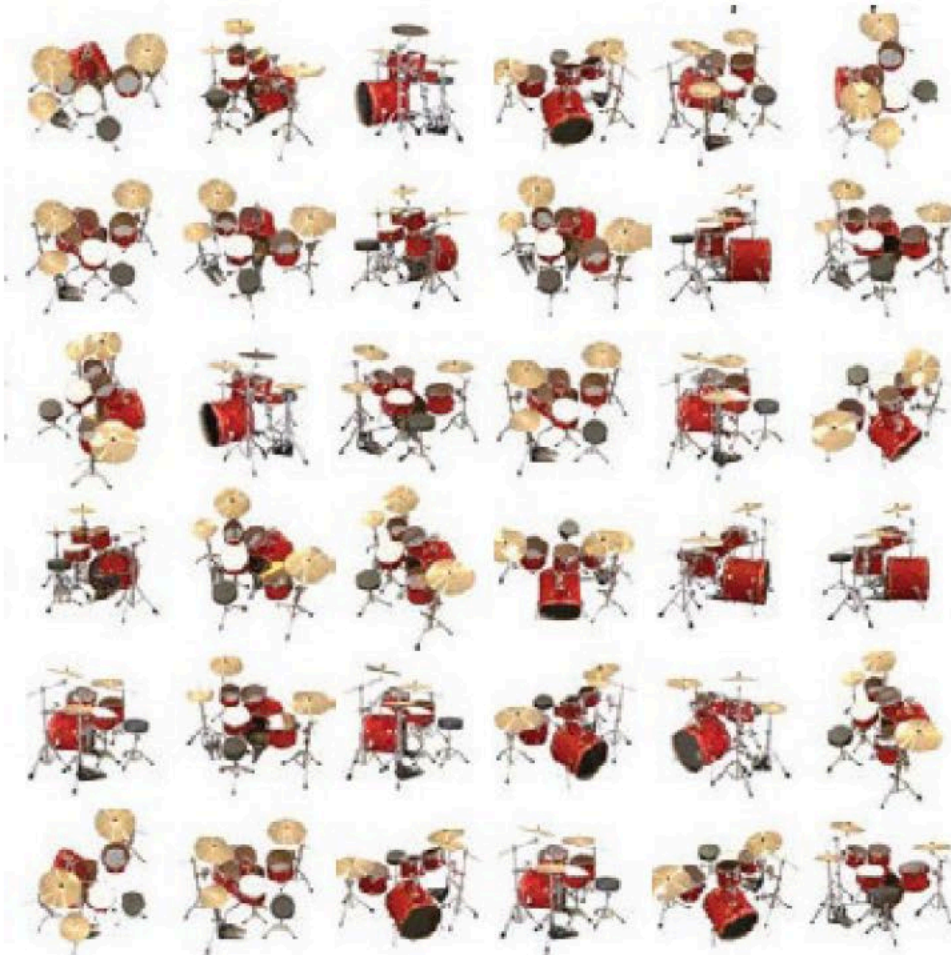




# NeRF: Neural radiance fields

[Mildenhall et al. 2020]

## Known data



NeRF can be considered as a computational method that attempts to infer a light field from a set of 2D images



# Take-home messages

Light fields capture spatial and angular information of a scene.

*This enables numerous applications, such as refocusing, depth extraction, scene reconstruction, viewpoint shifting...*

The concept builds on ideas that are >100 years old.

*Technological advances have enabled practical realization.*

*"We stand on the shoulders of giants."*

One step more towards full plenoptic imaging.

*Challenges: Efficient acquisition, higher spatio-angular resolution (and baseline), bandwidth, storage...*

# References and further reading

- Levoy and Hanrahan, “Light Field Rendering,” SIGGRAPH 1996.
- Gortler et al., “The Lumigraph,” SIGGRAPH 1996.  
The two papers introducing the light field.
- Ng et al., “Light field photography with a hand-held plenoptic camera,” Stanford TR 2005.  
The paper (re)-(re)-introducing the plenoptic camera, and the precursor to Lytro.
- Ng, “Fourier Slice Photography,” SIGGRAPH 2005.  
The paper on frequency-space analysis of refocusing and lightfield measurements.
- Wilburn et al., “High Performance Imaging Using Large Camera Arrays,” SIGGRAPH 2005.  
The camera array paper.
- Georgiev et al., “Unified Frequency Domain Analysis of Lightfield Cameras,” ECCV 2008.  
Make a lightfield camera from a pinhole array (and many other interesting stuff about lightfield cameras).
- Wu et al. “Light field image processing: an overview”, IEEE JSTSP 2017
- Jarabo et al. “How do people edit light fields?”, SIGGRAPH 2014.  
This last paper tackles the question of how to create “a new photoshop” for light fields.